



US012408451B2

(12) **United States Patent**
Umebayashi et al.

(10) **Patent No.:** **US 12,408,451 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SEMICONDUCTOR DEVICE, SOLID-STATE IMAGING DEVICE AND ELECTRONIC APPARATUS**

(58) **Field of Classification Search**

CPC H10F 39/182; H10F 39/018; H10F 39/18; H10F 39/199; H10F 39/8037;

(Continued)

(71) Applicant: **SONY GROUP CORPORATION**,
Tokyo (JP)

(56) **References Cited**

(72) Inventors: **Taku Umebayashi**, Kanagawa (JP);
Keiji Tatani, Kanagawa (JP); **Hajime Inoue**, Kanagawa (JP); **Ryuichi Kanamura**, Kumamoto (JP)

U.S. PATENT DOCUMENTS

9,007,503 B2 * 4/2015 Oike H04N 25/772
348/308

9,136,163 B2 9/2015 Kumano

(Continued)

(73) Assignee: **Sony Group Corporation**, Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

FOREIGN PATENT DOCUMENTS

CN 101355067 1/2009
CN 102054849 5/2011

(Continued)

(21) Appl. No.: **18/520,273**

(22) Filed: **Nov. 27, 2023**

(65) **Prior Publication Data**

US 2024/0096925 A1 Mar. 21, 2024

OTHER PUBLICATIONS

International Search Report prepared by the European Patent Office on Jan. 20, 2014, for International Application No. PCT/JP2013/006055.

(Continued)

Related U.S. Application Data

(63) Continuation of application No. 17/719,554, filed on Apr. 13, 2022, now Pat. No. 11,875,989, which is a
(Continued)

Primary Examiner — Dao H Nguyen

(74) *Attorney, Agent, or Firm* — SHERIDAN ROSS P.C.

(30) **Foreign Application Priority Data**

Oct. 18, 2012 (JP) 2012-230805
Apr. 22, 2013 (JP) 2013-089580

(51) **Int. Cl.**

H10F 39/18 (2025.01)

H10F 39/00 (2025.01)

H10F 39/12 (2025.01)

(52) **U.S. Cl.**

CPC **H10F 39/182** (2025.01); **H10F 39/018** (2025.01); **H10F 39/18** (2025.01);

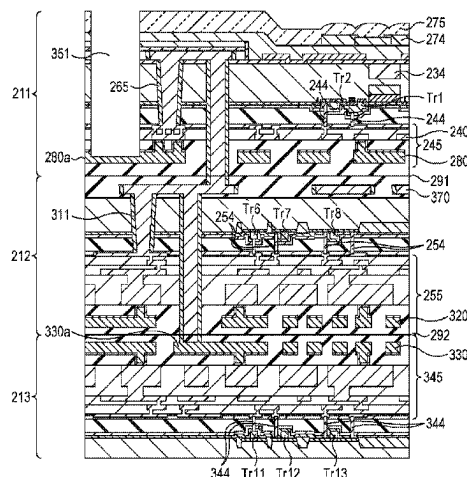
(Continued)

(57)

ABSTRACT

A semiconductor device including a first semiconductor section including a first wiring layer at one side thereof, the first semiconductor section further including a photodiode, a second semiconductor section including a second wiring layer at one side thereof, the first and second semiconductor sections being secured together, a third semiconductor section including a third wiring layer at one side thereof, the second and the third semiconductor sections being secured together such the first semiconductor section, second semiconductor section, and the third semiconductor section are stacked together, and a first conductive material electrically connecting at least two of (i) the first wiring layer, (ii) the second wiring layer, and (iii) the third wiring layer such that

(Continued)



the electrically connected wiring layers are in electrical communication.

15 Claims, 44 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/068,783, filed on Oct. 12, 2020, now Pat. No. 11,374,049, which is a continuation of application No. 16/555,067, filed on Aug. 29, 2019, now Pat. No. 10,840,290, which is a continuation of application No. 16/373,105, filed on Apr. 2, 2019, now Pat. No. 10,475,845, which is a continuation of application No. 16/042,094, filed on Jul. 23, 2018, now Pat. No. 10,535,700, which is a continuation of application No. 15/852,468, filed on Dec. 22, 2017, now Pat. No. 10,128,301, which is a continuation of application No. 15/403,154, filed on Jan. 10, 2017, now Pat. No. 9,917,131, which is a continuation of application No. 15/087,894, filed on Mar. 31, 2016, now Pat. No. 9,570,499, which is a continuation of application No. 14/434,288, filed as application No. PCT/JP2013/006055 on Oct. 10, 2013, now Pat. No. 9,431,450.

(52) U.S. Cl.

CPC *H10F 39/199* (2025.01); *H10F 39/8037* (2025.01); *H10F 39/8053* (2025.01); *H10F 39/8057* (2025.01); *H10F 39/8063* (2025.01); *H10F 39/809* (2025.01); *H10F 39/811* (2025.01)

(58) Field of Classification Search

CPC H10F 39/8053; H10F 39/8057; H10F 39/8063; H10F 39/809; H10F 39/811
See application file for complete search history.

(56) References Cited

U.S. PATENT DOCUMENTS

9,252,169 B2	2/2016	Kobayashi
9,385,152 B2	7/2016	Endo et al.
9,431,450 B2	8/2016	Umebayashi et al.
9,570,499 B2	2/2017	Umebayashi et al.
9,917,131 B2	3/2018	Umebayashi et al.
10,128,301 B2	11/2018	Umebayashi et al.
10,475,845 B2	11/2019	Umebayashi et al.
10,535,700 B2	1/2020	Umebayashi et al.
10,840,290 B2	11/2020	Umebayashi et al.
11,374,049 B2	6/2022	Umebayashi et al.
2009/0224345 A1	9/2009	Lee et al.
2010/0026865 A1	2/2010	Tivarus et al.
2010/0038802 A1	2/2010	Chen et al.
2010/0276572 A1	11/2010	Iwabuchi et al.
2011/0049336 A1 *	3/2011	Matsunuma H01L 27/14692 257/431
2011/0168434 A1 *	7/2011	Farooq H01L 21/2007 174/257
2012/0056288 A1	3/2012	Yoshihara et al.
2012/0062777 A1	3/2012	Shoji et al.
2012/0199930 A1	8/2012	Toshihko
2012/0241981 A1 *	9/2012	Hirano H01L 23/562 257/777
2013/0221473 A1	8/2013	Shimotsusa et al.
2013/0307103 A1	11/2013	Lin et al.
2014/0048853 A1	2/2014	Choi et al.
2015/0035109 A1	2/2015	Kataoka

2015/0091066 A1 *	4/2015	Lee H01L 25/0657 257/369
2018/0323231 A1	11/2018	Saito
2022/0246668 A1	8/2022	Umebayashi et al.

FOREIGN PATENT DOCUMENTS

CN	102110700	6/2011	
EP	2426718	3/2012	
EP	2426719	3/2012	
JP	2001-339057	* 7/2001	 H01L 27/146
JP	2006-147749	6/2006	
JP	2007-228460	9/2007	
JP	2009-176777	8/2009	
JP	2010-245506	10/2010	
JP	2011-096851	5/2011	
JP	2011-049445	10/2011	
JP	2012-015278	1/2012	
JP	2012-033894	2/2012	
JP	2012-054450	3/2012	
JP	2012-054876	3/2012	
JP	2012-064709	3/2012	
JP	2012-164870	8/2012	
KR	10-2001-0006328	1/2001	
KR	10-2008-0019652	3/2008	
KR	10-2011-0023758	3/2011	
TW	200709407	3/2007	
TW	200803484	1/2008	
WO	WO 2006/129762	12/2006	
WO	WO 2007/105478	9/2007	

OTHER PUBLICATIONS

Official Action (no English translation available) for Japanese Patent Application No. 2013-089580 mailed Jan. 26, 2017, 18 pages.

Official Action (no English translation available) for Japanese Patent Application No. 2013-089580 mailed Apr. 4, 2017, 5 pages.

Official Action (with English translation) for Chinese Patent Application No. 201380053203.0 dated Jan. 23, 2017, 19 pages.

Official Action (no English translation available) for Japanese Patent Application No. 2013-089580 dated Aug. 14, 2017, 11 pages.

Official Action for European Patent Application No. 13780221.1 dated Nov. 23, 2017, 8 pages.

Official Action (no translation available) for Japanese Patent Application No. 2017-091571, dated Mar. 15, 2018, 10 pages.

Official Action (no translation available) for Japanese Patent Application No. 2013-089580, dated Mar. 29, 2018, 33 pages.

Official Action (no translation available) for Japanese Patent Application No. 2018-139958, dated May 9, 2019, 3 pages.

Extended European Search Report for European Patent Application No. 19195353.8, dated Dec. 18, 2019, 13 pages.

Official Action (no English translation) for Japanese Patent Application No. 2018-139958, dated Jan. 7, 2020, 5 pages.

Official Action (with English translation) for Korean Patent Application No. 10-2015-7008512, dated Jan. 28, 2020, 13 pages.

Official Action for U.S. Appl. No. 14/434,288, dated Nov. 17, 2015, 15 pages.

Notice of Allowance for U.S. Appl. No. 14/434,288, dated Apr. 26, 2016, 7 pages.

Official Action for U.S. Appl. No. 15/087,894, dated May 20, 2016, 10 pages.

Notice of Allowance for U.S. Appl. No. 15/087,894, dated Sep. 28, 2016, 8 pages.

Official Action for U.S. Appl. No. 15/403,154, dated Mar. 8, 2017, 6 pages.

Notice of Allowance for U.S. Appl. No. 15/403,154, dated Oct. 24, 2017, 8 pages.

Official Action for U.S. Appl. No. 15/852,468, dated Mar. 9, 2018, 7 pages.

Notice of Allowance for U.S. Appl. No. 15/852,468, dated Jul. 2, 2018, 7 pages.

Official Action for U.S. Appl. No. 16/042,094, dated Apr. 18, 2019, 7 pages.

(56)

References Cited

OTHER PUBLICATIONS

Notice of Allowance for U.S. Appl. No. 16/042,094, dated Aug. 14, 2019, 7 pages.

Notice of Allowance for U.S. Appl. No. 16/373,105, dated Jul. 8, 2019, 10 pages.

Official Action for U.S. Appl. No. 16/555,067, dated May 14, 2020, 6 pages.

Notice of Allowance for U.S. Appl. No. 16/555,067, dated Jul. 16, 2020, 8 pages.

Notice of Allowance for U.S. Appl. No. 17/068,783, dated Feb. 18, 2022, 9 pages.

Official Action for U.S. Appl. No. 17/719,554, dated Apr. 12, 2023, 11 pages.

Notice of Allowance for U.S. Appl. No. 17/719,554, dated Aug. 31, 2023, 5 pages.

* cited by examiner

FIG. 1

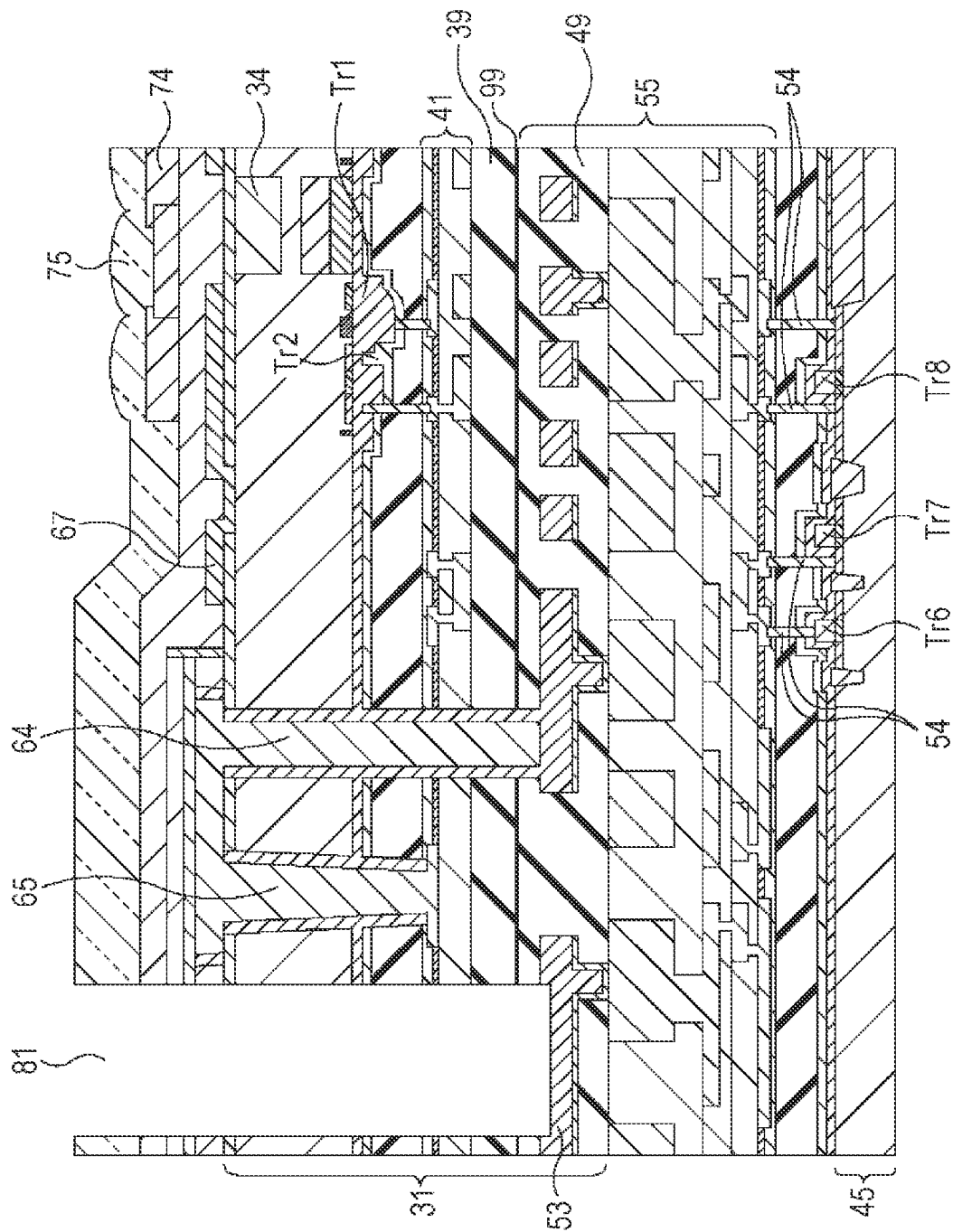


FIG. 2

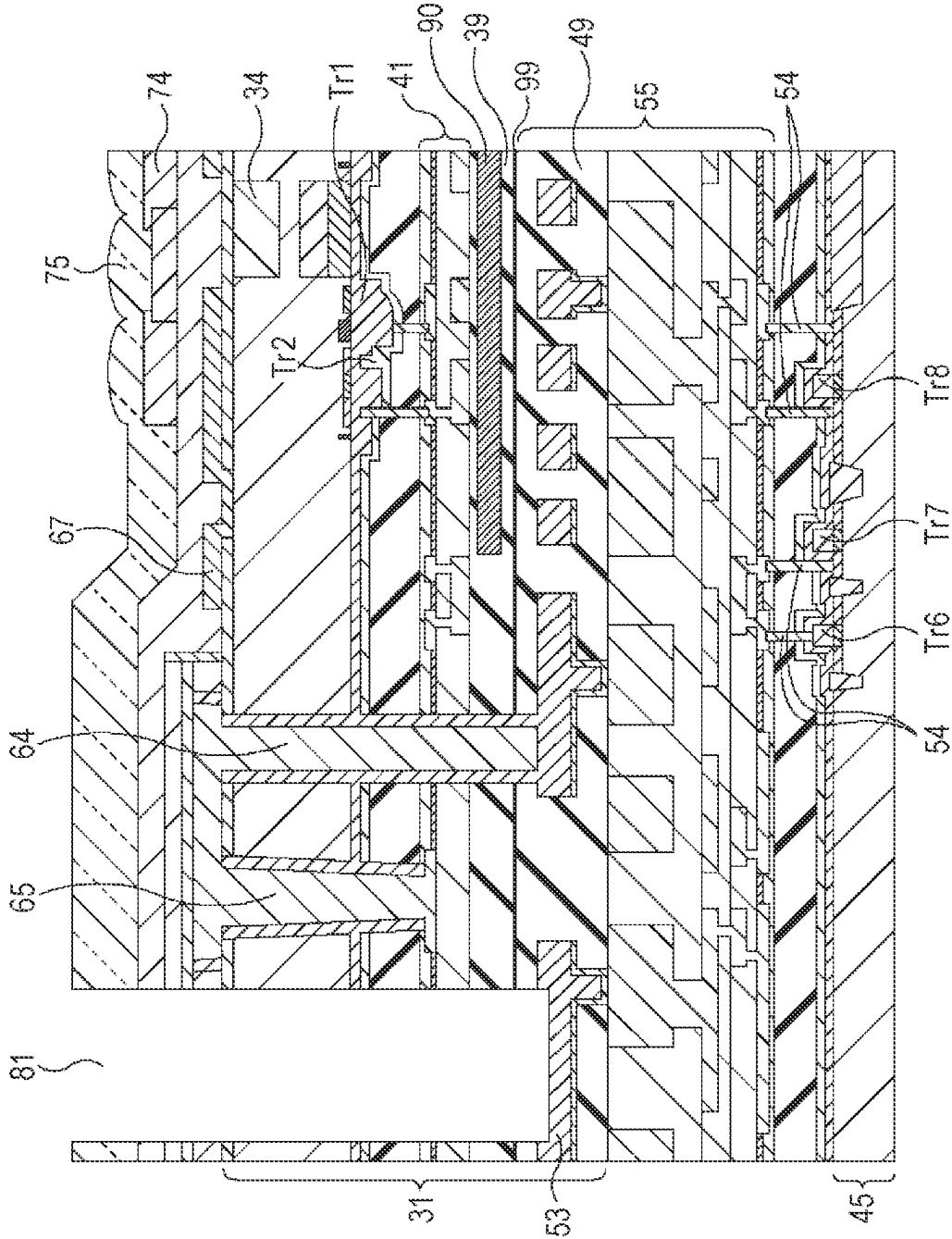


FIG. 3

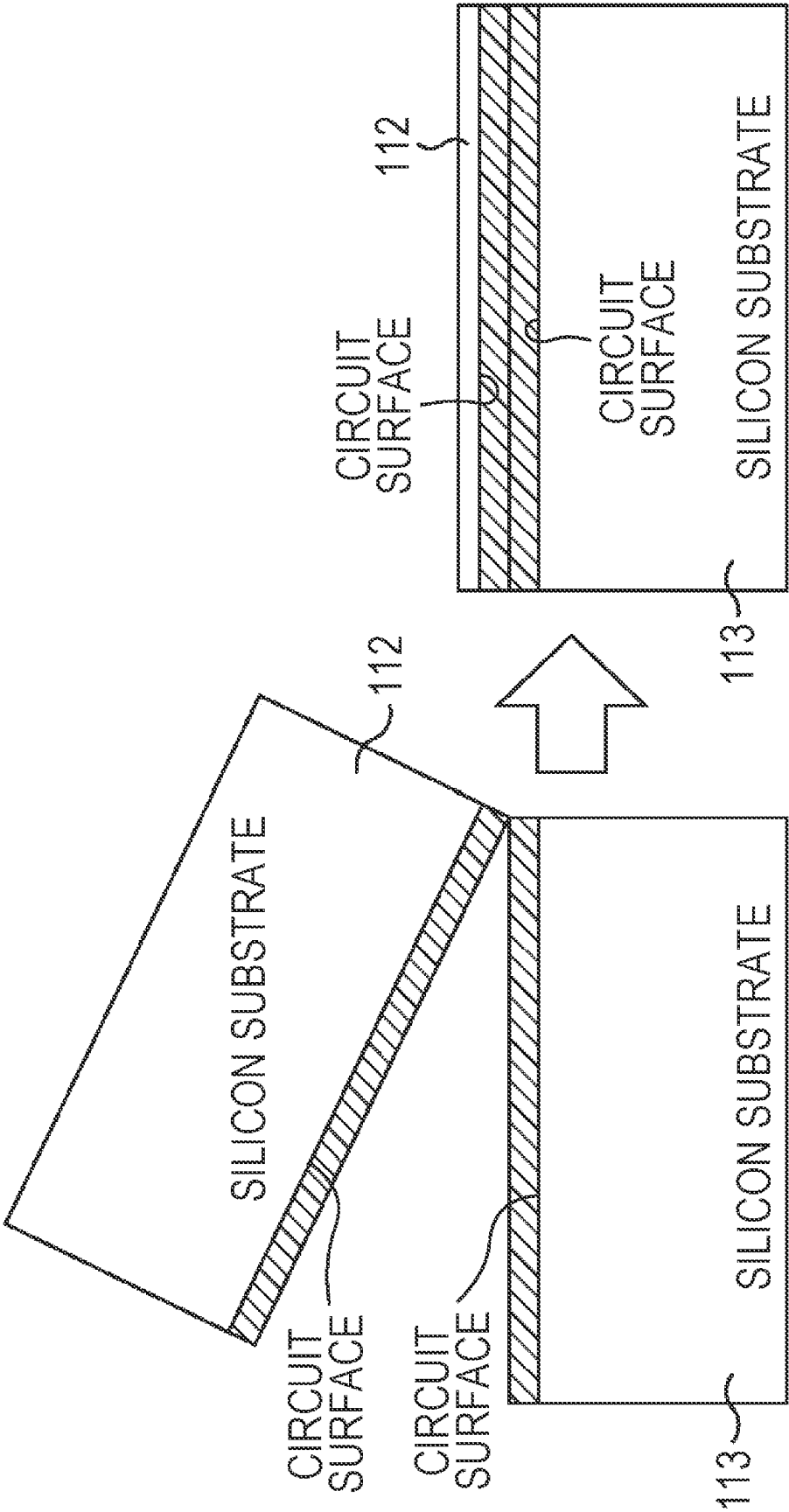


FIG. 4

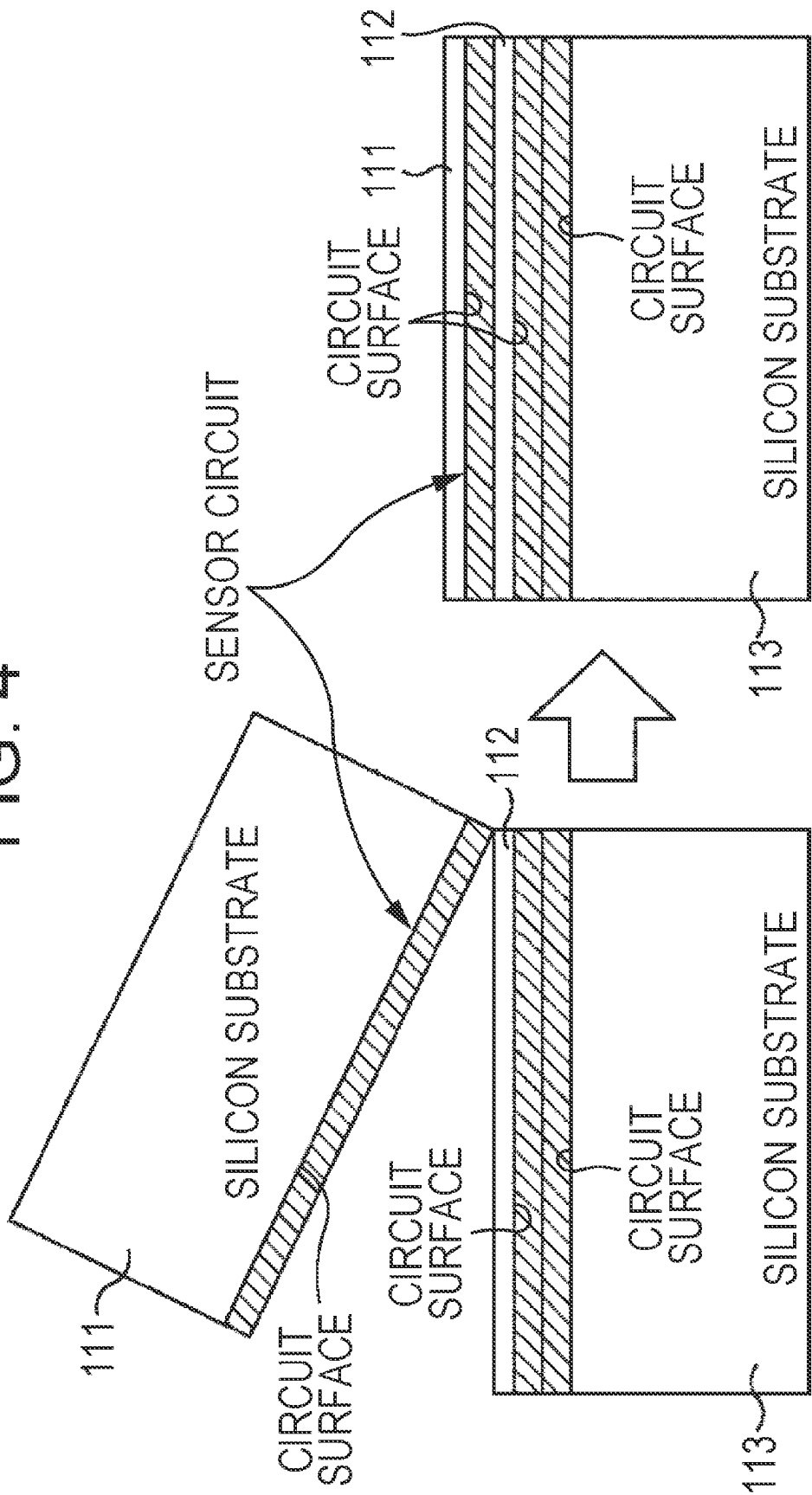


FIG. 5

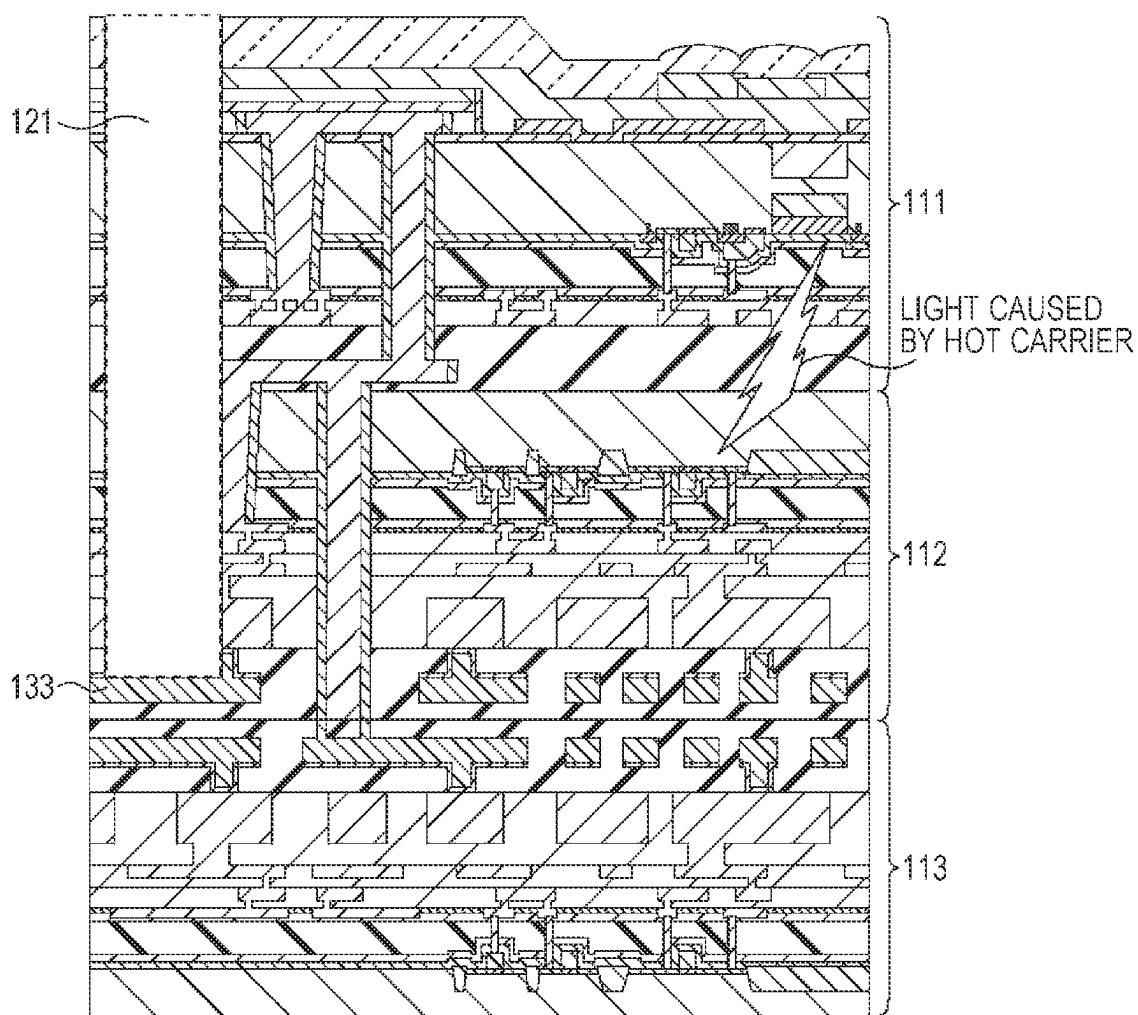


FIG. 6

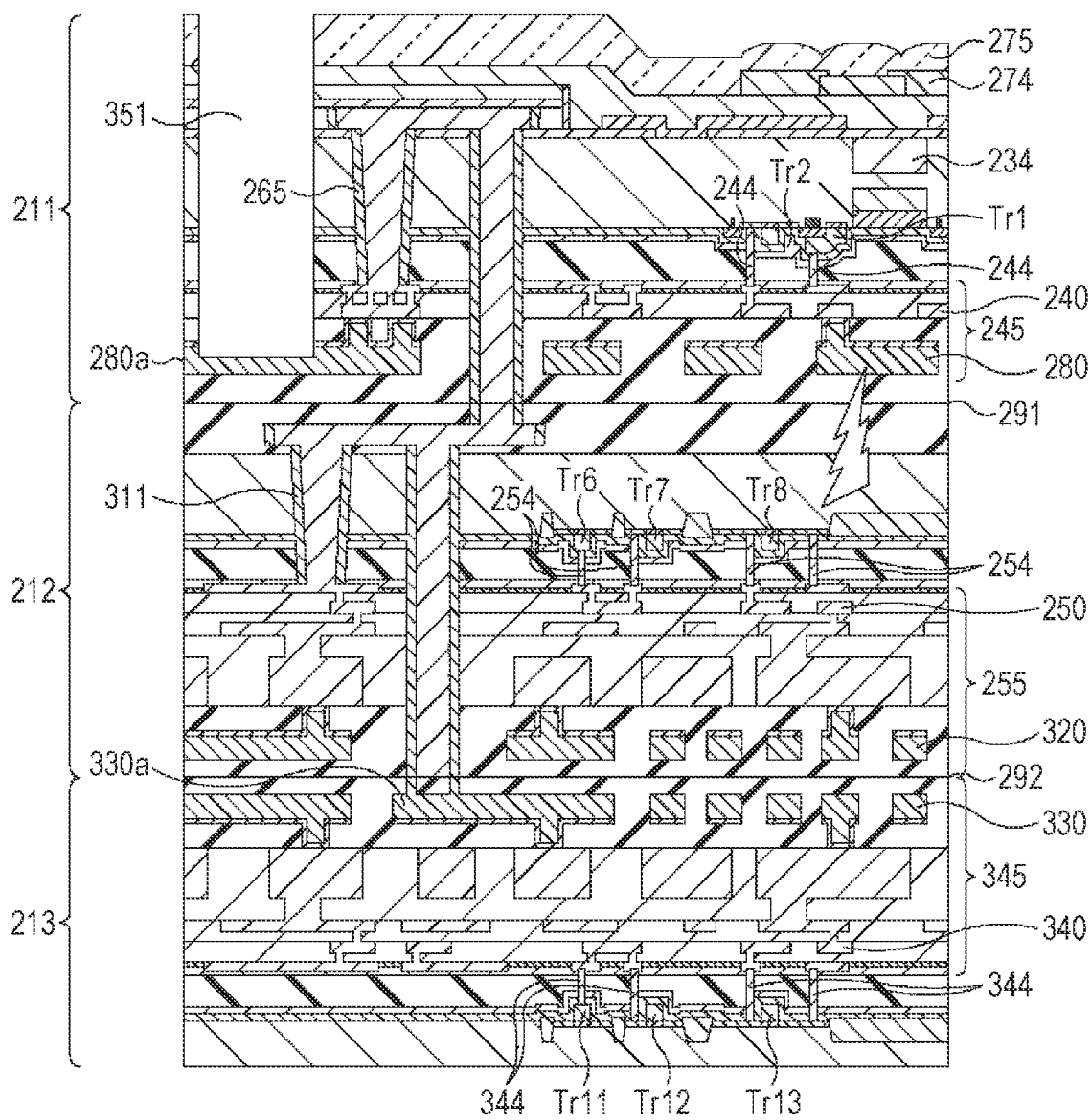


FIG. 7A

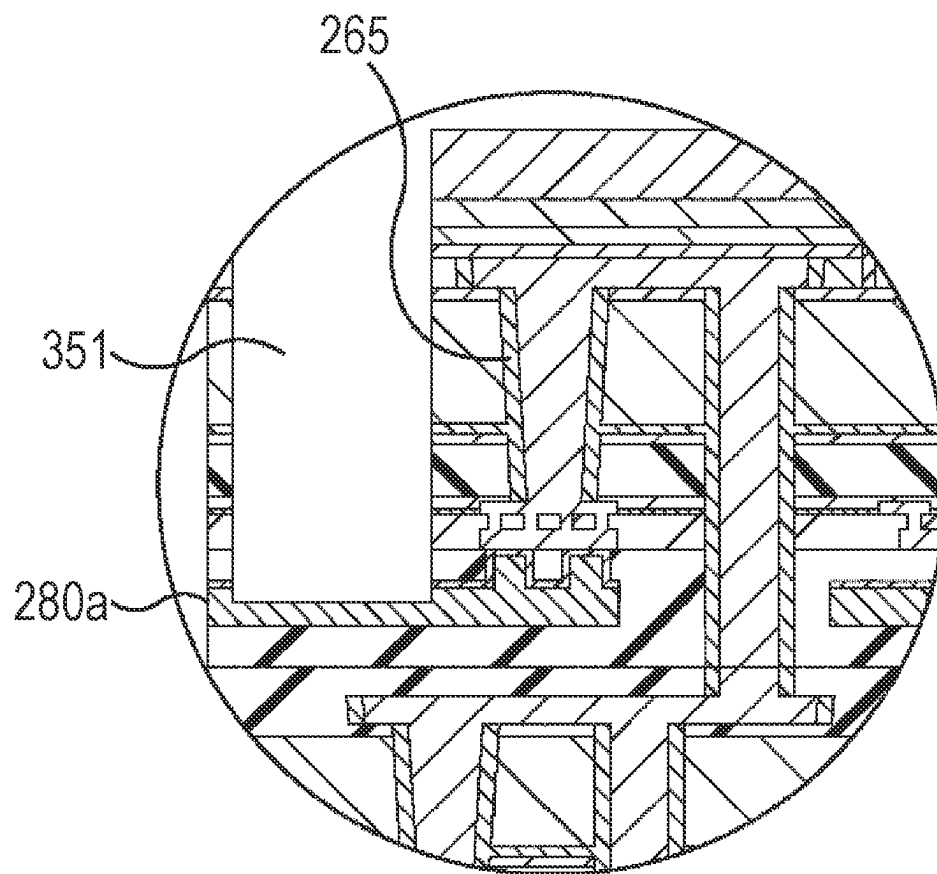


FIG. 7B

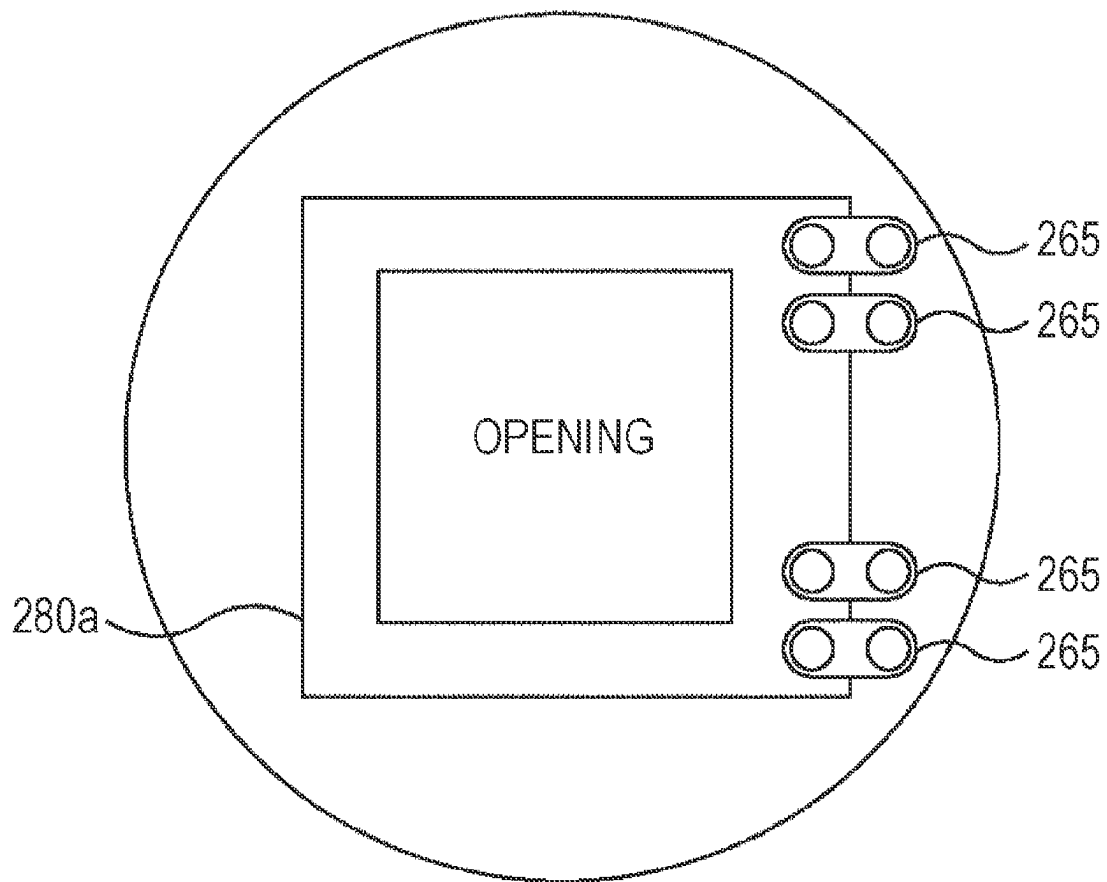


FIG. 8

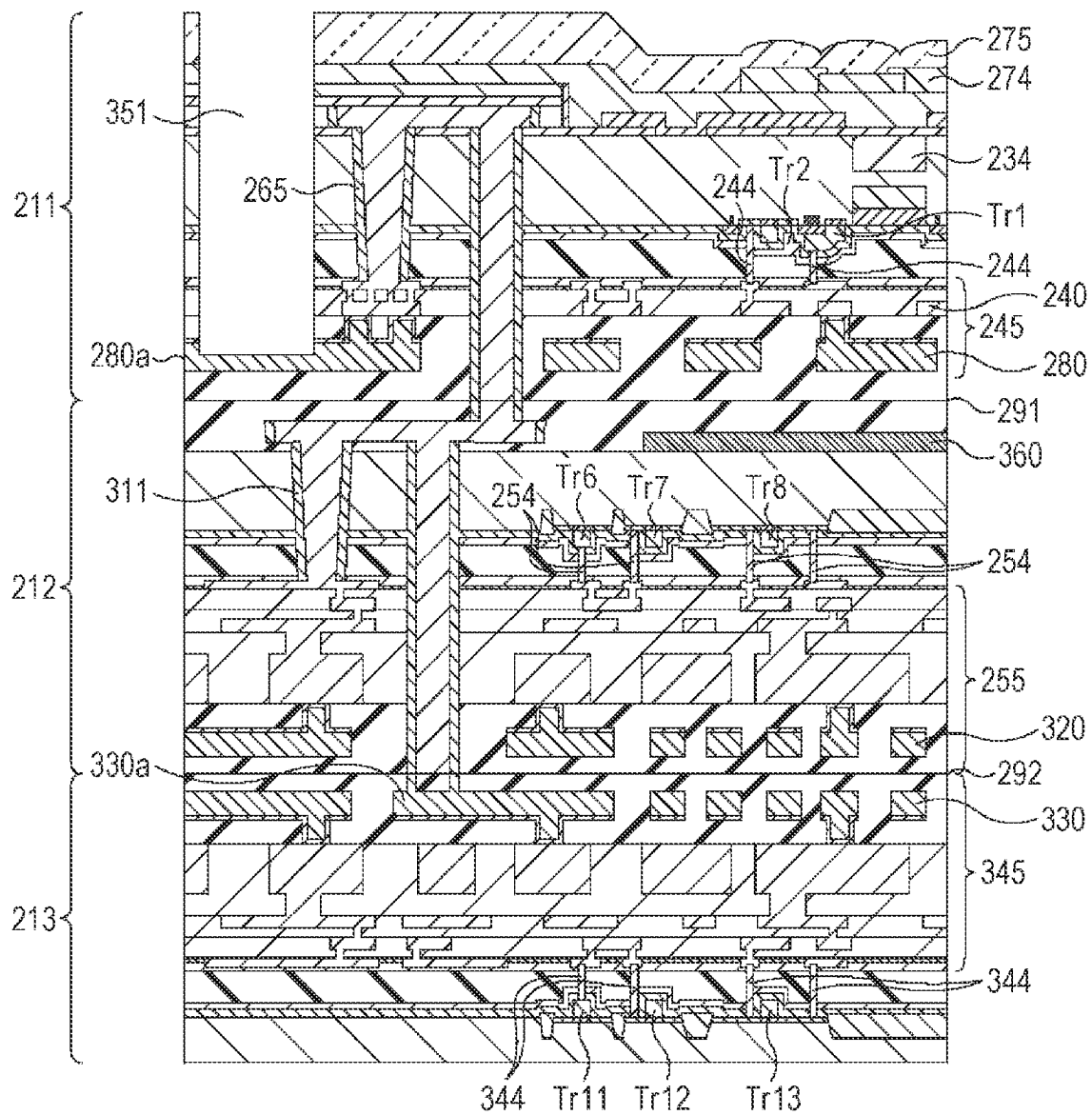


FIG. 9

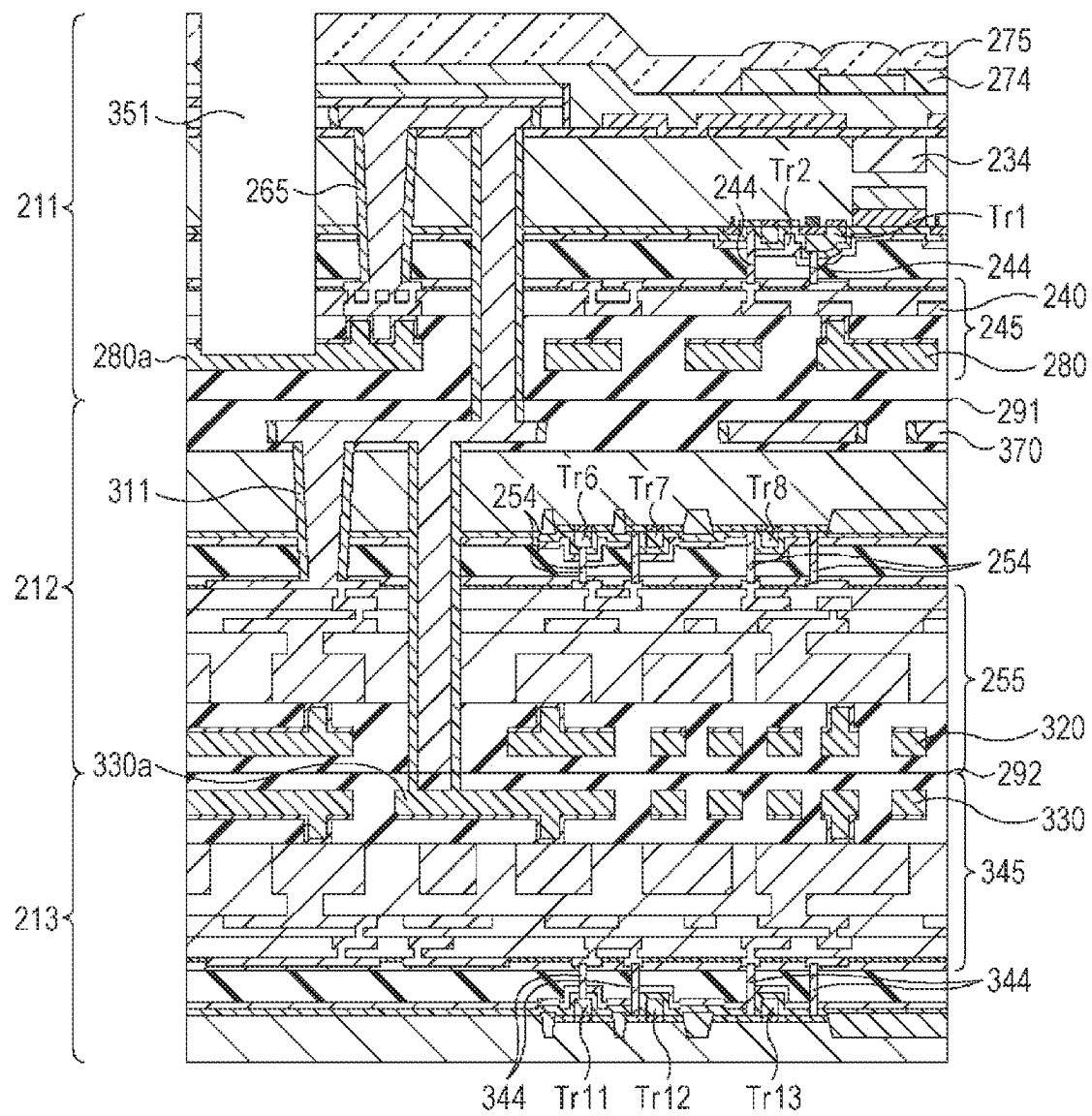


FIG. 10

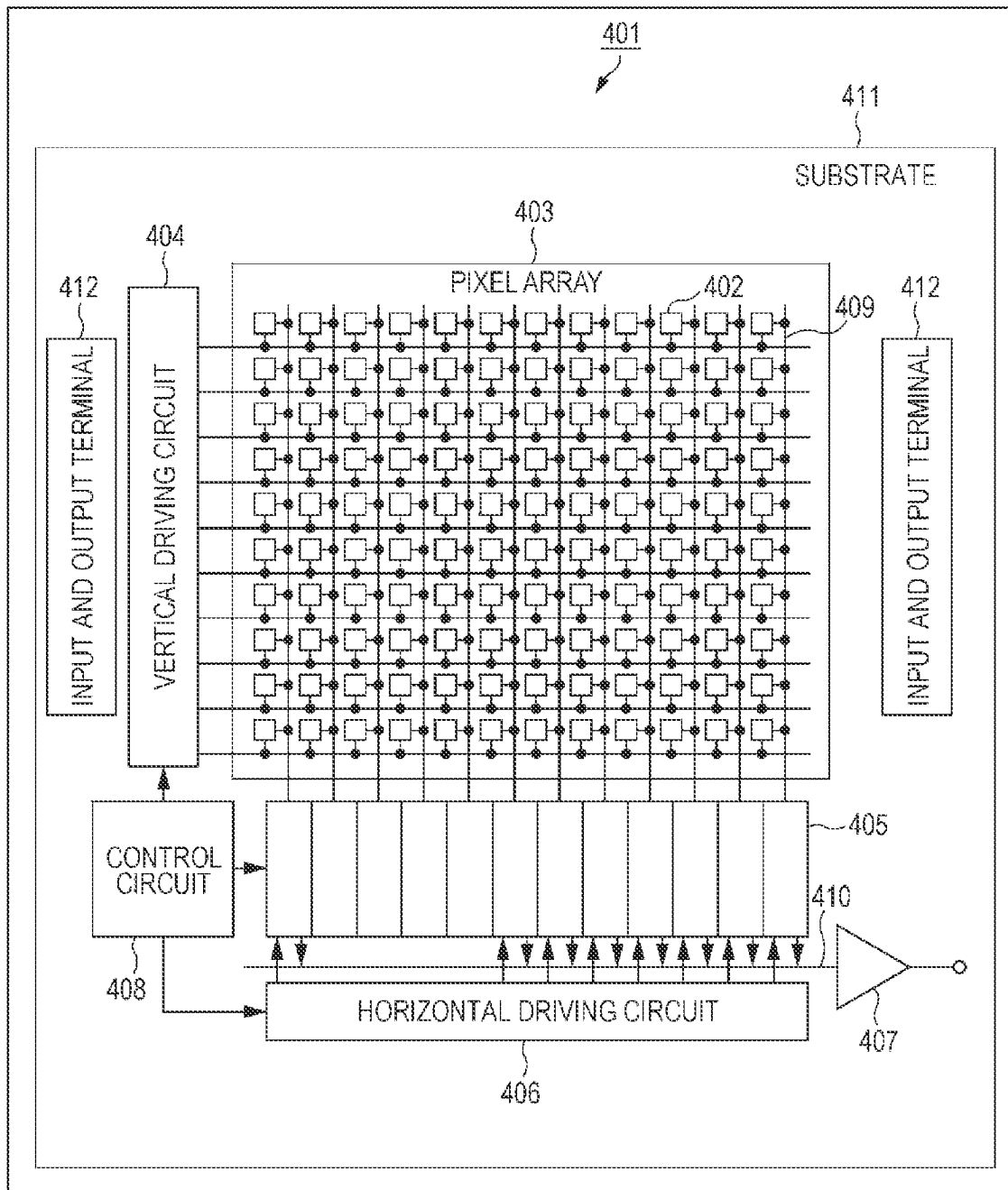


FIG. 11

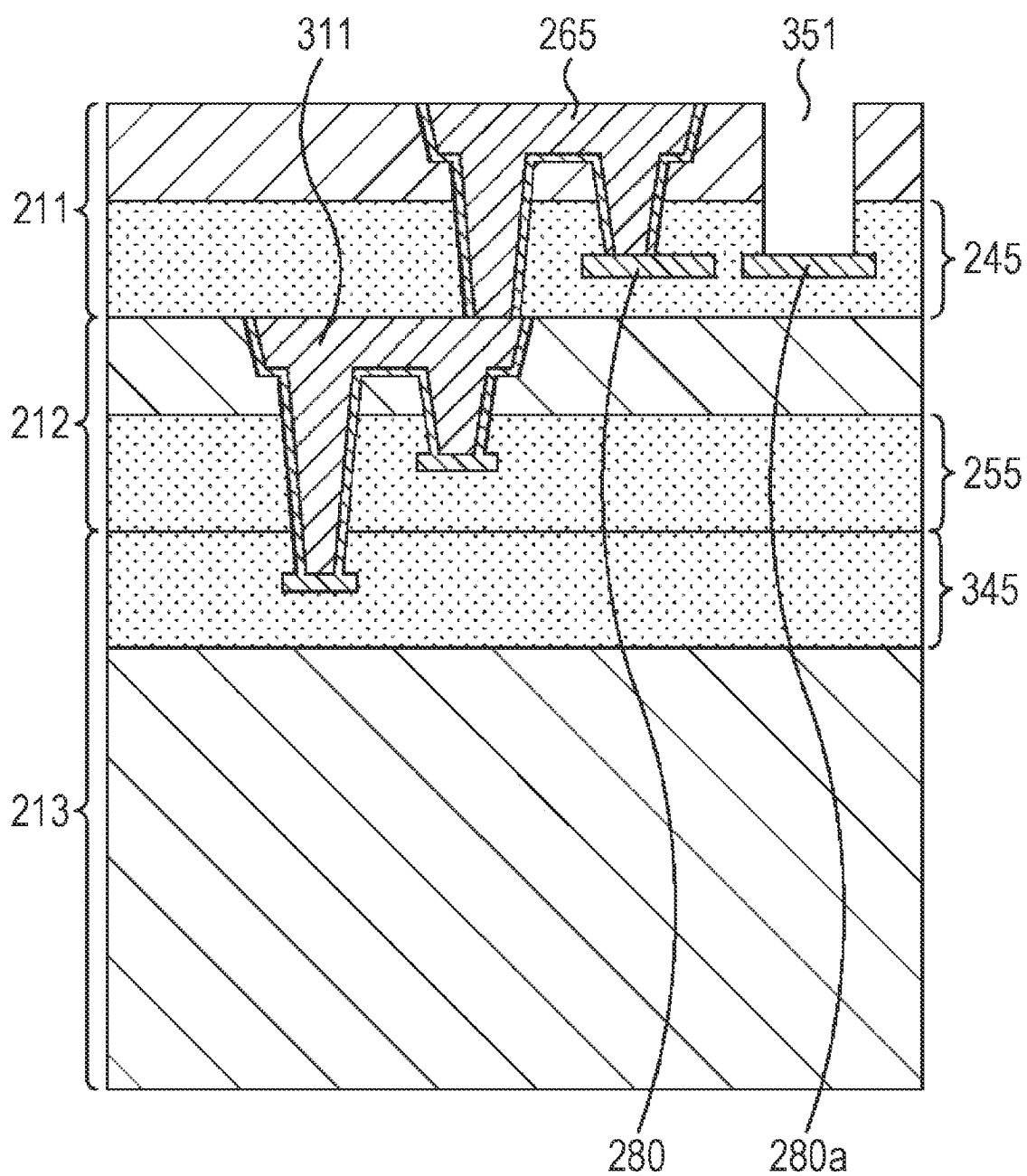


FIG. 12

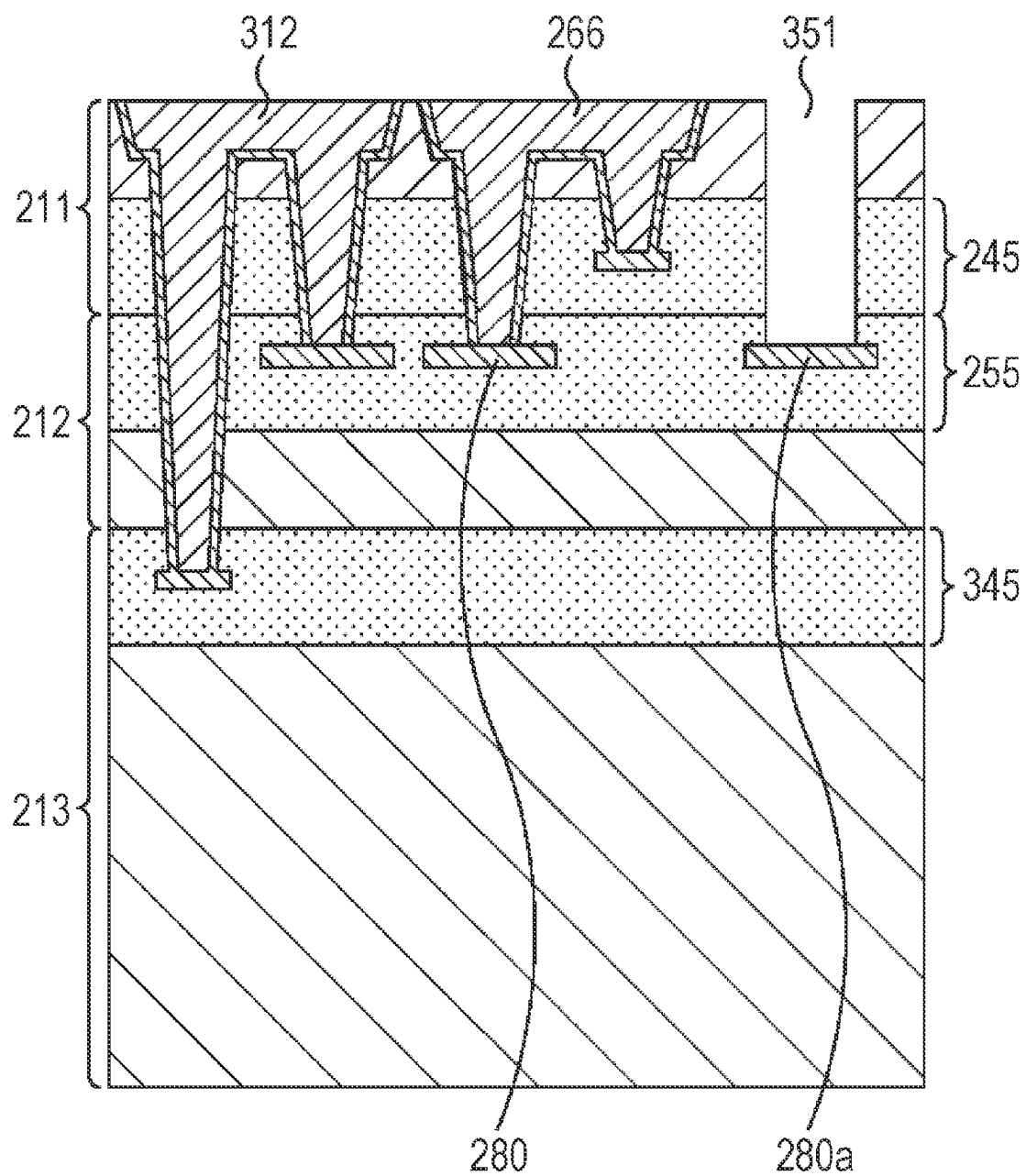


FIG. 13

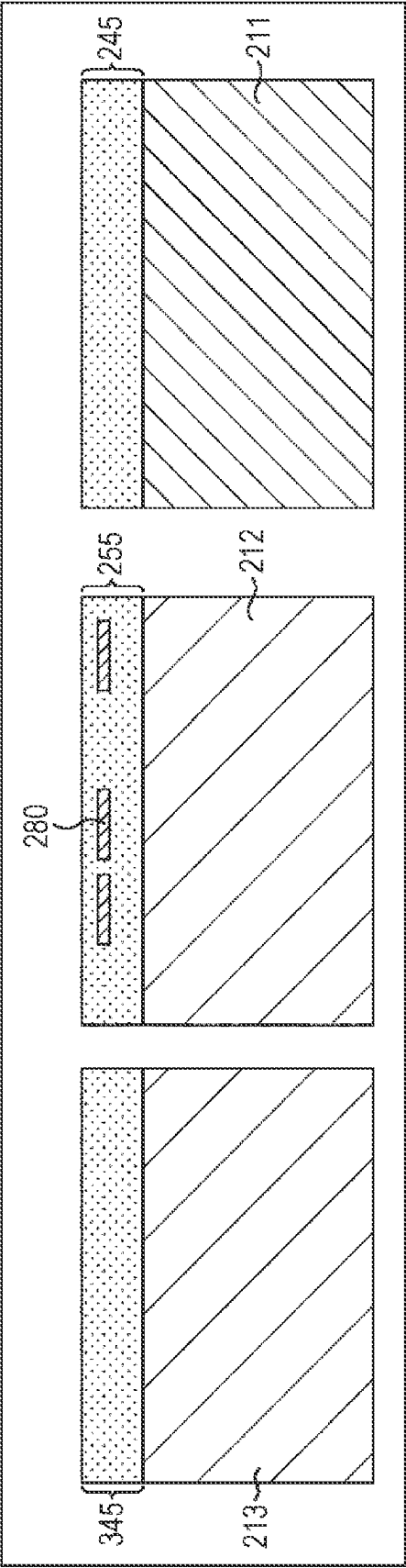


FIG. 14

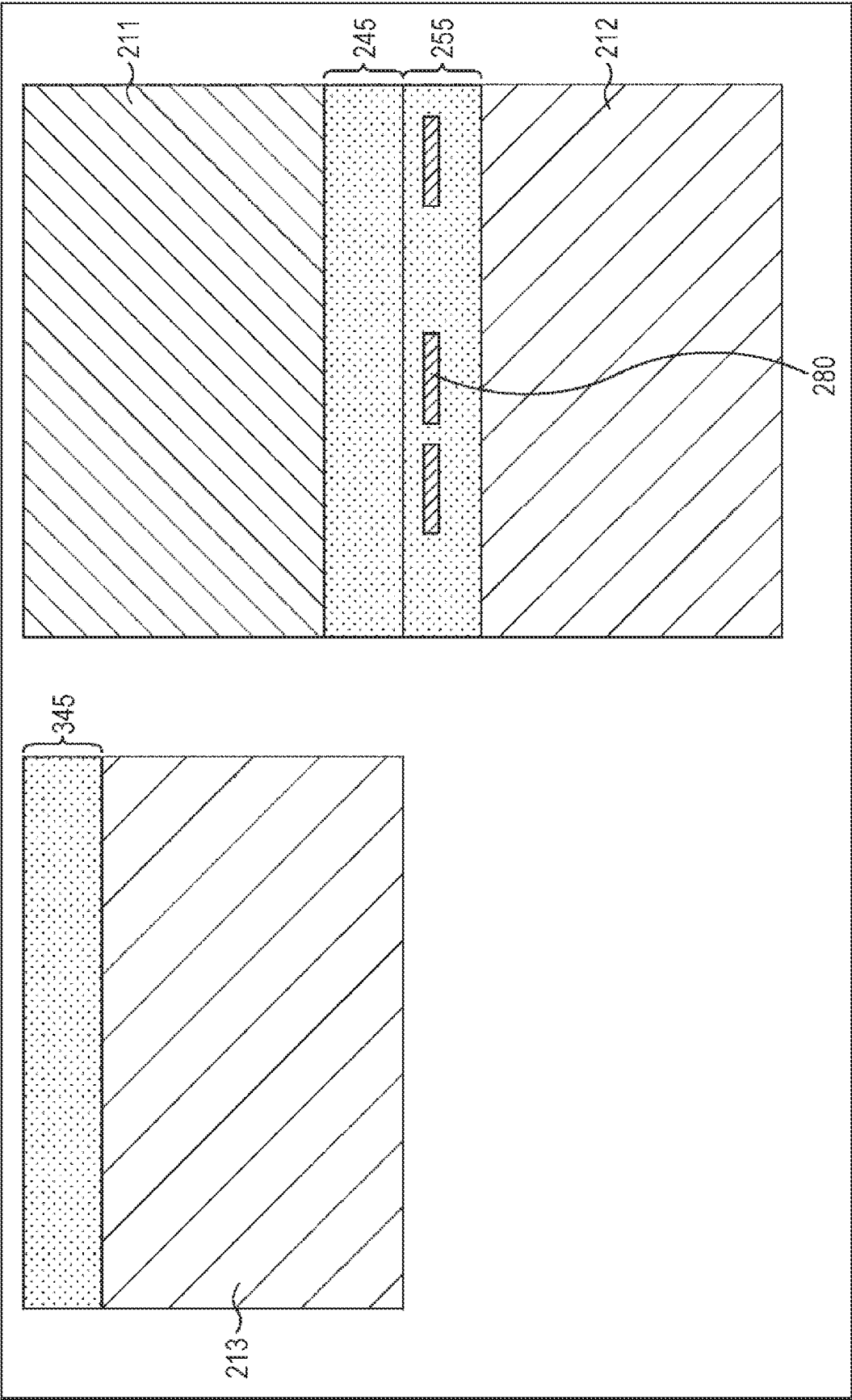


FIG. 15

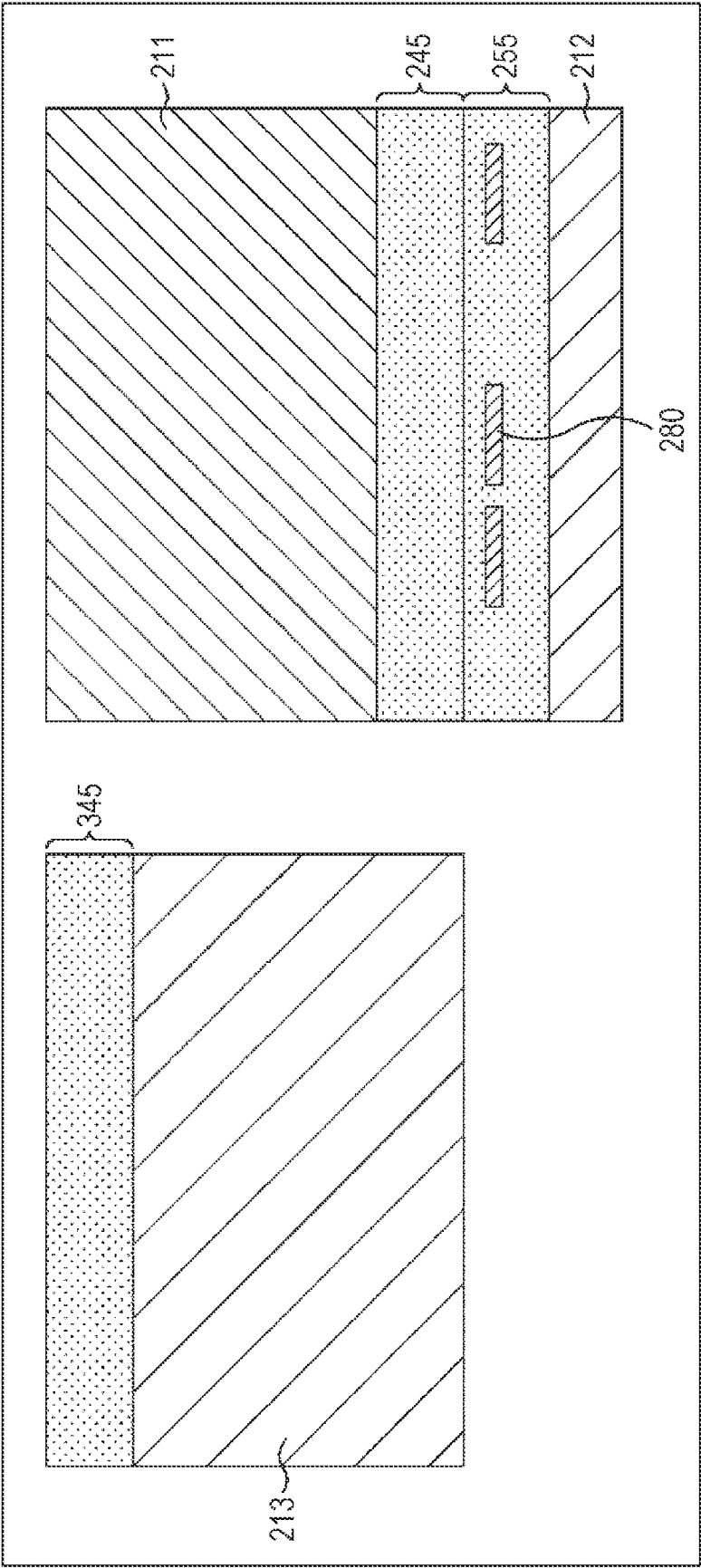


FIG. 16

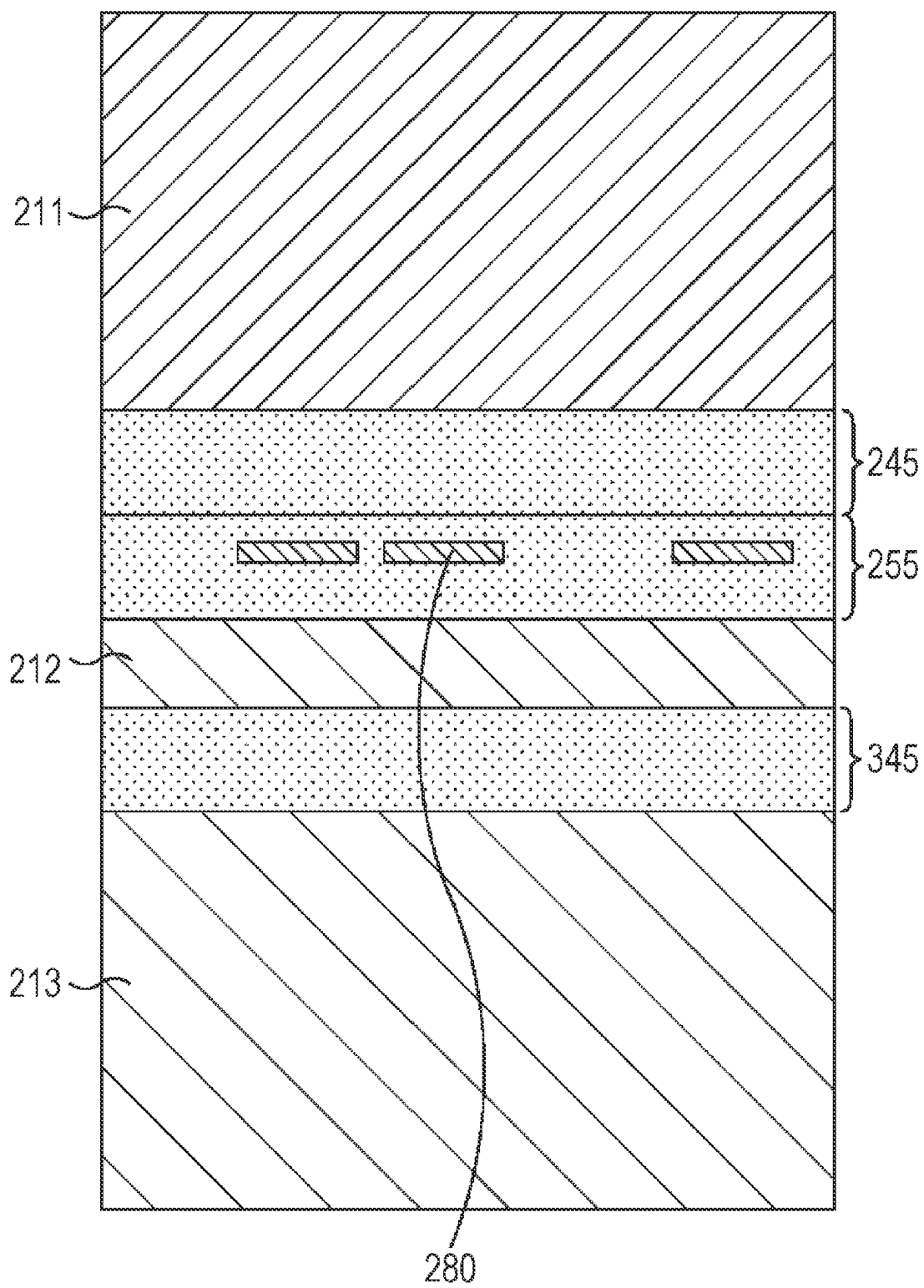


FIG. 17

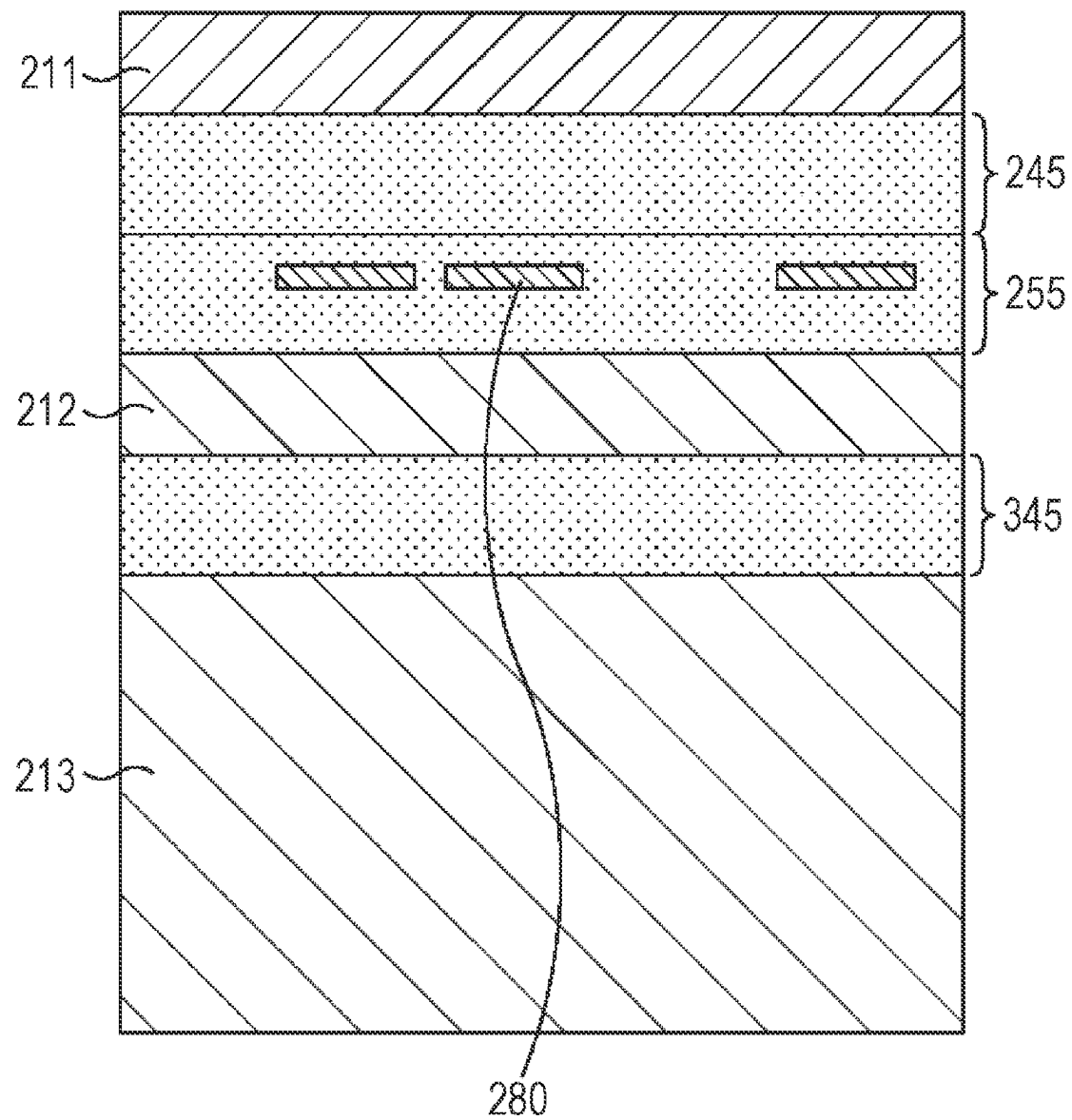


FIG. 18

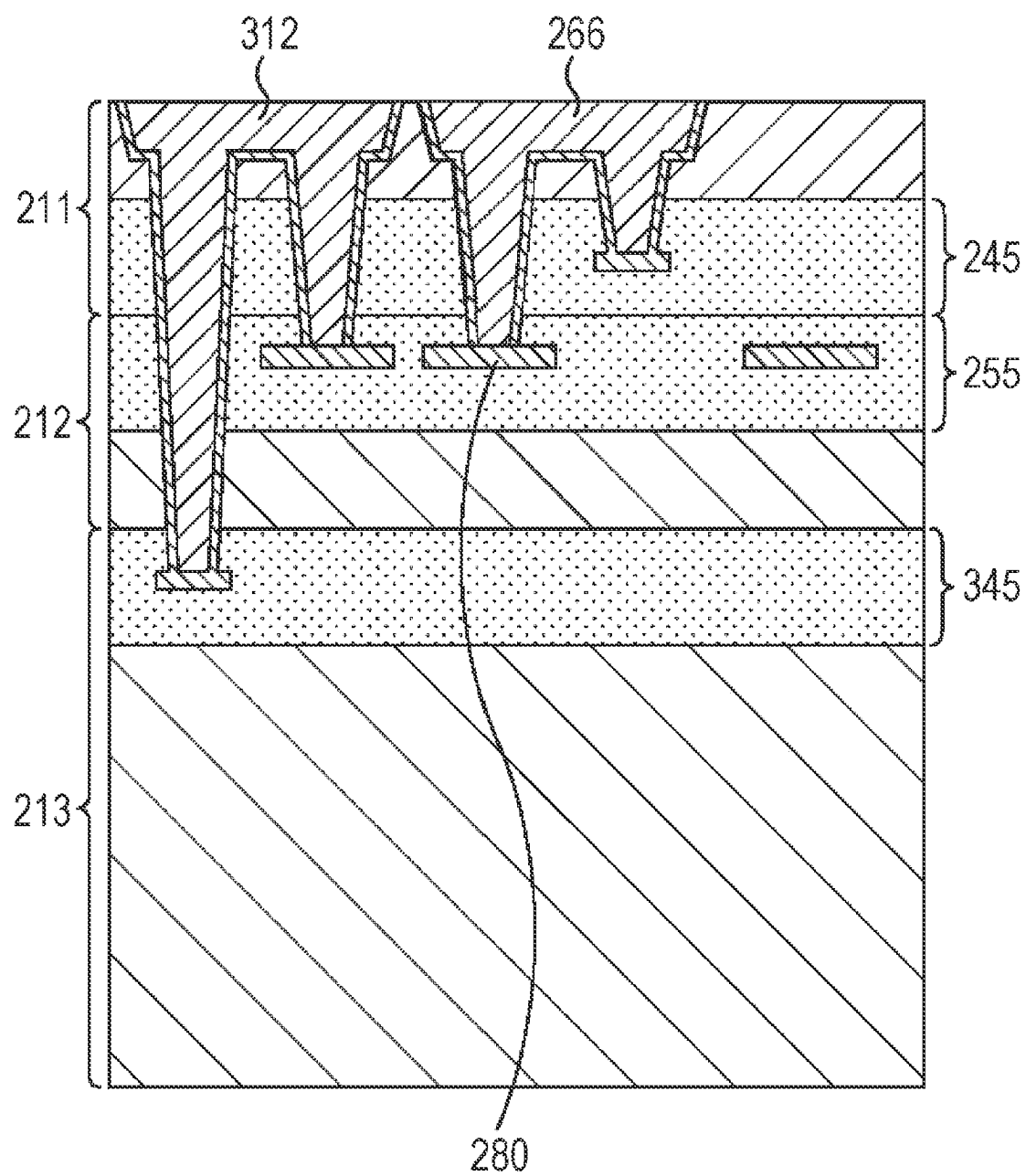


FIG. 19

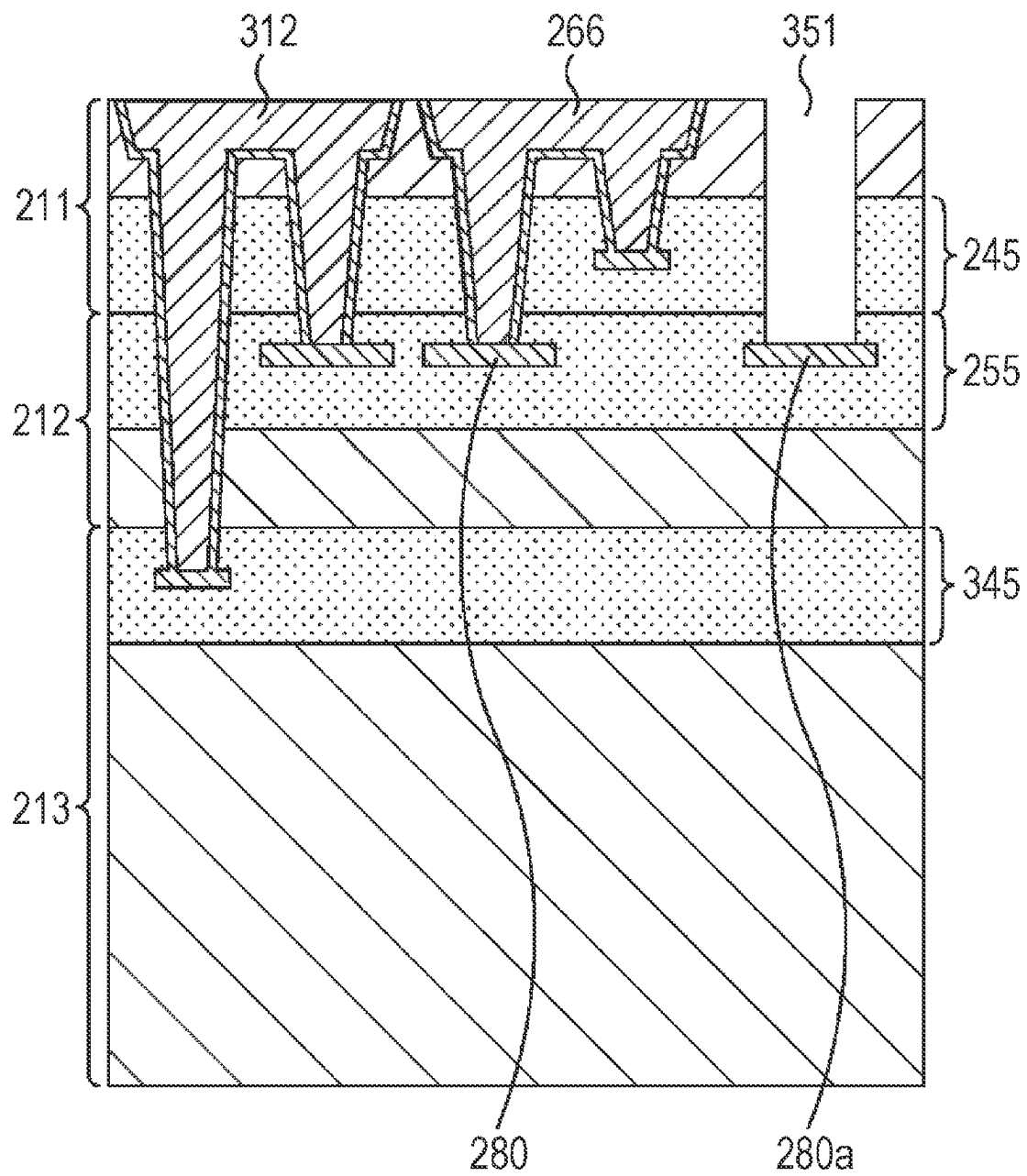


FIG. 20

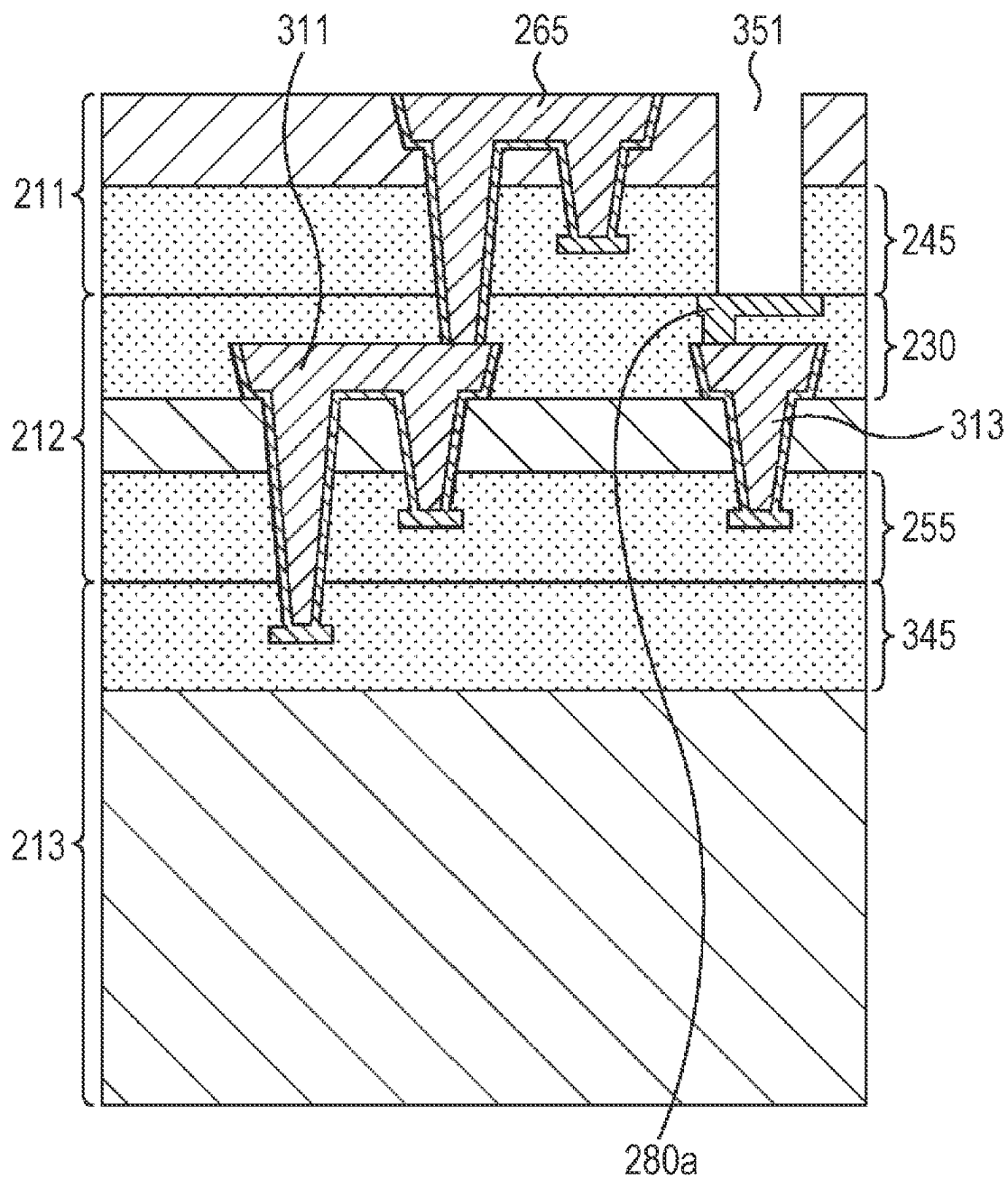


FIG. 21

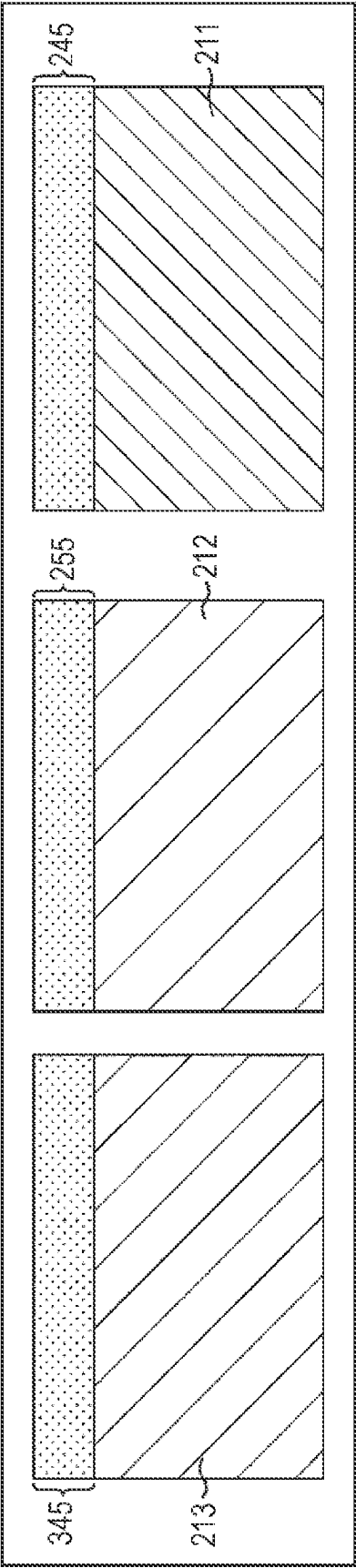


FIG. 22

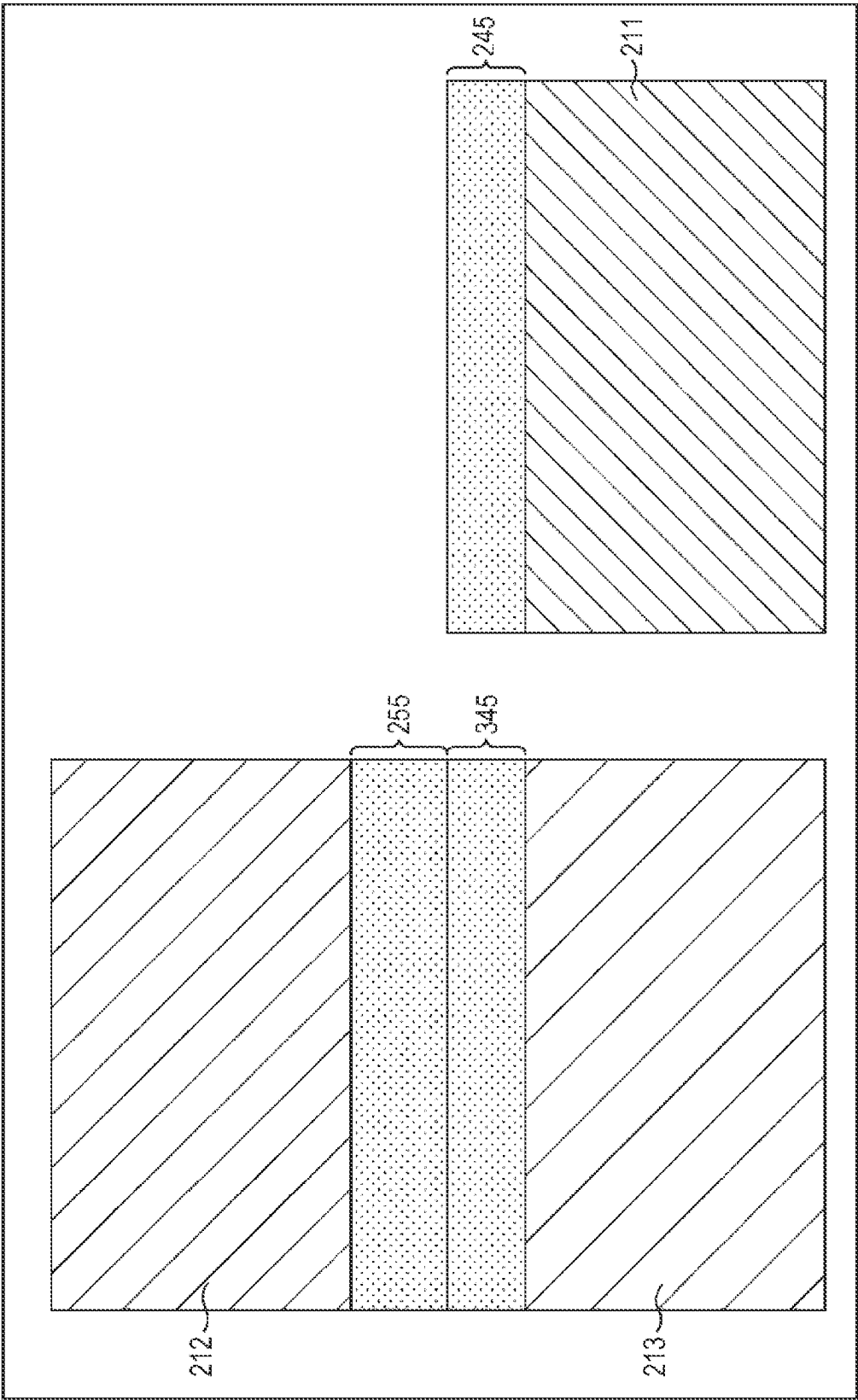


FIG. 23

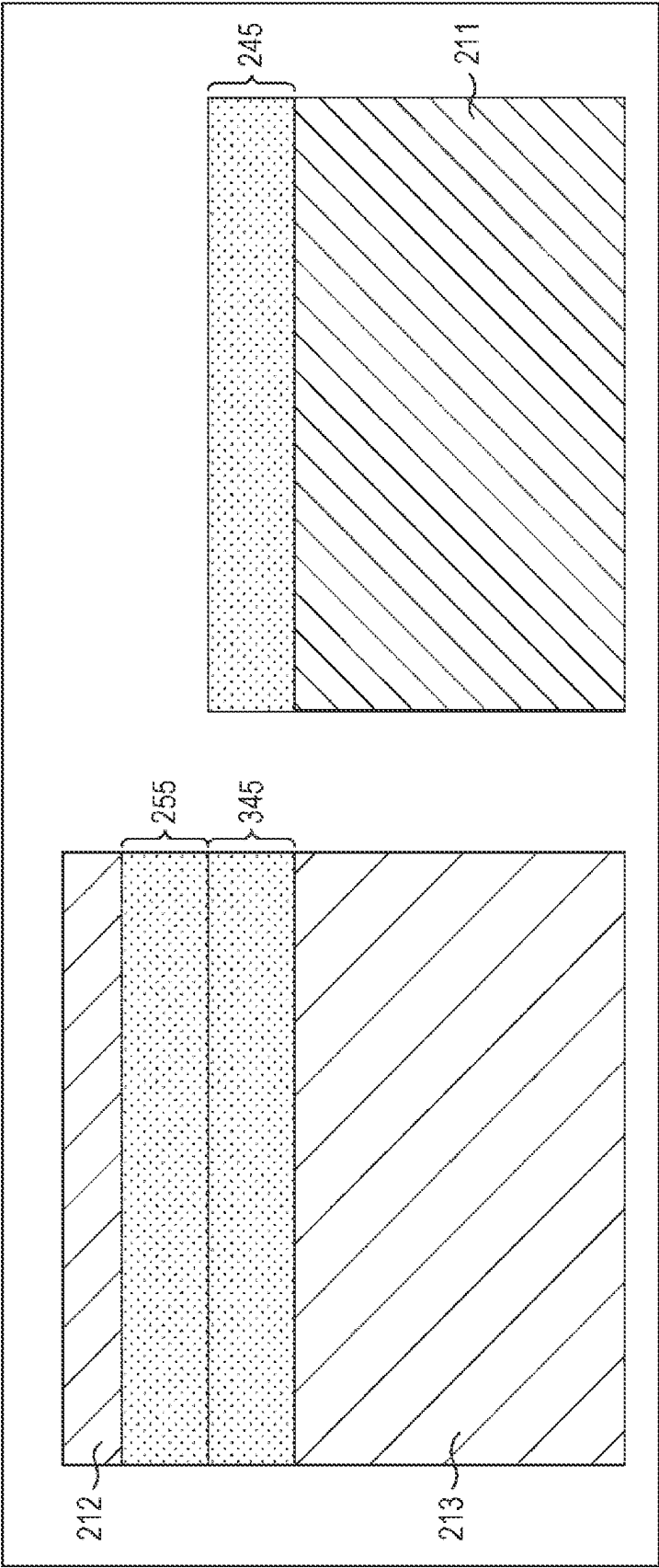


FIG. 24

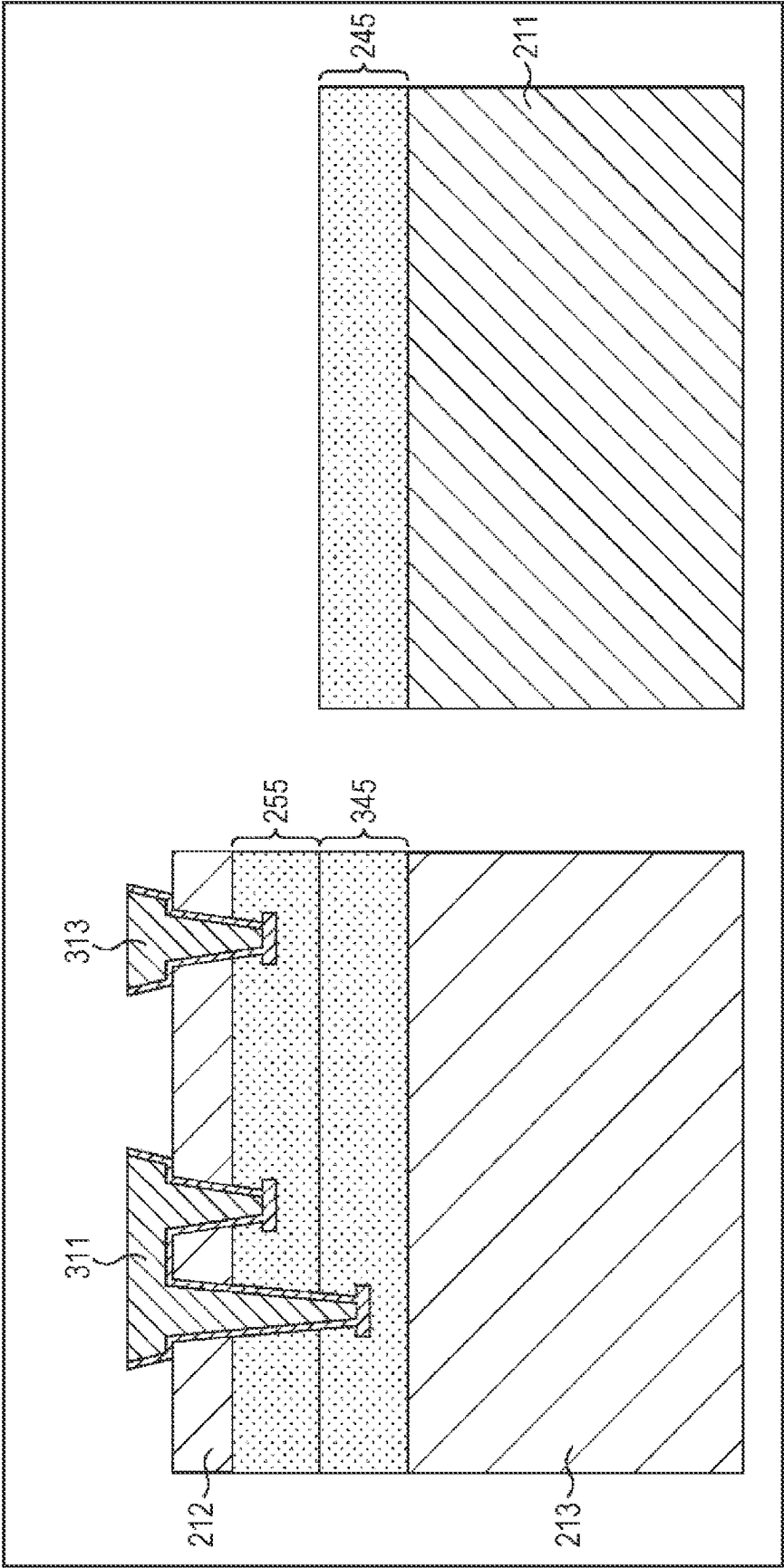


FIG. 25

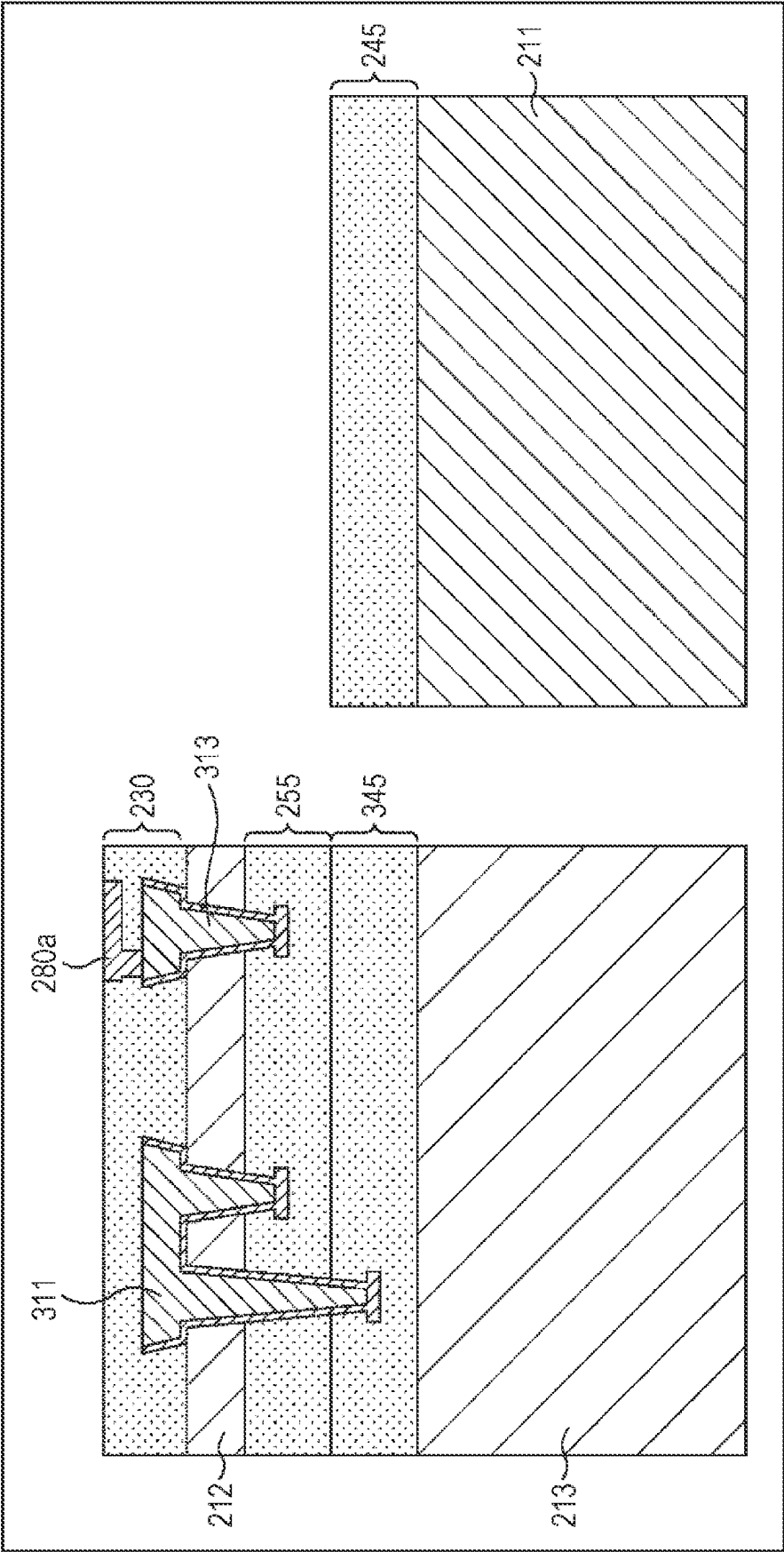


FIG. 26

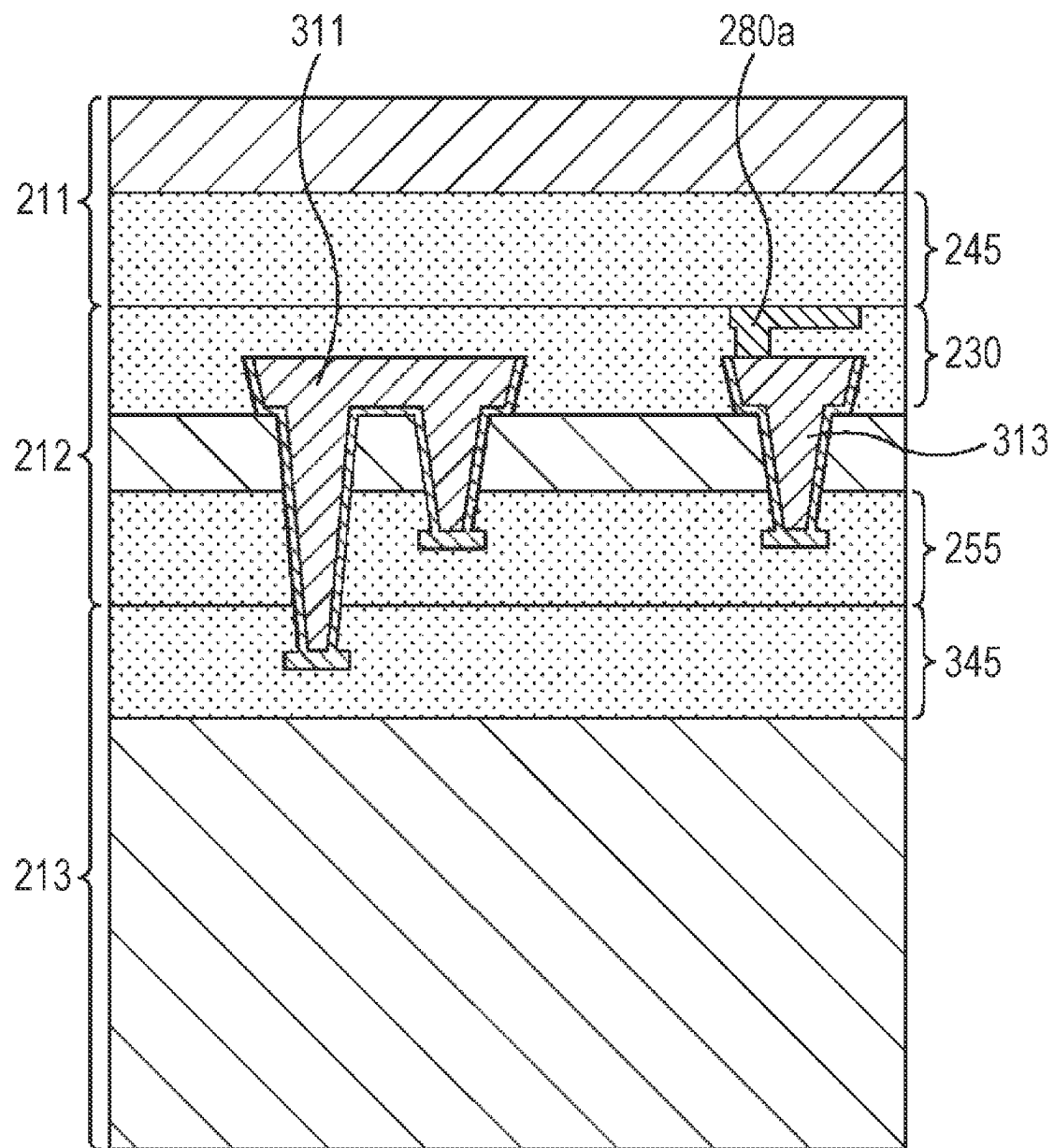


FIG. 27

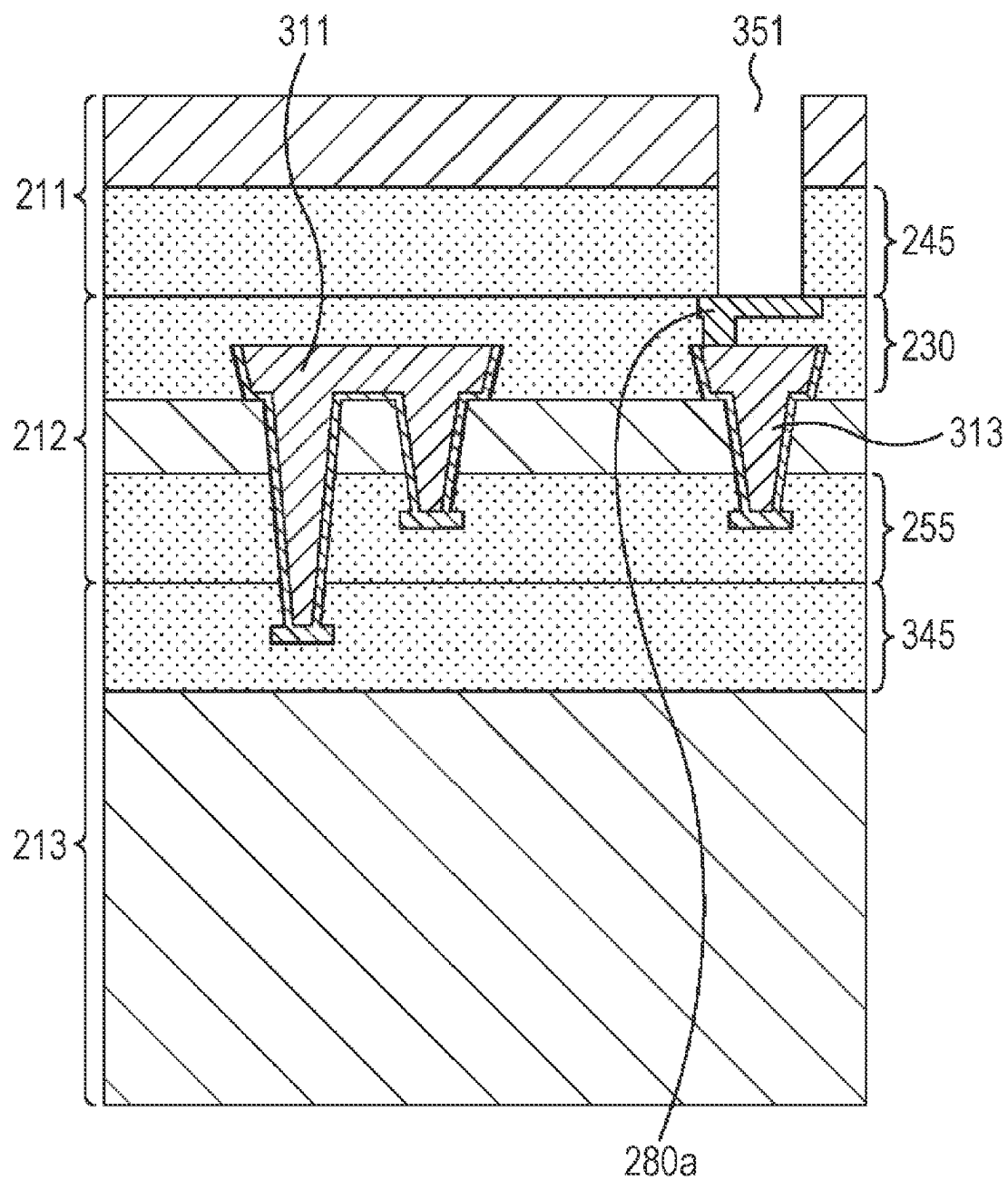


FIG. 28

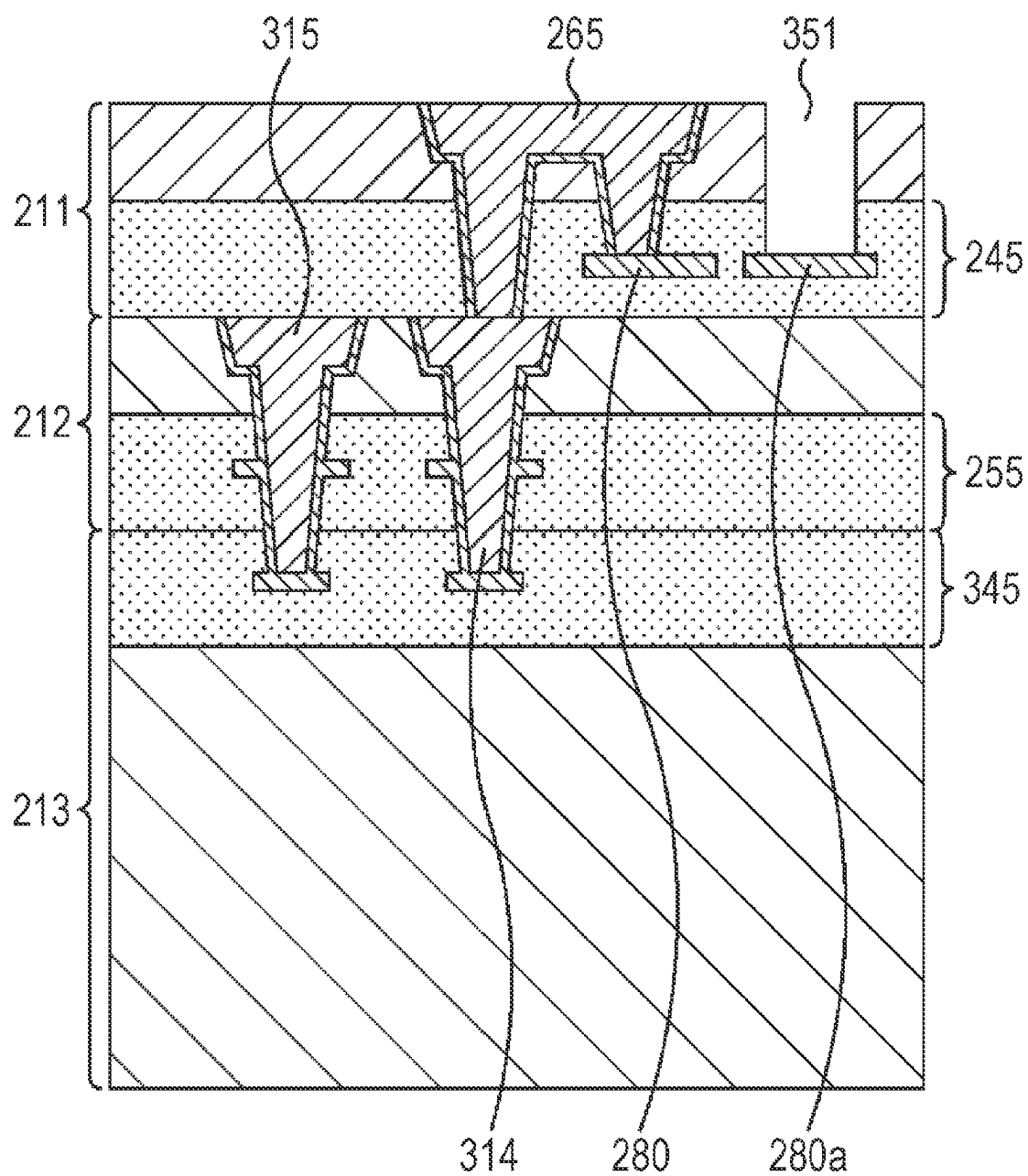


FIG. 29

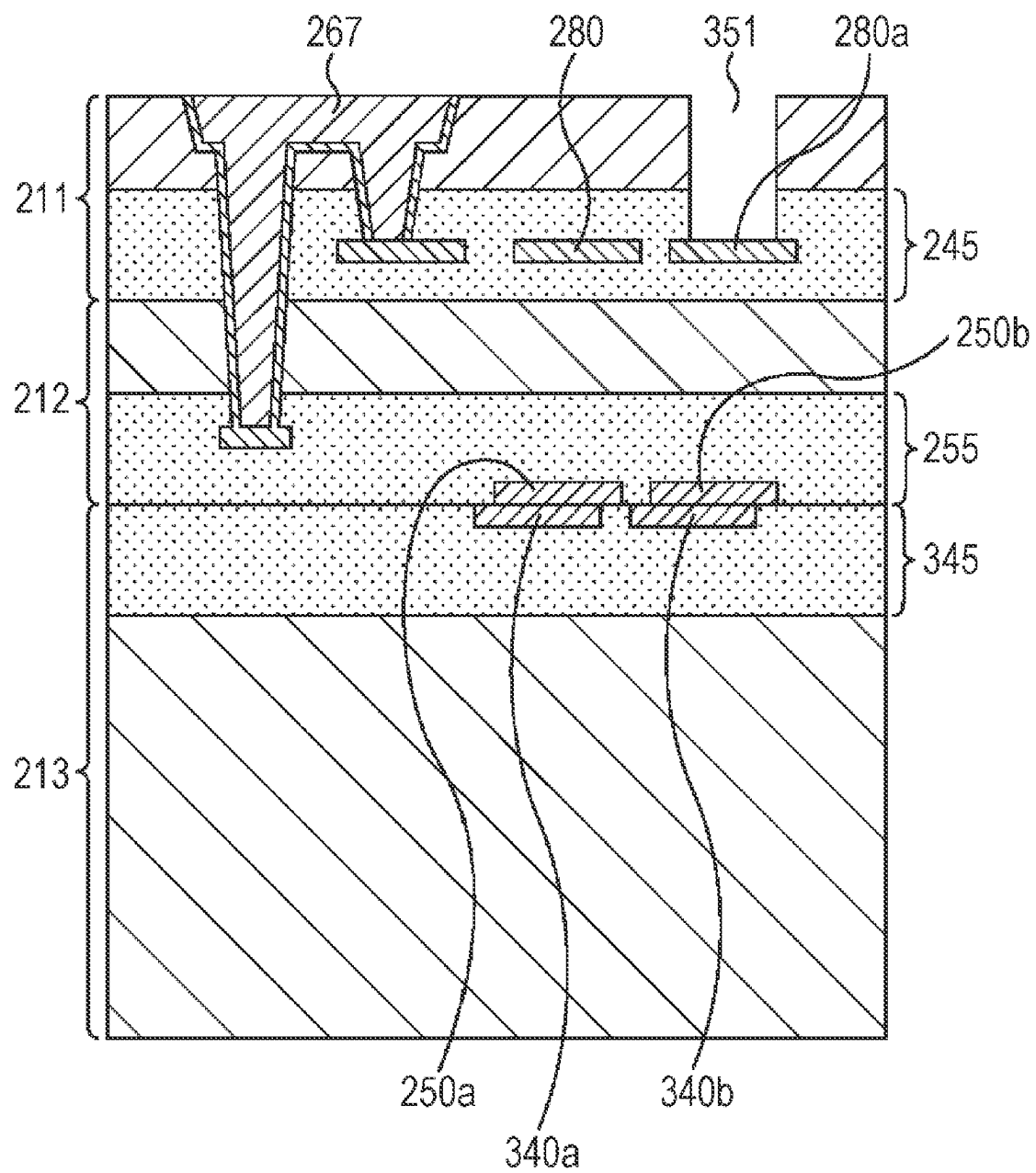
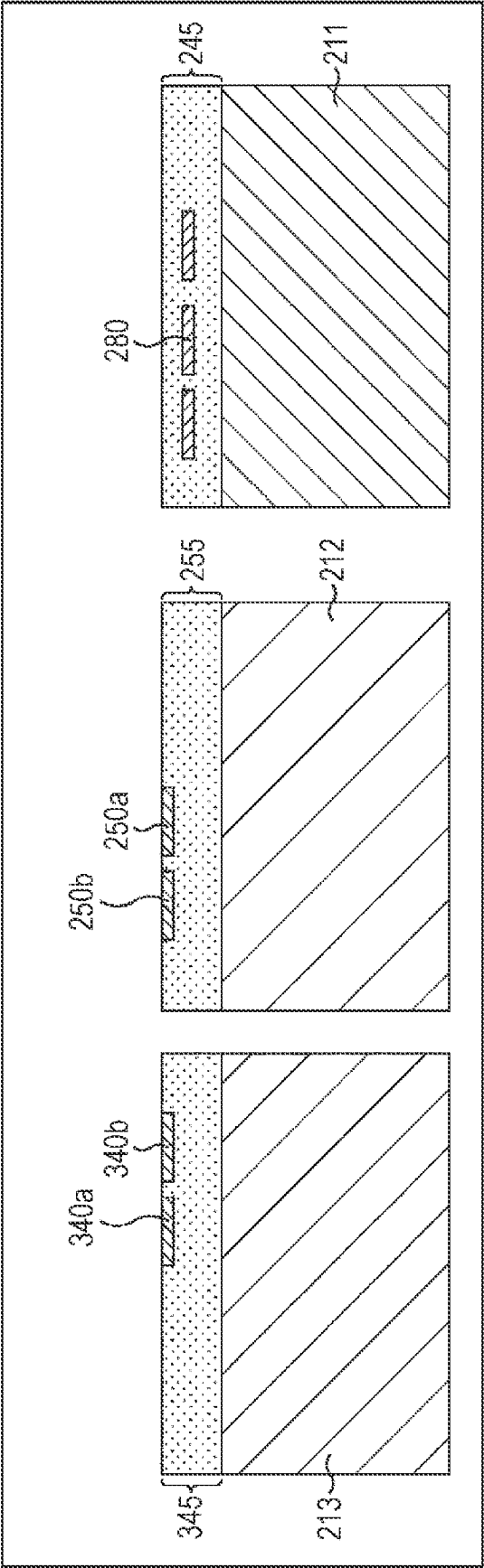


FIG. 30



3

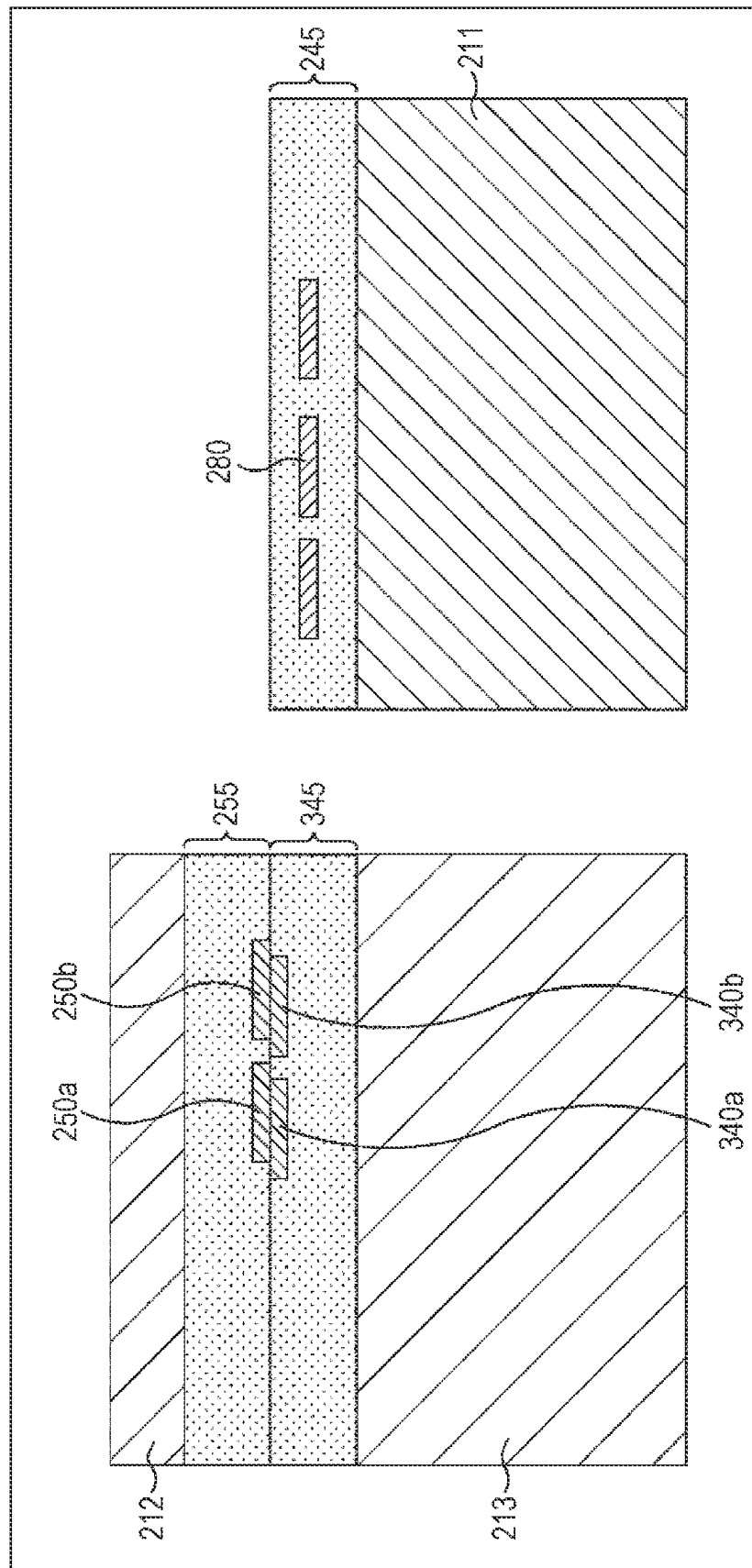


FIG. 32

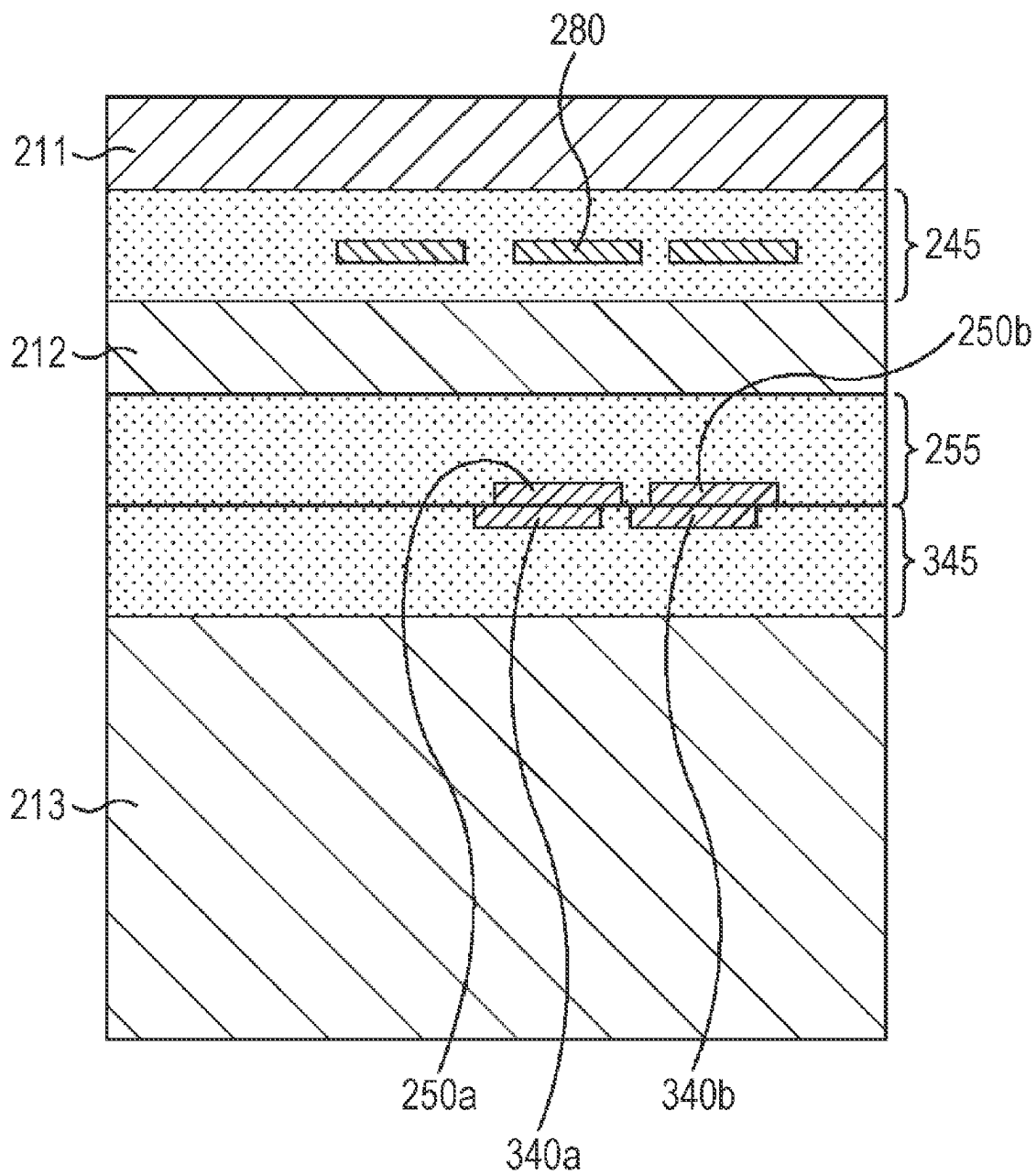


FIG. 33

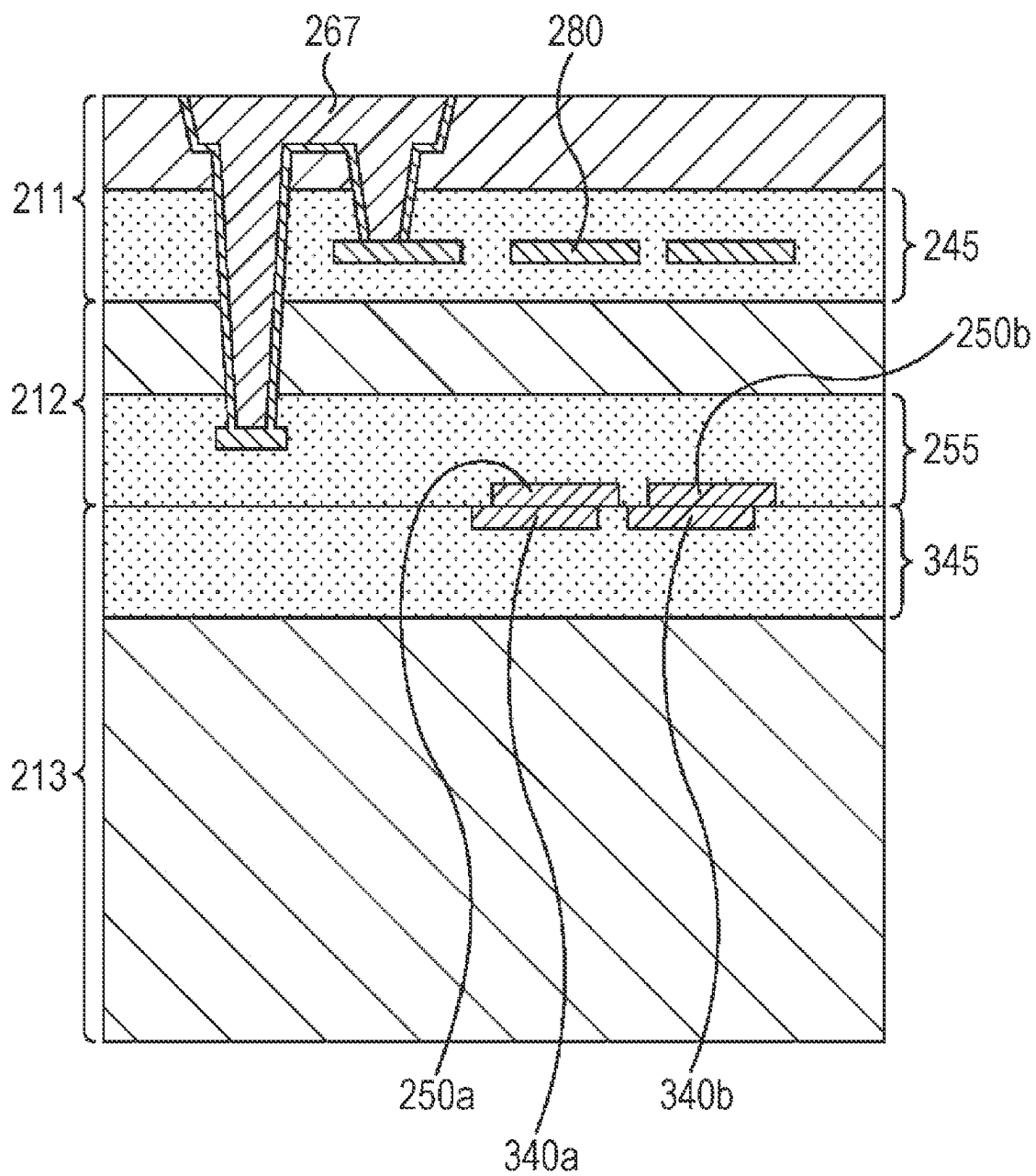
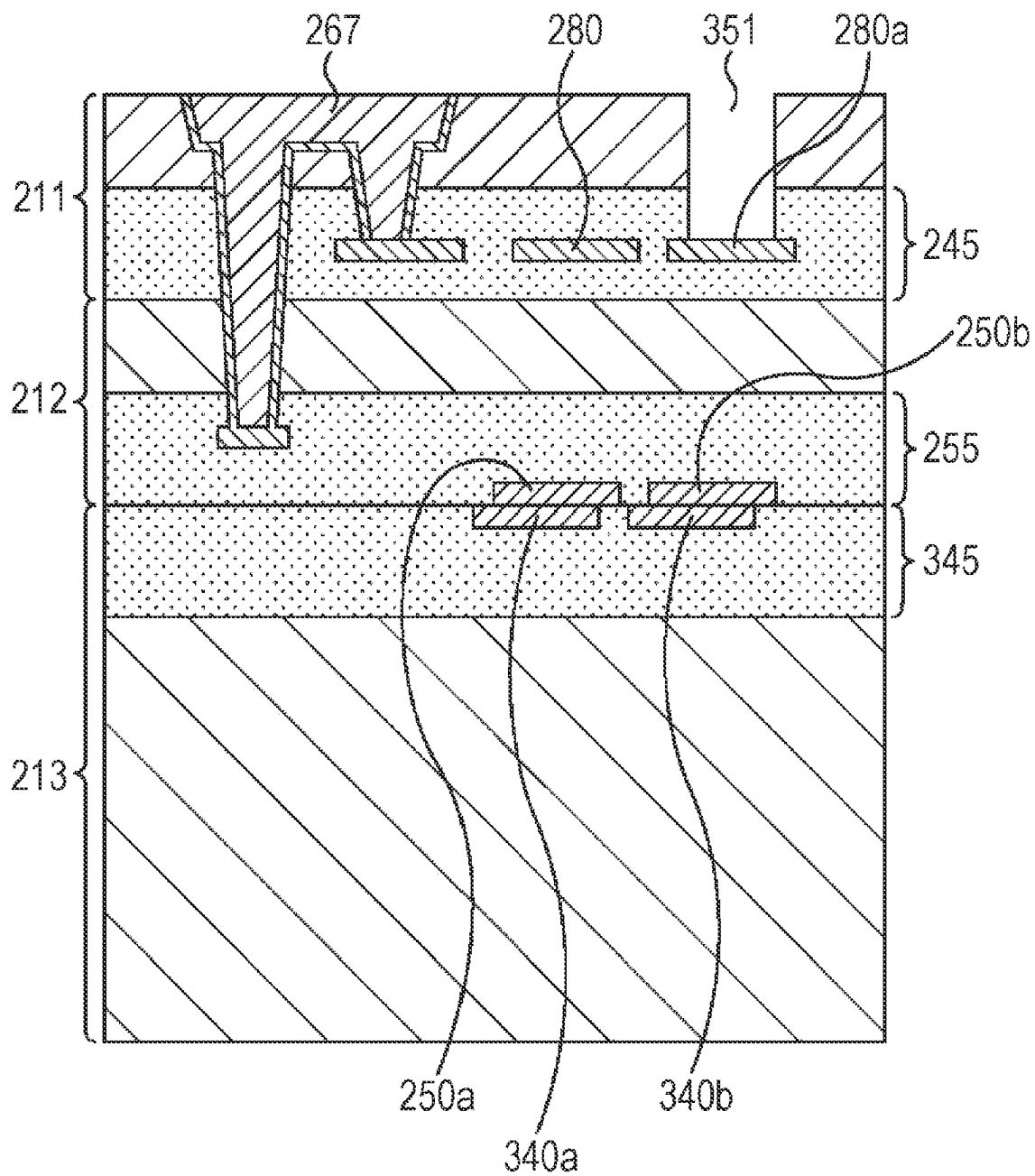


FIG. 34



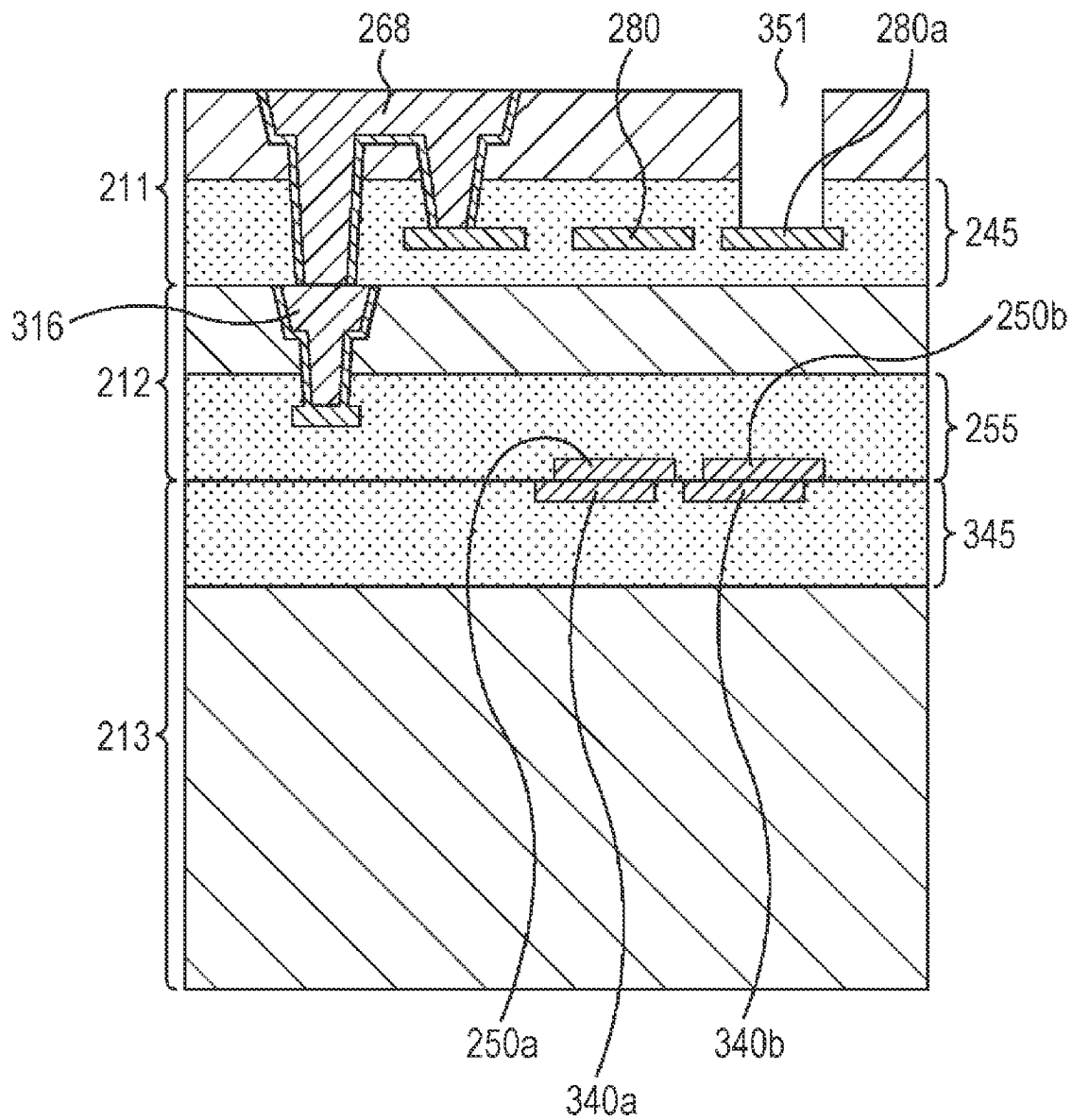


FIG. 36

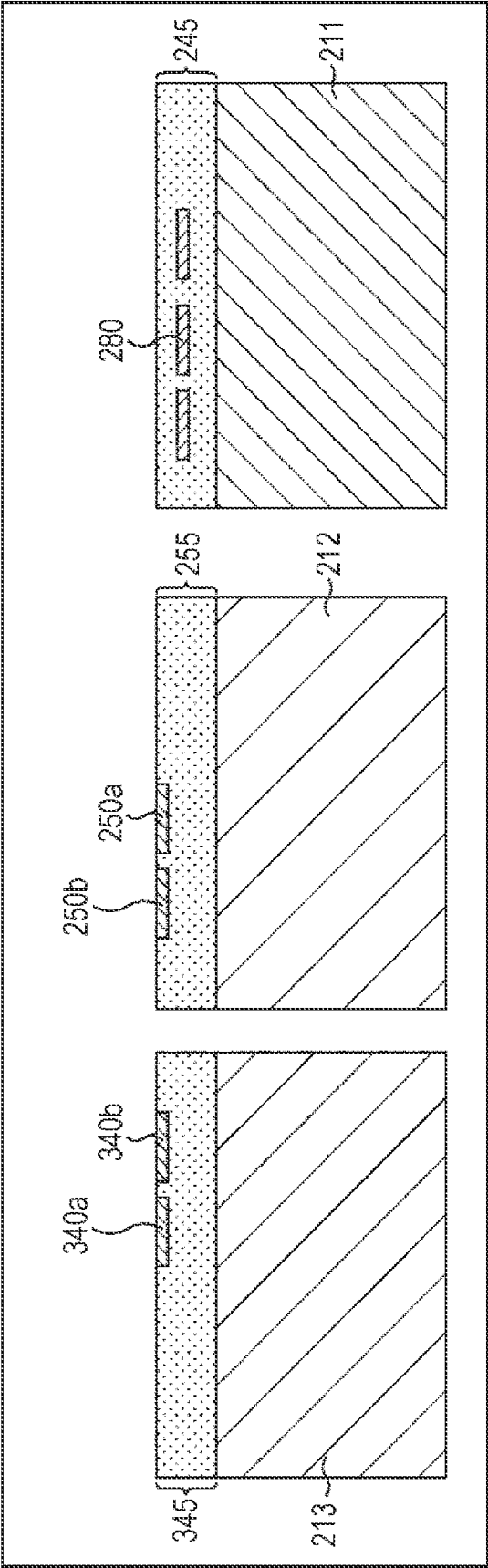


FIG. 37

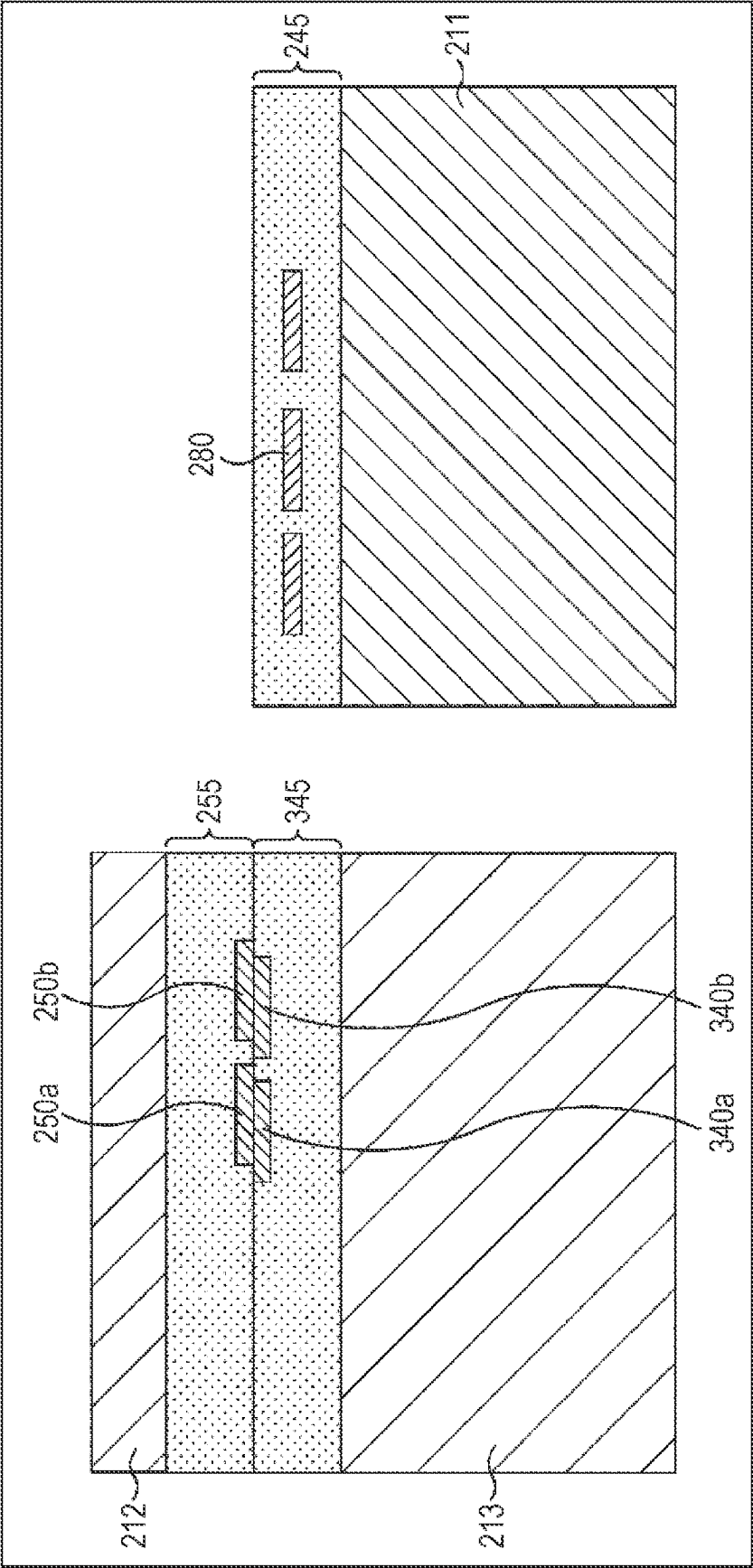


FIG. 38

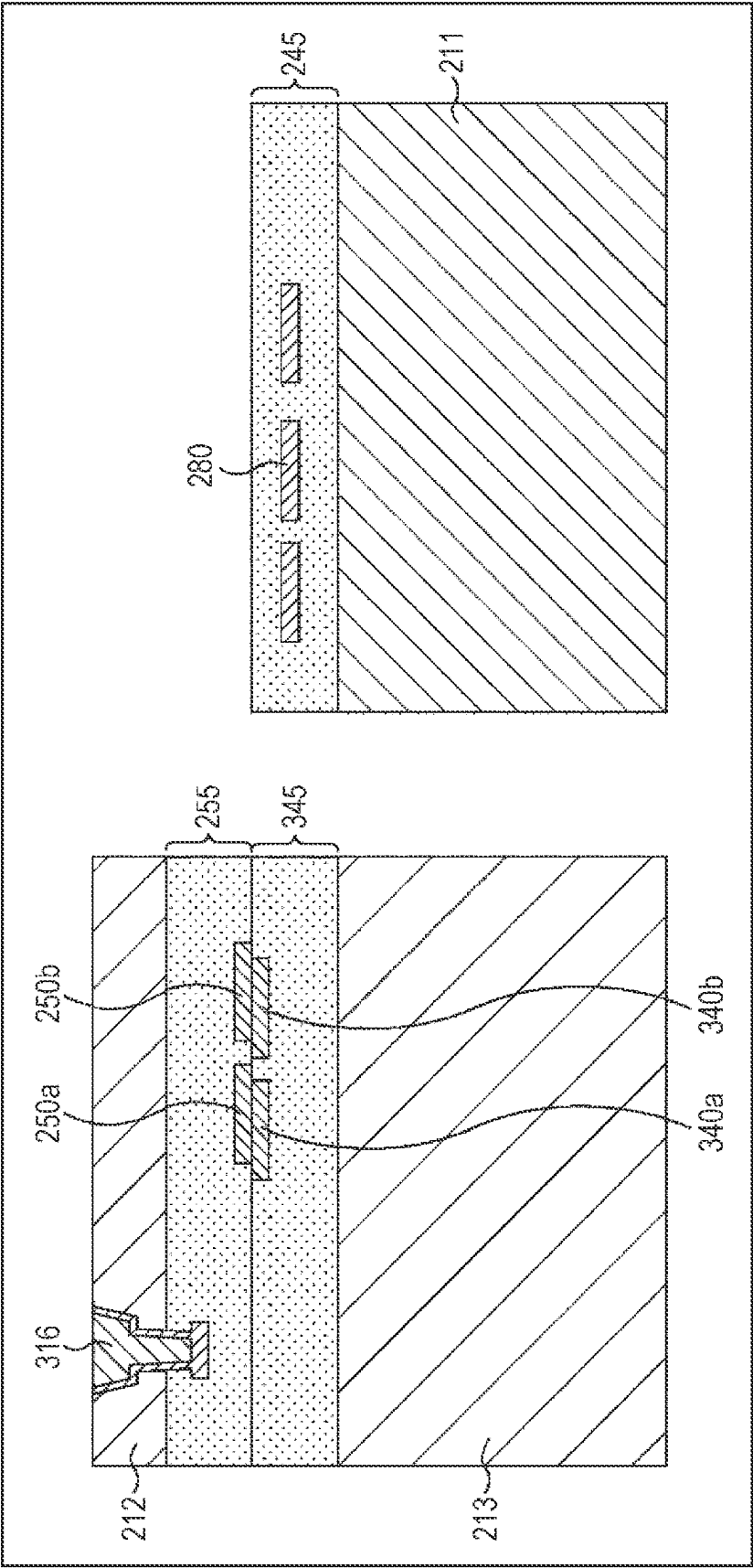


FIG. 39

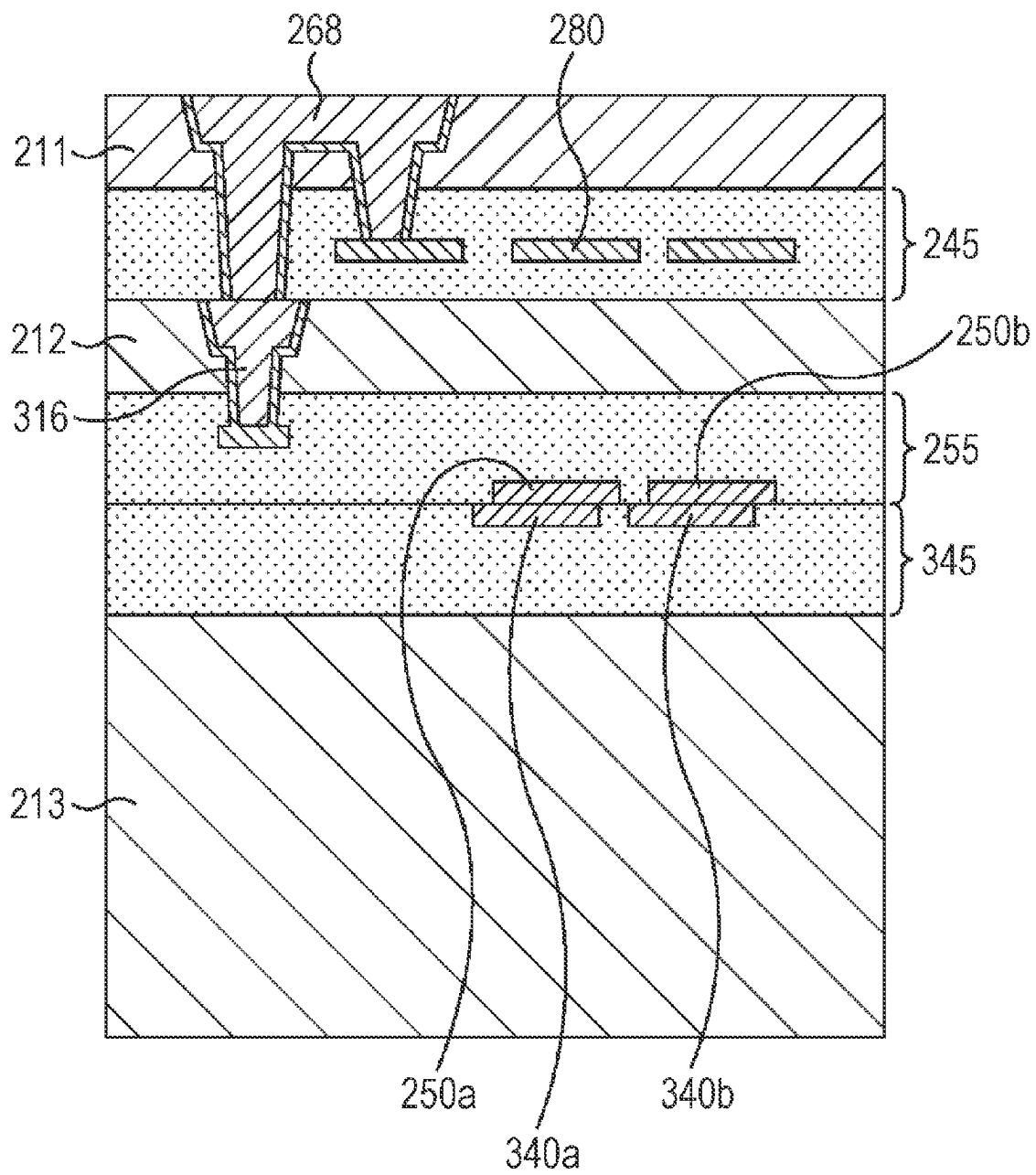


FIG. 40

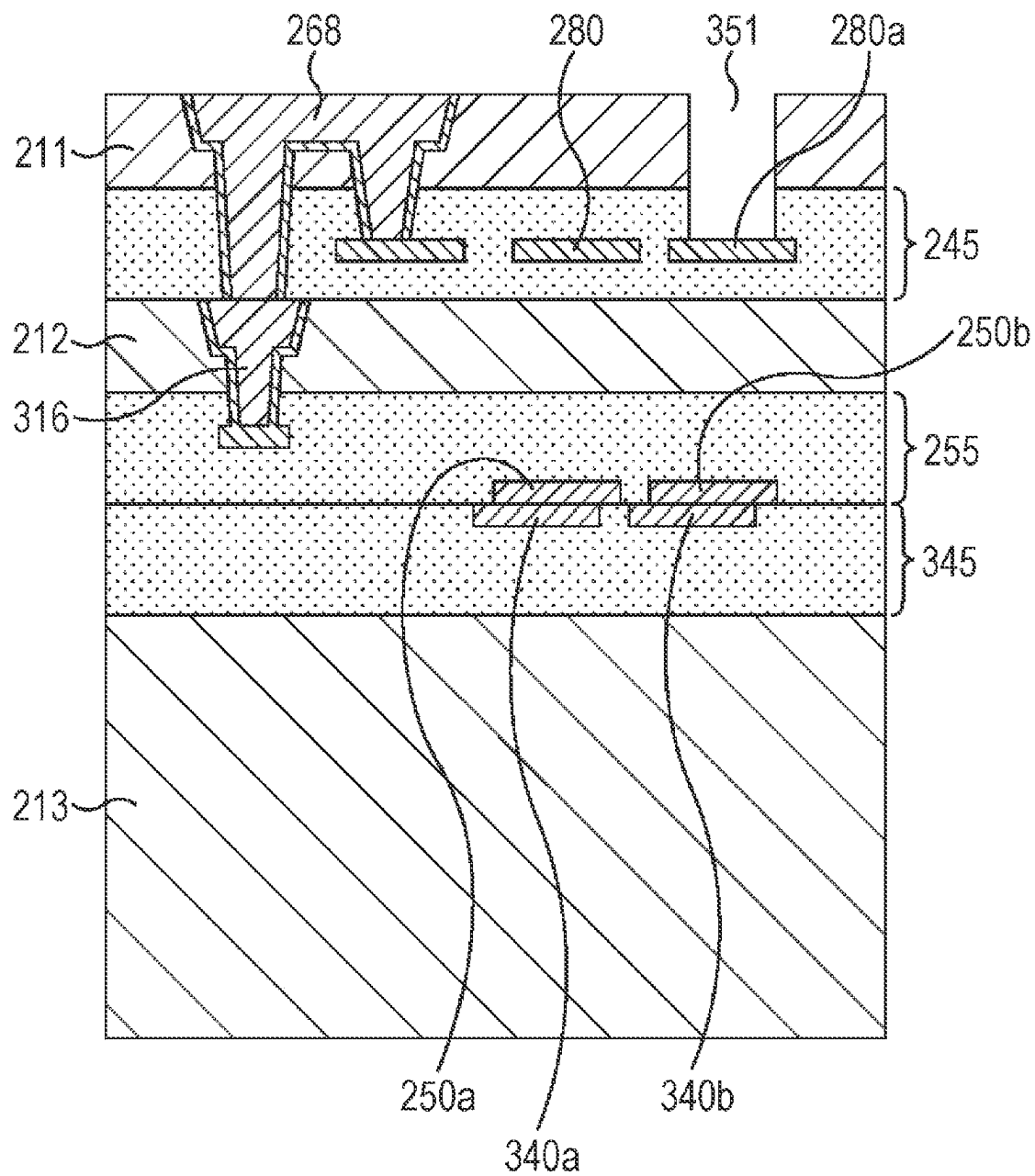


FIG. 41

ARRANGEMENT POSITION OF ALUMINIUM PAD	ELECTRICAL CONNECTION BETWEEN FIRST SEMICONDUCTOR SUBSTRATE AND SECOND SEMICONDUCTOR SUBSTRATE	ELECTRICAL CONNECTION BETWEEN SECOND SEMICONDUCTOR SUBSTRATE AND THIRD SEMICONDUCTOR SUBSTRATE
IN MULTILAYER WIRE LAYER OF FIRST SEMICONDUCTOR SUBSTRATE	TWIN CONTACT	TWIN CONTACT
		SHARED CONTACT
		DIRECT BONDING
	SHARED CONTACT	SHARED CONTACT
	DIRECT BONDING	DIRECT BONDING
IN MULTILAYER WIRE LAYER OF SECOND SEMICONDUCTOR SUBSTRATE	TWIN CONTACT	TWIN CONTACT
		SHARED CONTACT
		DIRECT BONDING
	SHARED CONTACT	SHARED CONTACT
	DIRECT BONDING	DIRECT BONDING
INSULATING FILM LAYER	TWIN CONTACT	TWIN CONTACT
		SHARED CONTACT
		DIRECT BONDING
	SHARED CONTACT	SHARED CONTACT
	DIRECT BONDING	DIRECT BONDING

FIG. 42

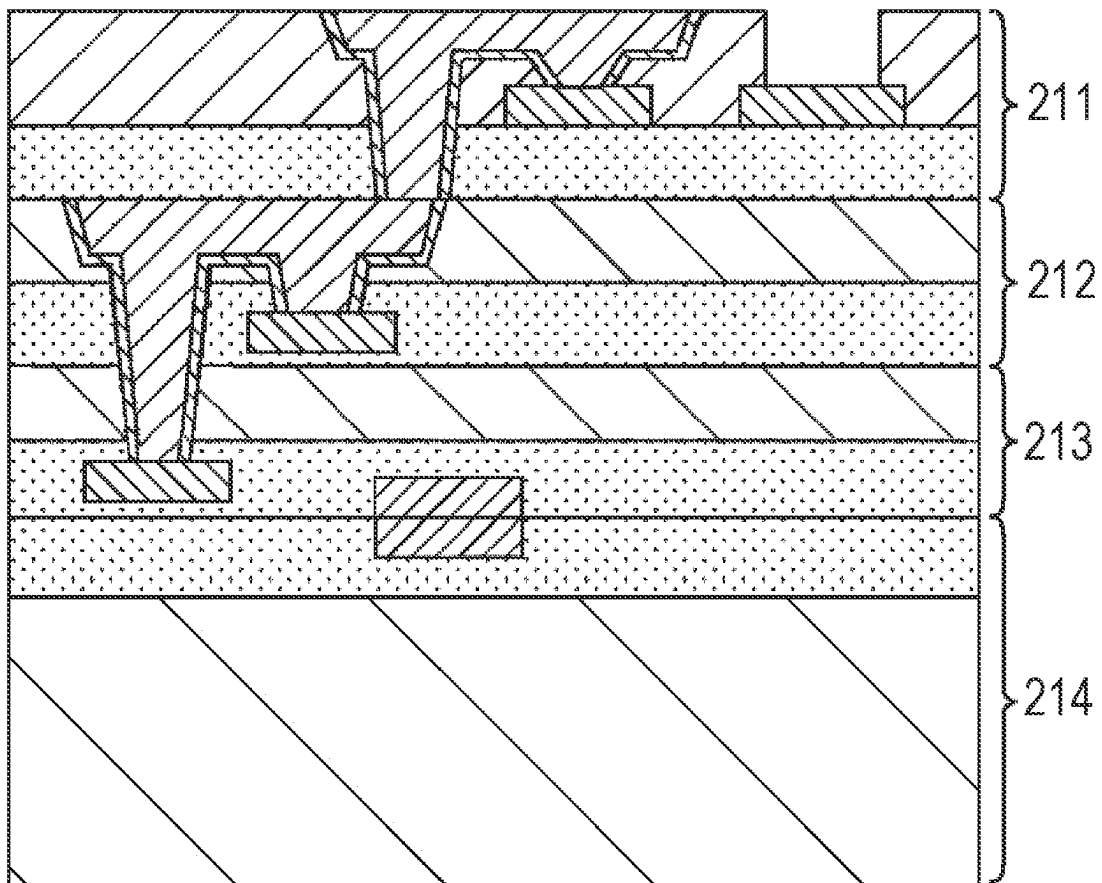
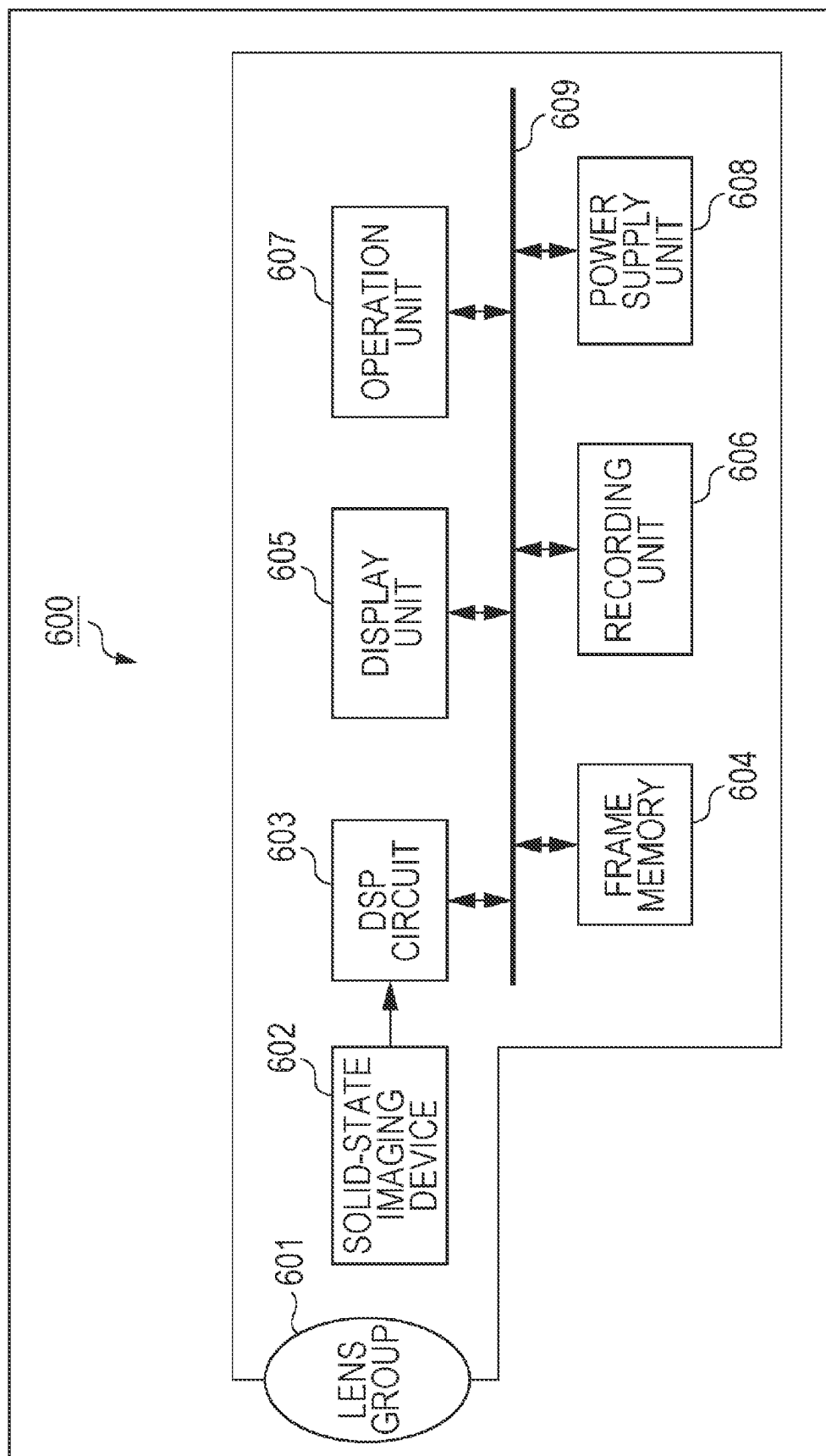


FIG. 43



SEMICONDUCTOR DEVICE, SOLID-STATE IMAGING DEVICE AND ELECTRONIC APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/719,554, filed Apr. 13, 2022, which is a continuation of U.S. patent application Ser. No. 17/068,783, filed Oct. 12, 2020, now U.S. Pat. No. 11,374,049, which is a continuation of U.S. patent application Ser. No. 16/555,067, filed Aug. 29, 2019, now U.S. Pat. No. 10,840,290, which is a continuation of U.S. patent application Ser. No. 16/373,105, filed Apr. 2, 2019, now U.S. Pat. No. 10,475,845, which is a continuation of U.S. patent application Ser. No. 16/042,094, filed Jul. 23, 2018, now U.S. Pat. No. 10,535,700, which is a continuation of U.S. patent application Ser. No. 15/852,468, filed Dec. 22, 2017, now U.S. Pat. No. 10,128,301, which is a continuation of U.S. patent application Ser. No. 15/403,154, filed Jan. 10, 2017, now U.S. Pat. No. 9,917,131, which is a continuation of U.S. patent application Ser. No. 15/087,894, filed Mar. 31, 2016, now U.S. Pat. No. 9,570,499, which is a continuation of U.S. patent application Ser. No. 14/434,288, filed Apr. 8, 2015, now U.S. Pat. No. 9,431,450, which is a national stage application under 35 U.S.C. 371 and claims the benefit of PCT Application No. PCT/JP2013/006055, filed Oct. 10, 2013, which claims priority to Japanese Patent Application Nos. JP 2012-230805 and JP 2013-089580, filed in the Japan Patent Office on Oct. 18, 2012, and Apr. 22, 2013, respectively, the entire disclosures of which are hereby incorporated herein by reference.

TECHNICAL FIELD

The present technology relates to a solid-state imaging device, and particularly to a solid-state imaging device capable of easily providing a high-quality stacked image sensor.

BACKGROUND ART

As solid-state imaging devices, there is an amplification type solid-state imaging device represented by a MOS type image sensor such as a complementary metal oxide semiconductor (CMOS). In addition, there is a charge transfer type solid-state imaging device represented by a charge coupled device (CCD) image sensor.

These solid-state imaging devices are frequently used in digital still cameras, digital video cameras, and the like. In recent years, as solid-state imaging devices have been mounted in mobile apparatuses, such as mobile phones and personal digital assistants (PDAs) with cameras, a MOS type image sensor has been frequently used from the viewpoint of having a low power supply voltage, low power consumption, and the like.

The MOS type solid-state imaging device includes a pixel array (pixel region) in which a plurality of unit pixels are arranged in a two-dimensional array form and a peripheral circuit region, and each of the unit pixels includes a photo diode which is a photoelectric conversion portion and a plurality of pixel transistors. A plurality of pixel transistors are formed of MOS transistors, and generally comprise three transistors including a transfer transistor, a reset transistor, an amplification transistor, or four transistors additionally including a selection transistor.

In addition, in the above-described solid-state imaging device, a stacked structure has been proposed in which a plurality of semiconductor substrates having different functions are stacked in an overlapping manner and are electrically connected to each other.

In the stacked structure, since each circuit can be formed optimally so as to correspond to the function of each semiconductor substrate, it is possible to easily realize high performance of a device.

For example, it is possible to manufacture a high performance solid-state imaging device by optimally forming a sensor circuit and a logical circuit so as to correspond to respective functions of a semiconductor substrate including the sensor circuit and a semiconductor substrate including the logical circuit in which a circuit processing signal is provided. At this time, through electrodes are provided in substrates of the semiconductor substrates, and thereby the plurality of semiconductor substrates are electrically connected to each other.

However, if a semiconductor device is formed by connecting different substrates to each other by using a connection conductor which penetrates through a substrate, it is necessary to form a connection hole while maintaining insulation in the deep substrate, and thus a practical use is difficult from the viewpoint of economic costs of a manufacturing process which is necessary in creating the connection hole and embedding the connection conductor.

On the other hand, for example, if a small contact hole of about 1 micrometer is to be formed, it is necessary to thin an upper substrate to the utmost limit. In this case, complex steps such as, attaching the upper substrate to a support substrate before being thinned and an increase in costs may result. In addition, in order to embed a connection conductor in a connection hole with a high aspect ratio, a CVD film having a good coatability property, such as tungsten (W), is necessarily used as a connection conductor, and thus materials to be used as a connection conductor may be limited.

Therefore, a manufacturing method of a semiconductor device such as a solid-state imaging device has been proposed which achieves a high performance by sufficiently exhibiting each performance, mass productivity, and a reduction in costs (for example, refer to PTL 1).

PTL 1 has proposed a stacked structure in which a support substrate of a rear surface type image sensor is stacked as a logical circuit, and a plurality of connection contacts are provided from the top by using a thinning step of the image sensor.

CITATION LIST

Patent Literature

[PTL 1]
Japanese Unexamined Patent Application Publication No. 2010-245506

SUMMARY OF INVENTION

Technical Problem

It is desirable to easily provide a high-quality stacked image sensor.

Solution to Problem

In accordance with at least one embodiment of the present invention, a semiconductor device is provided, the semicon-

3

ductor device comprising a first semiconductor section including a first wiring layer at one side thereof, the first semiconductor section further including a photodiode, a second semiconductor section including a second wiring layer at one side thereof, the first and second semiconductor sections being secured together, a third semiconductor section including a third wiring layer at one side thereof, the second and the third semiconductor sections being secured together such the first semiconductor section, second semiconductor section, and the third semiconductor section are stacked together, and a first conductive material electrically connecting at least two of (i) the first wiring layer, (ii) the second wiring layer, and (iii) the third wiring layer such that the electrically connected wiring layers are in electrical communication.

In accordance with at least one embodiment of the present invention, a backside illumination type solid-state imaging device is provided, the solid-state imaging device comprising a first semiconductor section including a first wiring layer at one side thereof, the first semiconductor section further including a circuit region and a pixel region, a second semiconductor section including a second wiring layer at one side thereof, the first and second semiconductor sections being secured together, a third semiconductor section including a third wiring layer at one side thereof, the second and the third semiconductor sections being secured together such the first semiconductor section, second semiconductor section, and the third semiconductor section are stacked together, and a first conductive material electrically connecting at least two of (i) the first wiring layer, (ii) the second wiring layer, and (iii) the third wiring layer such that the electrically connected wiring layers are in electrical communication.

In accordance with at least one embodiment of the present invention, an electronic apparatus is provided, the electronic apparatus comprising an optical unit, and a solid-state imaging device, the solid state image device including a first semiconductor section including a first wiring layer at one side thereof, the first semiconductor section further including a circuit region and a pixel region, a second semiconductor section including a second wiring layer at one side thereof, the first and second semiconductor sections being secured together, a third semiconductor section including a third wiring layer at one side thereof, the second and the third semiconductor sections being secured together such the first semiconductor section, second semiconductor section, and the third semiconductor section are stacked together, and a first conductive material electrically connecting at least two of (i) the first wiring layer, (ii) the second wiring layer, and (iii) the third wiring layer such that the electrically connected wiring layers are in electrical communication.

Other systems, methods, features, and advantages of the present invention will be or will become apparent to one with skill in the art upon examination of the following figures and detailed description. It is intended that all such additional systems, methods, features, and advantages be within this description, be within the scope of the invention, and be protected by the accompanying claims.

Advantageous Effects of the Invention

According to the present technology, it is possible to easily provide a high-quality stacked image sensor.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a cross-sectional view illustrating a configuration of a pixel portion of a stacked solid-state imaging device in the related art.

4

FIG. 2 is a cross-sectional view illustrating another configuration of the pixel portion of the stacked solid-state imaging device in the related art.

FIG. 3 is a diagram illustrating a manufacturing method of a three-layer stacked solid-state imaging device.

FIG. 4 is a diagram illustrating a manufacturing method of the three-layer stacked solid-state imaging device.

FIG. 5 is a cross-sectional view illustrating a configuration of a pixel portion of a solid-state imaging device with a three-layer stacked structure manufactured according to FIGS. 3 and 4.

FIG. 6 is a cross-sectional view illustrating a configuration according to an embodiment of a pixel portion of a solid-state imaging device to which the present technology is applied.

FIG. 7A is an enlarged view in the vicinity of the pad hole.

FIG. 7B is a diagram in which the aluminum pad is viewed from the top of the pad hole.

FIG. 8 is a cross-sectional view illustrating a configuration according to another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

FIG. 9 is a cross-sectional view illustrating a configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

FIG. 10 is a diagram illustrating a schematic configuration of a solid-state imaging device to which the present technology is applied.

FIG. 11 is a schematic diagram of the cross-sectional view related to the configuration of the pixel portion of the solid-state imaging device shown in FIG. 6.

FIG. 12 is a schematic diagram of the cross-sectional view illustrating the configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

FIG. 13 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 12.

FIG. 14 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 12.

FIG. 15 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 12.

FIG. 16 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 12.

FIG. 17 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 12.

FIG. 18 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 12.

FIG. 19 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 12.

FIG. 20 is a schematic diagram of the cross-sectional view illustrating the configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

FIG. 21 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 20.

FIG. 22 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 20.

FIG. 23 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 20.

FIG. 24 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 20.

FIG. 25 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 20.

FIG. 26 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 20.

FIG. 27 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 20.

FIG. 28 is a schematic diagram of the cross-sectional view illustrating the configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

FIG. 29 is a schematic diagram of the cross-sectional view illustrating the configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

FIG. 30 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 29.

FIG. 31 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 29.

FIG. 32 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 29.

FIG. 33 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 29.

FIG. 34 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 29.

FIG. 35 is a schematic diagram of the cross-sectional view illustrating the configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

FIG. 36 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 35.

FIG. 37 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 35.

FIG. 38 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 35.

FIG. 39 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 35.

FIG. 40 is a diagram illustrating a manufacturing process of the solid-state imaging device shown in FIG. 35.

FIG. 41 is a diagram illustrating a combination of configurations which may be employed as an embodiment of a solid-state imaging device to which the present technology is applied.

FIG. 42 is a schematic diagram of a cross-sectional view illustrating a configuration of a pixel portion of a solid-state imaging device to which the present technology is applied in a case of employing a four-layer structure.

FIG. 43 is a block diagram illustrating a configuration example of an electronic apparatus to which the present technology is applied.

DESCRIPTION OF EMBODIMENTS

Hereinafter, embodiments of the technology disclosed herein will be described with reference to the drawings.

First, problems of the related art will be described.

As solid-state imaging devices, there is an amplification type solid-state imaging device represented by a MOS type image sensor such as a complementary metal oxide semiconductor (CMOS). In addition, there is a charge transfer type solid-state imaging device represented by a charge coupled device (CCD).

These solid-state imaging devices are frequently used in digital still cameras, digital video cameras, and the like. In recent years, as solid-state imaging devices have been mounted in mobile apparatuses such as, mobile phones and personal digital assistants (PDAs) with cameras, a MOS type image sensor has been frequently used from the viewpoint of a low power supply voltage, power consumption, and the like.

The MOS type solid-state imaging device includes a pixel array (pixel region) in which a plurality of unit pixels are

arranged in a two-dimensional array and a peripheral circuit region, and each of the unit pixels includes a photo diode which is a photoelectric conversion portion and a plurality of pixel transistors. A plurality of pixel transistors are formed of MOS transistors, and generally comprise three transistors including a transfer transistor, a reset transistor, an amplification transistor, or four transistors additionally including a selection transistor.

In addition, in the above-described solid-state imaging device, a stacked structure has been proposed in which a plurality of semiconductor substrates having different functions are stacked in an overlapping manner and are electrically connected to each other.

In the stacked structure, since each circuit can be formed optimally so as to correspond to the function of each semiconductor substrate, it is possible to easily realize high performance of a device.

For example, it is possible to manufacture a high performance solid-state imaging device by optimally forming a sensor circuit and a logical circuit so as to correspond to respective functions of a semiconductor substrate including the sensor circuit and a semiconductor substrate including the logical circuit in which a circuit processing signals is provided. At this time, through electrodes are provided in substrates of the semiconductor substrates, and thereby the plurality of semiconductor substrates are electrically connected to each other.

FIG. 1 is a cross-sectional view illustrating a configuration of a pixel portion of a stacked solid-state imaging device in the related art.

A solid-state imaging device related to this pixel portion includes a rear surface irradiation type CMOS image sensor which is formed by stacking a first semiconductor substrate and a second semiconductor substrate. In other words, the solid-state imaging device shown in FIG. 1 has a two-layer stacked structure.

As shown in FIG. 1, an image sensor, that is, a pixel array (hereinafter, referred to as a pixel region) and a control region is formed in each region of the first semiconductor substrate 31.

That is, a photodiode (PD) 34 which is a photoelectric conversion portion of each pixel is formed in each region of the semiconductor substrate (for example, a silicon substrate) 31, and a source/drain region of each pixel transistor is formed in a semiconductor well region thereof.

A gate electrode is formed on a substrate surface forming the pixel via a gate insulating film, and a pixel transistor Tr1 and a pixel transistor Tr2 are formed by the gate electrode and the source/drain region corresponding thereto.

The pixel transistor Tr1 adjacent to the photodiode (PD) 34 corresponds to a transfer transistor, and the source/drain region corresponds to a floating diffusion (FD).

Next, an interlayer insulating film 39 of the first layer is formed on the surface of the first semiconductor substrate 31, then a connection hole is formed in the interlayer insulating film 39, and connection conductors which are connected to necessary transistors are formed therein.

Successively, a metal wire of a plurality of layers (in this example, two layers) is formed via the interlayer insulating film 39 so as to be connected to each connection conductor, thereby forming a multilayer wire layer 41. The metal wire is formed of a copper (Cu) wire. Generally, each copper wire (metal wire) is covered with a barrier metal film for preventing diffusion of Cu. For this reason, a protective film which is a cap film of the copper wire is formed on the multilayer wire layer 41.

The first semiconductor substrate **31** having the pixel region and the control region is formed through the steps hitherto.

On the other hand, for example, a logical circuit having a signal processing circuit which controls the pixel region or controls communication with an external device and is related to signal processing is formed in each region of the second semiconductor substrate **45**. In other words, a plurality of MOS transistor **Tr6**, MOS transistor **Tr7**, and MOS transistor **Tr8** which form the logical circuit so as to be separated by element separation regions are formed in a p type semiconductor well region on the front surface side of the semiconductor substrate (for example, a silicon substrate) **45**.

Next, an interlayer insulating film **49** of the first layer is formed on the surface of the second semiconductor substrate **45**, then a connection hole is formed in the interlayer insulating film **49**, and connection conductors **54** which are connected to necessary transistors are formed therein.

Successively, a metal wire of a plurality of layers, in this example, four layers, is formed via the interlayer insulating film **49** so as to be connected to each connection conductor **54**, thereby forming a multilayer wire layer **55**.

The metal wire is formed of a copper (Cu) wire. A protective film which is a cap film of the copper wire (metal wire) is formed on the multilayer wire layer **55**. However, the uppermost layer of the multilayer wire layer **55** is formed of an aluminum pad which may be an electrode.

The second semiconductor substrate **45** having the logical circuit is formed through the steps hitherto.

In addition, the first semiconductor substrate **31** and the second semiconductor substrate **45** are joined to each other such that the multilayer wire layer **41** and the multilayer wire layer **55** face each other at a bonding surface **99**. The joining is performed using, for example, plasma bonding and adhesive bonding.

In addition, grinding and polishing are performed from a rear surface **31b** side of the first semiconductor substrate **31** so as to thin the first semiconductor substrate **31**, and the rear surface of the first semiconductor substrate **31** is a light incident surface in a rear surface irradiation type solid-state imaging device.

Through connection holes, which penetrate through the first semiconductor substrate **31** from the rear surface side and reach the aluminum pad of the uppermost layer of the multilayer wire layer **55** of the second semiconductor substrate **45**, are formed at necessary positions of the thinned first semiconductor substrate **31**. In addition, a connection hole, which reaches the wire of the first layer of the first semiconductor substrate **31** side from the rear surface side, is formed in the first semiconductor substrate **31** around the through connection hole.

Next, a through connection conductor **64** and a connection conductor **65** are embedded in the through connection holes. The through connection conductor **64** and the connection conductor **65** may use, for example, a metal such as copper (Cu) or tungsten (W).

As described above, since the logical circuit performing signal processing is formed in the second semiconductor substrate **45**, it is necessary to input and output signals by connecting electrodes of the respective transistors to signal lines. That is, the logical circuit is operated based on input and output of signals with an external device. Therefore, the aluminum pad **53** of the second semiconductor substrate **45** is an electrode for external connection.

For this reason, a pad hole **81** which penetrates through the first semiconductor substrate **31** is formed so as to be

wire-bonded to the aluminum pad **53** of the second semiconductor substrate, thereby exposing the aluminum pad **53**.

Next, an insulating protective film is formed on the entire rear surface of the first semiconductor substrate **31** such that a light blocking film **67** is formed in a region which it is necessary to block light from reaching. For example, a metal film such as tungsten may be used as the light blocking film **67**.

Successively, a planarized film is formed on the light blocking film **67**, on-chip color filters **74** of, for example, red (R), green (G), and blue (B) corresponding to the respective pixels are formed on the planarized film, and an on-chip micro lens **75** is formed thereon.

In addition, the pad hole **81** is formed in the first semiconductor substrate **31** so as to reach the aluminum pad **53** which is an electrode used for transmission, reception, and the like of signals with an external apparatus or circuit, from the rear surface side (light receiving surface side) of the first semiconductor substrate **31**.

Thus, the process for the stacked semiconductor structure is completed. In other words, the pixel region and the control region are formed in the first semiconductor substrate **31**, and the logical circuit is formed in the second semiconductor substrate **45**.

Subsequently, the stacked semiconductor structure is divided into substrates which correspond to a substrate of the rear surface irradiation type solid-state imaging device.

In addition, in the solid-state imaging device with the stacked structure, it is also necessary to take into consideration the influence of noise and the like due to hot carriers. The hot carriers may be high-speed electrons having high kinetic energy that come out of a transistor and generate light when the high-speed electrons impact silicon atoms.

In the solid-state imaging device with the stacked structure, transistors are provided in the second semiconductor substrate separately from the first semiconductor substrate in which the PD is provided. For this reason, light generated by hot carriers coming out of the transistors of the second semiconductor substrate penetrates from the rear side (the opposite side to the light receiving surface) of the PD of the first semiconductor substrate, thereby causing noise.

For this reason, in the solid-state imaging device with the stacked structure, in order to block light caused by the hot carriers for example, a preventative measure, such as providing a light blocking body is performed.

FIG. 2 is a cross-sectional view illustrating another configuration of the pixel portion of the stacked solid-state imaging device in the related art.

In the example shown in FIG. 2, a light blocking body **90** is formed under the PD **34** in the first semiconductor substrate **31**. Thus, light caused by hot carriers coming out of the MOS transistor **Tr6**, the MOS transistor **Tr7**, and the MOS transistor **Tr8** of the second semiconductor substrate **45** is blocked.

Alternatively, a shape of the copper wire of the multilayer wire layer **55** may be changed, or the like, so as to block light caused by hot carriers coming out of the MOS transistor **Tr6**, the MOS transistor **Tr7**, and the MOS transistor **Tr8**.

As described above with reference to FIGS. 1 and 2, in the solid-state imaging device with the two-layer stacked structure, the pad hole **81** is provided so as to be electrically connected to an external device, and light caused by the hot carriers is blocked using the light blocking body **90** or a shape of the copper wire of the multilayer wire layer **55**.

A three-layer stacked solid-state imaging device has been developed. The three-layer stacked solid-state imaging device includes, for example, a third semiconductor sub-

strate in which a memory circuit is formed in addition to the first semiconductor substrate in which the pixel region and the control region (hereinafter, referred to as a sensor circuit) and the second semiconductor substrate in which the logical circuit is formed.

The three-layer stacked solid-state imaging device is manufactured, for example, as shown in FIGS. 3 and 4.

First, as shown in FIG. 3, a second semiconductor substrate 112 and a third semiconductor substrate 113 are joined together such that circuit surfaces thereof face each other. In addition, interlayer films of the two semiconductor substrates are joined to each other. Further, the second semiconductor substrate 112 is thinned.

Next, as shown in FIG. 4, a first semiconductor substrate 111 is joined onto the thinned second semiconductor substrate 112 in a state where the rear surface thereof face upward. In addition, interlayer films of the two semiconductor substrates are joined to each other. Further, the first semiconductor substrate 111 is thinned.

As above, in a case where a stacked image sensor is formed using the three-layer stacked structure, it is necessary for the sensor circuit having a light receiving portion to incorporate light, and thus the sensor circuit is disposed in the uppermost part, and the two logical circuit and memory circuit are stacked in lower layers thereof.

In addition, when the circuits are stacked, it is preferable that a support substrate for thinning a silicon substrate not be used. For this reason, in manufacturing circuits, first, the circuit surfaces of two semiconductor substrates of the lower layers are joined together so as to face each other, and the semiconductor substrate (the second semiconductor substrate 112) of the second layer is thinned. Thereafter, the semiconductor substrate (the first semiconductor substrate 111) of the uppermost layer is joined thereto so as to be stacked thereon as a rear surface type, and is further thinned.

However, in this way, the inventors have found that the following problems occur in the three-layer stacked structure.

FIG. 5 is a cross-sectional view illustrating a configuration of a pixel portion of a three-layer stacked solid-state imaging device manufactured according to FIGS. 3 and 4.

A first problem in the three-layer stacked structure in FIG. 5 is that a pad hole is too deep. In FIG. 5, a pad hole 121 which is deeper than the pad hole 81 of FIG. 1 is formed.

In other words, in the three-layer stacked structure, as described with reference to FIGS. 3 and 4, the circuit surface of the second semiconductor substrate 112 is joined to the circuit surface of the third semiconductor substrate so as to face each other. For this reason, an aluminum pad of the uppermost layer of the multilayer wire layer of the second semiconductor substrate is distant from the light receiving surface of the first semiconductor substrate 111, and thus the aluminum pad 133 (an electrode for external connection) of the second semiconductor substrate is not exposed unless an opening is formed so as to penetrate through the first semiconductor substrate and, further, substantially penetrate through the second semiconductor substrate.

In order to open the deep pad hole 121, thickening of a resist is necessary. If the resist is thickened in order to open the deep pad hole 121, curing of the resist after dry etching is problematic.

For example, since an on-chip micro lens using an organic material has already been formed on the first semiconductor substrate when the opening is formed, the resist is removed using a solution, but the cured resist tends to remain in a residual state and thus impedes light incidence to the lens.

In addition, in a case where the deep pad hole 121 is opened, a deposited substance occurring as a result of the dry etching is also problematic.

Particularly, a deposited substance which is attached to the surface of the aluminum pad 133 or the sidewall of the pad hole 121 and is not removed, for example, absorbs humidity so as to generate fluorine ions after the three-layer stacked structure is completed, and thus causes a defect (corrosion) in which the aluminum pad metal melts.

As above, in FIGS. 3 and 4, it is difficult to perform a manufacturing process of a solid-state imaging device due to the deep pad hole.

A second problem in the three-layer stacked structure in FIG. 5 is that it is difficult to block light caused by hot carriers.

In other words, in using the three-layer stacked structure, as described with reference to FIGS. 3 and 4, the circuit surface of the second semiconductor substrate 112 is joined to the circuit surface of the third semiconductor substrate so as to face each other. For this reason, a transistor of the second semiconductor substrate faces the first semiconductor substrate without using the multilayer wire layer. Therefore, for example, in the same manner as a case of the two-layer stacked structure, light caused by hot carriers is unable to be blocked by the copper wire of the multilayer wire layer of the second semiconductor substrate.

Therefore, in the present technology, it is not necessary to provide a deep pad hole; therefore, light caused by hot carriers can be easily blocked.

FIG. 6 is a cross-sectional view illustrating a configuration according to an embodiment of a pixel portion of a solid-state imaging device to which the present technology is applied. A solid-state imaging device related to this pixel portion includes a rear surface irradiation type CMOS image sensor formed by stacking a first semiconductor substrate, a second semiconductor substrate, and a third semiconductor substrate. In other words, the solid-state imaging device related to the pixel portion shown in FIG. 6 has a three-layer stacked structure.

In addition, the solid-state imaging device includes, for example, a first semiconductor substrate provided with a sensor circuit, a second semiconductor substrate provided with a logical circuit, and a third semiconductor substrate provided with a memory circuit. The logical circuit and the memory circuit are respectively operated based on input and output of signals with an external device.

As shown in FIG. 6, a photodiode (PD) 234 which is a photoelectric conversion portion of each pixel is formed in the semiconductor substrate (for example, a silicon substrate) 211, and a source/drain region of each pixel transistor is formed in a semiconductor well region thereof.

A gate electrode is formed on a substrate surface forming the pixel via a gate insulating film, and a pixel transistor Tr1 and a pixel transistor Tr2 are formed by the gate electrode and the source/drain region corresponding thereto.

The pixel transistor Tr1 adjacent to the photodiode (PD) 234 corresponds to a transfer transistor, and the source/drain region corresponds to a floating diffusion (FD).

In addition, an interlayer insulating film is formed in the first semiconductor substrate 211, connection hole are formed in the interlayer insulating film, and connection conductors 244 connected to the pixel transistor Tr1 and the pixel transistor Tr2 are formed therein.

In addition, in order to be connected to each connection conductor 244, a metal wire 240 of a plurality of layers is formed so as to form a multilayer wire layer 245. The metal wire 240 is formed of a copper (Cu) wire. Generally, each

11

copper wire is covered with a barrier metal film for preventing diffusion of Cu. For this reason, a protective film which is a cap film of the copper wire is formed on the multilayer wire layer **245**.

In addition, an aluminum pad **280** which is an electrode for external connection is formed in the lowermost layer of the multilayer wire layer **245** of the first semiconductor substrate **211**. In other words, the aluminum pad **280** is formed at a position closer to an adhesive surface **291** with the second semiconductor substrate **212** than the copper wire **240**. This electrode for external connection is used as one end of a wire related to input and output of a signal with an external device. In addition, although the electrode has been described as being made of aluminum, the electrode may be made of other metals.

In addition, a contact **265** which is used for electrical connection to the second semiconductor substrate **212** is formed in the first semiconductor substrate **211**. The contact **265** is connected to a contact **311** of the second semiconductor substrate **212**, described later, and is also connected to an aluminum pad **280a** of the first semiconductor substrate **211**.

In addition, a pad hole **351** is formed in the first semiconductor substrate **211** so as to reach the aluminum pad **280a** from the rear surface side (the light receiving surface side) of the first semiconductor substrate **211**.

FIGS. 7A and 7B are diagrams illustrating a configuration of the pad hole **351** and the aluminum pad **280a**. FIG. 7A is an enlarged view of the vicinity of the pad hole **351**, and FIG. 7B is a diagram in which the aluminum pad **280a** is viewed from the top of the pad hole **351**.

As shown in FIG. 7B, a plurality of contacts **265** are arranged and are connected to an end of the aluminum pad **280a** so as to reduce contact resistance.

Returning to FIG. 6, in the same manner as the case described with reference to FIG. 1, an insulating protective film is formed on the entire rear surface in the first semiconductor substrate **211** so as to form a light block film in a region which it is necessary to block light from reaching. Further, on-chip color filters **274** which correspond to the respective pixels are formed on the planarized film, and an on-chip micro lens **275** is formed thereon.

Meanwhile, a logical circuit is formed in the second semiconductor substrate **212**. In other words, a MOS transistor **Tr6**, a MOS transistor **Tr7**, and a MOS transistor **Tr8** which comprise a plurality of transistors forming the logical circuit, are formed in a p type semiconductor well region of the semiconductor substrate (for example, a silicon substrate) **212**.

In addition, connection conductors **254** which are connected to the MOS transistor **Tr6**, the MOS transistor **Tr7**, and the MOS transistor **Tr8** are formed in the second semiconductor substrate **212**.

In addition, a metal wire **250** of a plurality of layers is formed, and a multilayer wire layer **255** is formed so as to be connected to the respective connection conductors **254**.

The metal wire is formed of a copper (Cu) wire. A protective film which is a cap film of the copper wire (metal wire) **250** is formed on the multilayer wire layer **255**.

In addition, an aluminum pad **320** which is an electrode is formed in the lowermost layer of the multilayer wire layer **255** of the second semiconductor substrate **212**.

Further, a contact **311** which is used for electrical connection to the first semiconductor substrate **211** and the third semiconductor substrate **213** is formed in the second semiconductor substrate **212**. The contact **311** is connected to the

12

contact **265** of the first semiconductor substrate **211** and is also connected to an aluminum pad **330a** of the third semiconductor substrate **213**.

In addition, a memory circuit is formed in the third semiconductor substrate **213**. In other words, a MOS transistor **Tr11**, a MOS transistor **Tr12**, and a MOS transistor **Tr13** which are a plurality of transistors forming the memory circuit are formed in a p type semiconductor well region of the semiconductor substrate (for example, a silicon substrate) **213**.

In addition, connection conductors **344** which are connected to the MOS transistor **Tr11**, the MOS transistor **Tr12**, and the MOS transistor **Tr13** are formed in the third semiconductor substrate **213**.

In addition, a metal wire **340** of a plurality of layers is formed, and a multilayer wire layer **345** is formed so as to be connected to the respective connection conductors **344**.

The metal wire is formed of a copper (Cu) wire. A protective film which is a cap film of the copper wire (metal wire) **340** is formed on the multilayer wire layer **345**.

In addition, an aluminum pad **330** which is an electrode is formed in the uppermost layer of the multilayer wire layer **345**.

In the solid-state imaging device shown in FIG. 6, the contact **265** and the contact **311** are provided, and thus input and output of signals can be performed between the first semiconductor substrate **211** to the third semiconductor substrate **213** via the aluminum pad **280a**.

Also in the solid-state imaging device shown in FIG. 6, as described with reference to FIGS. 3 and 4, interlayer films of the second semiconductor substrate **212** and the third semiconductor substrate **213** are joined to each other at an adhesive surface **292**. In addition, the interlayer films of the second semiconductor substrate **212** and the first semiconductor substrate **211** are joined to each other at the adhesive surface **291**.

In other words, as described with reference to FIGS. 3 and 4, first, the two semiconductor substrates of the lower layers are joined together such that the circuit surfaces thereof face each other, and the semiconductor substrate (the second semiconductor substrate **212**) of the second layer is thinned. Next, the semiconductor substrate (the first semiconductor substrate **211**) of the uppermost layer is joined thereto so as to be stacked thereon as a rear surface type, and is further thinned. At this time, after the upper layer of the contact **311** is planarized, the first semiconductor substrate **211** is joined to the second semiconductor substrate **212** as a rear surface type.

In this way, when the circuits are stacked, a support substrate for thinning the silicon substrate is not used.

In the present technology, the aluminum pad **280** is also provided in the first semiconductor substrate **211**. In addition, an electrode for an external electrode is not provided in the second semiconductor substrate **212** having the logical circuit in which input and output of signals to and from an external device are necessary, or the third semiconductor substrate **213** having the memory circuit, and an electrode (the aluminum pad **280a**) for an external connection is provided in the first semiconductor substrate **211** having the sensor circuit.

In this way, it is possible to expose the electrode for external connection without deepening the pad hole **351**.

In addition, in the present technology, since the aluminum pad **280** is also provided in the first semiconductor substrate **211**, it is possible to block light caused by hot carriers coming out of each transistor of the second semiconductor substrate **212** by the aluminum pad **280**.

13

As above, in the present technology, it is not necessary to provide a deep pad hole, and it is possible to easily prevent the influence of noise and the like due to hot carriers by blocking the light caused by hot carriers.

In addition, and with respect to FIG. 6, although the aluminum pad 320 is provided in the second semiconductor substrate and the aluminum pad 330 is provided in the third semiconductor substrate 213, the aluminum pad 320 and the aluminum pad 330 may not be provided. For example, if the contact 311 is to be directly connected to the copper wire 340 of the third semiconductor substrate 213, it is unnecessary to provide the aluminum pad 320 and the aluminum pad 330.

In addition, a shape of the contact which electrically connects the semiconductor substrates to each other is not limited to the shapes of the contact 265 and the contact 311. In addition, since a hole for forming a contact can be formed before an on-chip micro lens is formed, even the problem associated with a deep pad hole is not an issue. For example, a contact may be provided which penetrates through the second semiconductor substrate so as to connect the copper wire of the first semiconductor substrate to the copper wire of the third semiconductor substrate.

Alternatively, a light blocking body which blocks light caused by hot carriers may be formed.

FIG. 8 is a cross-sectional view illustrating a configuration according to another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied. In the same manner as in FIG. 6, a solid-state imaging device related to this pixel portion includes a rear surface type CMOS image sensor formed by stacking a first semiconductor substrate, a second semiconductor substrate, and a third semiconductor substrate. In other words, the solid-state imaging device related to the pixel portion shown in FIG. 8 also has a three-layer stacked structure.

In the example shown in FIG. 8, a light blocking body 360 is disposed in an interlayer film which is the uppermost layer of the second semiconductor substrate 212 in FIG. 8. Thus, it is possible to more reliably block light caused by hot carriers coming out of each transistor of the second semiconductor substrate 212.

In addition, since the aluminum pad 280 is formed in the first semiconductor substrate 211, the light blocking body 360 is not disposed in the first semiconductor substrate 211 and is disposed in the interlayer film of the second semiconductor substrate 212.

The other constituent elements in FIG. 8 are the same as in the case described with reference to FIG. 6, and thus detailed description thereof will be omitted.

Alternatively, a copper wire may be formed in the interlayer film which is the uppermost layer of the second semiconductor substrate 212 in FIG. 8, and light caused by hot carriers may be blocked by a combination of the aluminum pad and the copper wire.

FIG. 9 is a cross-sectional view illustrating a configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied. In the same manner as in FIG. 6, a solid-state imaging device related to this pixel portion includes a rear surface irradiation type CMOS image sensor formed by stacking a first semiconductor substrate, a second semiconductor substrate, and a third semiconductor substrate. In other words, the solid-state imaging device related to the pixel portion shown in FIG. 9 also has a three-layer stacked structure.

14

In the example shown in FIG. 9, a copper wire 370 is disposed in an interlayer film which is the uppermost layer of the second semiconductor substrate 212 in FIG. 9.

A part of the contact 311 is formed in the interlayer film which is the uppermost layer of the second semiconductor substrate 212 in FIG. 9. For example, if the copper wire 370 is further formed in the interlayer film when the contact 311 is formed, a solid-state imaging device with the configuration shown in FIG. 9 can be obtained.

When light is blocked by a combination of the copper wire 370 and the aluminum pad 280, it is possible to more reliably block light caused by hot carriers coming out of each transistor of the second semiconductor substrate 212. In addition, in an embodiment of the configuration shown in FIG. 9, additional freedom in designing wires related to the aluminum pad 280 may be utilized, as compared with a case of blocking light only by the aluminum pad 280, for example, as shown in FIG. 6.

The other constituent elements in FIG. 9 are the same as in the case described with reference to FIG. 6, and thus detailed description thereof will be omitted.

FIG. 10 is a diagram illustrating a schematic configuration of a solid-state imaging device to which the present technology is applied. This solid-state imaging device 401 includes, for example, a CMOS image sensor.

The solid-state imaging device 401 of FIG. 10 includes a pixel region (a so-called pixel array) 403 in which pixels 402 including a plurality of photoelectric conversion portions are regularly arranged in a two-dimensional array on the semiconductor substrate 411 and a peripheral circuit portion.

Each of the pixels 402 includes, for example, a photodiode which is the photoelectric conversion portion and a plurality of pixel transistors (so-called MOS transistors).

In addition, the pixels 402 may have a shared pixel structure. This pixel shared structure is formed by a plurality of photodiodes, a plurality of transfer transistors, a single shared floating diffusion, and another shared transistor.

The peripheral circuit portion includes a vertical driving circuit 404, a column signal processing circuit 405, a horizontal driving circuit 406, an output circuit 407, a control circuit 408, and the like.

The control circuit 408 receives an input clock and data for commanding an operation mode and the like, and outputs data such as internal information of the solid-state imaging device. In other words, the control circuit 408 generates a clock signal used as a reference of operations of the vertical driving circuit 404, the column signal processing circuit 405, the horizontal driving circuit 406, and the like, and control signals on the basis of a vertical synchronization signal, a horizontal synchronization signal, and a master clock. In addition, these signals are input to the vertical driving circuit 404, the column signal processing circuit 405, the horizontal driving circuit 406, and the like.

The vertical driving circuit 404 including, for example, shift registers, selects a pixel driving line, and supplies a pulse for driving the pixels to the selected pixel driving line so as to drive the pixels in the unit of a row. In other words, the vertical driving circuit 404 sequentially selectively scans the respective pixels 402 of the pixel region 403 in the vertical direction in the unit of a row, and supplies a pixel signal based on signal charge which is generated according to a light receiving amount in, for example, the photodiode which is a photoelectric conversion portion of each pixel 402, to the column signal processing circuit 405 via a vertical signal line 409.

The column signal processing circuit 405 is disposed, for example, for each column of the pixels 402, and performs a

15

signal process such as noise removal on signals output from the pixels **402** of one row for each pixel column. In other words, the column signal processing circuit **405** performs signal processes such as CDS for removing fixed pattern noise unique to the pixels **402**, signal amplification, and AD conversion. In an output end of the column signal processing circuit **405**, a horizontal selection switch (not shown) is provided so as to be connected to a horizontal signal line **410**.

The horizontal driving circuit **406** includes, for example, shift registers, sequentially outputs horizontal scanning pulses so as to sequentially select the respective column signal processing circuits **405**, thereby outputting a pixel signal from each of the column signal processing circuits **405** to the horizontal signal line **410**.

The output circuit **407** performs a signal process on the signals which are sequentially supplied from the respective column signal processing circuits **405** via the horizontal signal line **410** so as to be output. For example, only buffering may be performed, or black level adjustment, column disparity correction, a variety of digital signal processes, and the like may be performed. An input and output terminal **412** sends and receives signals to and from an external device.

The solid-state imaging device **401** shown in FIG. **10** includes a rear surface irradiation type CMOS image sensor with a three-layer stacked structure. For example, the pixels **402** shown in FIG. **10** are sensor circuits formed in the first semiconductor substrate, and the peripheral circuits are logical circuits formed in the second semiconductor substrate or memory circuits formed in the third semiconductor substrate.

In the above-described embodiment, and as previously described, the aluminum pad **280** is formed in the lowermost layer of the multilayer wire layer **245** of the first semiconductor substrate **211**. However, for example, in a case where the aluminum pad **280** is disposed in the first semiconductor substrate **211**, it is necessary to provide an electro-static discharge (ESD) circuit which is a circuit for protecting circuits of the first semiconductor substrate **211** from over-current, and this increases the number of steps.

In addition, in the above example described with reference to FIG. **6**, it is possible to achieve an effect in which light caused by hot carriers is blocked by the aluminum pad **280** disposed in the first semiconductor substrate **211**. However, since the multilayer wire layer **245** of the first semiconductor substrate **211** includes three wire layers, it is difficult to dispose the aluminum pad **280** so as to block light caused by hot carriers without restricting a shape of the copper wire **240**.

For example, since the multilayer wire layer **255** of the second semiconductor substrate **212** includes six wire layers, if the aluminum pad **280** is disposed in the second semiconductor substrate **212**, it is easy to dispose the aluminum pad **280** so as to block light caused by hot carriers without restricting a shape of the metal wire **250**.

In addition, in the above-described embodiment, the contact **265** used for electrical connection between the first semiconductor substrate **211** and the second semiconductor substrate **212** has a configuration in which conductors embedded in two through holes which penetrate through the first semiconductor substrate **211** in the vertical direction are connected to each other on the light receiving surface (the uppermost surface in FIG. **9**) of the first semiconductor substrate **211**. This contact is also referred to as a twin contact. The contact **311** used for electrical connection

16

between the second semiconductor substrate **212** and the third semiconductor substrate **213** is also a twin contact.

However, since it is necessary to provide two through holes for the twin contact, the number of manufacturing steps increases, and the area occupying the substrate also increases.

For example, when a contact is formed which penetrates through the first semiconductor substrate **211** from the uppermost side of the first semiconductor substrate **211** in FIG. **9** and reaches the wire inside the multilayer wire layer **255** of the second semiconductor substrate, a part of which reaches the wire inside the multilayer wire layer **245** of the first semiconductor substrate **211**, only a single through hole is provided and thereby the first semiconductor substrate **211** can be electrically connected to the second semiconductor substrate **212**. This contact is also referred to as a shared contact.

When the shared contact is used for electrical connection between the semiconductor substrates, manufacturing steps can be simplified and thus the area occupying the substrate can be reduced as compared with a case of using the twin contact.

In addition, when semiconductor substrates are joined together, a method in which the copper wires of the multilayer wire layers are directly bonded to each other has been utilized. If the copper wires of the multilayer wire layers are directly bonded to each other, it is unnecessary to provide a contact for electrical connection between semiconductor substrates, manufacturing steps can be further simplified, and thus the area occupying the substrate can be reduced. In addition, the method of directly bonding the copper wires to each other is also referred to as direct bonding.

FIG. **11** is a schematic diagram of the cross-sectional view related to the configuration of the pixel portion of the solid-state imaging device shown in FIG. **6**. As shown in FIG. **11**, the pad hole **351** is formed in the first semiconductor substrate **211** so as to reach the aluminum pad **280a** from the rear surface side (light receiving surface side) of the first semiconductor substrate **211**. In addition, the aluminum pad **280** is formed in the multilayer wire layer **245** of the first semiconductor substrate **211**.

Further, in the configuration shown in FIG. **11**, the multilayer wire layer **255** of the second semiconductor substrate faces the third semiconductor substrate **213** side (the lower side of FIG. **11**), and the first semiconductor substrate **211** is joined to the second semiconductor substrate **212**.

In addition, in the configuration shown in FIG. **11**, the contact **265** used for electrical connection between the first semiconductor substrate **211** and the second semiconductor substrate **212** and the contact **311** used for electrical connection between the second semiconductor substrate **212** and the third semiconductor substrate **213** are provided. The contact **265** and the contact **311** are respectively formed by a twin contact.

FIG. **12** is a schematic diagram of the cross-sectional view illustrating the configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

In the configuration shown in FIG. **12**, unlike in the case of FIG. **11**, the multilayer wire layer **255** of the second semiconductor substrate faces the first semiconductor substrate **211** side (the upper side in FIG. **12**), and the first semiconductor substrate **211** is joined to the second semiconductor substrate **212**.

In addition, in the configuration shown in FIG. **12**, unlike in the case of FIG. **11**, the aluminum pad **280** is provided in the multilayer wire layer **255** of the second semiconductor

17

substrate **212**. Further, the pad hole **351** is formed in the first semiconductor substrate **211** so as to reach the aluminum pad **280a** from the rear surface side (light receiving surface side) of the first semiconductor substrate **211**.

As shown in FIG. **12**, the multilayer wire layer **255** of the second semiconductor substrate **212** faces the first semiconductor substrate side, and thus light caused by hot carriers can be blocked by the multilayer wire layer **255**. In addition, the aluminum pad **280** is disposed in the multilayer wire layer **255** including the six wire layers, and thus it is easy to dispose the aluminum pad **280** so as to block light caused by hot carriers without restricting a shape of the metal wire **250**.

In addition, since the aluminum pad **280** is provided in the multilayer wire layer **255** of the second semiconductor substrate **212**, it is unnecessary to provide an ESD circuit in the first semiconductor substrate **211** (because an ESD circuit is preferably formed in the second semiconductor substrate), and thus it is possible to manufacture a solid-state imaging device at a low cost.

In addition, in the configuration shown in FIG. **12**, the contact **266** used for electrical connection between the first semiconductor substrate **211** and the second semiconductor substrate **212** and the contact **312** used for electrical connection between the second semiconductor substrate **212** and the third semiconductor substrate **213** are provided. The contact **266** and the contact **312** are respectively formed by a twin contact.

In a case of the configuration shown in FIG. **12**, unlike in the case of FIG. **11**, the contact **312** penetrates through the first semiconductor substrate **211** and the second semiconductor substrate **212** and reaches the multilayer wire layer **345** of the third semiconductor substrate **213**.

Next, a manufacturing process of the solid-state imaging device shown in FIG. **12** will be described.

First, as shown in FIG. **13**, the first semiconductor substrate **211**, the second semiconductor substrate **212**, and the third semiconductor substrate **213** are prepared which are respectively provided with the multilayer wire layers. As shown in FIG. **13**, the first semiconductor substrate **211** is provided with the multilayer wire layer **245**, the second semiconductor substrate **212** is provided with the multilayer wire layer **255**, and the third semiconductor substrate **213** is provided with the multilayer wire layer **345**.

In addition, as shown in FIG. **13**, the aluminum pad **280** is formed in the multilayer wire layer **255** of the second semiconductor substrate **212**.

Next, as shown in FIG. **14**, the first semiconductor substrate **211** is joined to the second semiconductor substrate **212**. At this time, the first semiconductor substrate **211** is joined to the second semiconductor substrate **212** such that the multilayer wire layer **245** and the multilayer wire layer **255** face each other.

In addition, as shown in FIG. **15**, the second semiconductor substrate **212** is thinned. In FIG. **15**, the width of the second semiconductor substrate **212** in the vertical direction of FIG. **15** is reduced.

Next, as shown in FIG. **16**, the third semiconductor substrate **213** is joined to the second semiconductor substrate **212**. At this time, the second semiconductor substrate **212** is joined to the third semiconductor substrate **213** such that the multilayer wire layer **345** of the third semiconductor substrate faces upward in FIG. **16**.

In addition, as shown in FIG. **17**, the first semiconductor substrate **211** is thinned. In FIG. **17**, the width of the first semiconductor substrate **211** in the vertical direction of FIG. **17** is reduced.

18

Next, as shown in FIG. **18**, the contact **312** and the contact **266** are formed. At this time, a hole is provided which reaches the multilayer wire layer **245** from the light receiving surface of the first semiconductor substrate **211**, and a hole is provided which reaches the aluminum pad **280** of the multilayer wire layer **255** from the light receiving surface, thereby forming the contact **266**. In addition, a hole is provided which reaches the aluminum pad **280** of the multilayer wire layer **255** from the light receiving surface of the first semiconductor substrate **211**, and a hole is provided which reaches the multilayer wire layer **345** from the light receiving surface, thereby forming the contact **312**.

In addition, as shown in FIG. **19**, the pad hole **351** is formed so as to reach the aluminum pad **280a** from the rear surface side (light receiving surface side) of the first semiconductor substrate **211**.

In this way, the solid-state imaging device described with reference to FIG. **12** is manufactured. Thus, it is possible to block light caused by hot carriers by the multilayer wire layer **255**. In addition, the aluminum pad **280** is disposed in the multilayer wire layer **255** including the six wire layers, and thus it is easy to locate the aluminum pad **280** so as to block light caused by hot carriers without restricting a shape of the metal wire **250**. Further, since the aluminum pad **280** is provided in the multilayer wire layer **255** of the second semiconductor substrate **212**, it is unnecessary to provide an ESD circuit in the first semiconductor substrate **211** (because an ESD circuit is preferably formed in the second semiconductor substrate), and thus it is possible to manufacture a solid-state imaging device at a low cost.

FIG. **20** is a schematic diagram of the cross-sectional view illustrating the configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

In the configuration shown in FIG. **20**, in the same manner as in the case of FIG. **11**, the multilayer wire layer **255** of the second semiconductor substrate faces the third semiconductor substrate **213** side (the lower side of FIG. **20**), and the first semiconductor substrate **211** is joined to the second semiconductor substrate **212**.

In addition, in the configuration shown in FIG. **20**, in the same manner as in the case of FIG. **11**, the contact **265** used for electrical connection between the first semiconductor substrate **211** and the second semiconductor substrate **212** and the contact **311** used for electrical connection between the second semiconductor substrate **212** and the third semiconductor substrate **213** are provided. The contact **265** and the contact **311** are respectively formed by a twin contact.

In addition, in the configuration shown in FIG. **20**, unlike in the case of FIG. **11**, an insulating film layer **230** is formed between the first semiconductor substrate **211** and the second semiconductor substrate **212**. Further, the aluminum pad **280a** is disposed in the insulating film layer **230**, and the aluminum pad **280a** is connected to a contact **313** which is connected to the multilayer wire layer **255** of the second semiconductor substrate **212**.

Furthermore, in the configuration shown in FIG. **20**, the pad hole **351** is formed in the first semiconductor substrate **211** so as to reach the aluminum pad **280a** inside the insulating film layer **230** from the rear surface side (light receiving surface side) of the first semiconductor substrate **211**.

In a case of the configuration shown in FIG. **20**, since the aluminum pad **280** is provided in the insulating film layer **230**, it is unnecessary to provide an ESD circuit in the first semiconductor substrate **211** (because an ESD circuit is

19

preferably formed in the second semiconductor substrate), and thus it is possible to manufacture a solid-state imaging device at a low cost.

Next, a manufacturing process of the solid-state imaging device shown in FIG. 20 will be described.

First, as shown in FIG. 21, the first semiconductor substrate 211, the second semiconductor substrate 212, and the third semiconductor substrate 213 are prepared which are respectively provided with the multilayer wire layers. As shown in FIG. 21, the first semiconductor substrate 211 is provided with the multilayer wire layer 245, the second semiconductor substrate 212 is provided with the multilayer wire layer 255, and the third semiconductor substrate 213 is provided with the multilayer wire layer 345.

In addition, as shown in FIG. 21, the aluminum pad 280 is not formed in the multilayer wire layer 255 of the second semiconductor substrate 212.

Next, as shown in FIG. 22, the second semiconductor substrate 212 is joined to the third semiconductor substrate 213. At this time, the second semiconductor substrate 212 is joined to the third semiconductor substrate 213 such that the multilayer wire layer 255 and the multilayer wire layer 345 face each other.

In addition, as shown in FIG. 23, the second semiconductor substrate 212 is thinned. In FIG. 23, the width of the second semiconductor substrate 212 in the vertical direction of FIG. 23 is reduced.

Next, as shown in FIG. 24, the contact 311 and the contact 313 are formed. At this time, a hole is provided which reaches the multilayer wire layer 345 from the upper surface of the second semiconductor substrate 212 in FIG. 24, and a hole is provided which reaches the multilayer wire layer 255 from the upper surface of the second semiconductor substrate 212 in FIG. 24, thereby forming the contact 311. In addition, a hole is provided which reaches the multilayer wire layer 255 from the upper surface of the second semiconductor substrate 212 in FIG. 24, thereby forming the contact 313.

In addition, as shown in FIG. 25, the aluminum pad 280a is formed, and the insulating film layer 230 is formed. As shown in FIG. 25, the aluminum pad 280a is formed so as to be connected to the upper end of the contact 313 in FIG. 25. In addition, the insulating film layer 230 is formed around the aluminum pad 280a in the upper surface of the second semiconductor substrate 212 in FIG. 25.

Next, as shown in FIG. 26, the first semiconductor substrate 211 is joined to the second semiconductor substrate 212 (more accurately, the insulating film layer 230). At this time, the first semiconductor substrate 211 is joined to the second semiconductor substrate 212 such that the multilayer wire layer 245 comes into contact with the insulating film layer 230.

In addition, the first semiconductor substrate 211 is thinned. In FIG. 26, the width of the first semiconductor substrate 211 in the vertical direction of FIG. 26 is reduced.

In addition, as shown in FIG. 27, the pad hole 351 is formed so as to reach the aluminum pad 280a from the rear surface side (light receiving surface side) of the first semiconductor substrate 211. Successively, a hole is provided which reaches the multilayer wire layer 245 from the light receiving surface of the first semiconductor substrate 211, and a hole is provided which reaches the contact 311 from the light receiving surface, thereby forming the contact 265.

In this way, the solid-state imaging device described with reference to FIG. 20 is manufactured. Further, since the aluminum pad 280 is provided in the insulating film layer 230, it is unnecessary to provide an ESD circuit in the first

20

semiconductor substrate 211 (because an ESD circuit is preferably formed in the second semiconductor substrate), and thus it is possible to manufacture a solid-state imaging device at a low cost.

FIG. 28 is a schematic diagram of the cross-sectional view illustrating the configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

In the configuration shown in FIG. 28, in the same manner as in the case of FIG. 11, the pad hole 351 is formed in the first semiconductor substrate 211 so as to reach the aluminum pad 280a from the rear surface side (light receiving surface side) of the first semiconductor substrate 211. In addition, the aluminum pad 280 is formed in the multilayer wire layer 245 of the first semiconductor substrate 211.

In the configuration shown in FIG. 28, in the same manner as in the case of FIG. 11, the multilayer wire layer 255 of the second semiconductor substrate faces the third semiconductor substrate 213 side (the lower side of FIG. 28), and the first semiconductor substrate 211 is joined to the second semiconductor substrate 212.

In addition, in the configuration shown in FIG. 28, in the same manner as in the case of FIG. 11, the contact 265 used for electrical connection between the first semiconductor substrate 211 and the second semiconductor substrate 212 is provided. The contact 265 is formed by a twin contact.

In the configuration shown in FIG. 28, unlike in the case of FIG. 11, the contact 311 used for electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213 is not provided. On the other hand, the contact 314 and the contact 315 used for electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213 are provided.

Each of the contact 314 and the contact 315 is formed by providing a through hole which penetrates through the second semiconductor substrate 212 and reaches the multilayer wire layer 345 of the third semiconductor substrate 213 and embedding a conductor therein. In other words, each of the contact 314 and the contact 315 connects the multilayer wire layer 255 of the second semiconductor substrate 212 to the multilayer wire layer 345 of the third semiconductor substrate 213 by providing only a single through hole.

In other words, each of the contact 314 and the contact 315 is formed by a shared contact.

In the configuration shown in FIG. 28, it is possible to simplify manufacturing steps and to reduce the area occupying the substrate by using the shared contact.

Although the description has been provided here of a case where the shared contact is used for electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213, the shared contact may be used for electrical connection between the first semiconductor substrate 211 and the second semiconductor substrate 212.

In addition, similarly, in the solid-state imaging device with the configuration described with reference to FIG. 11, 12 or 20, the shared contact may also be used for electrical connection between the first semiconductor substrate 211 and the second semiconductor substrate 212, or electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213.

In other words, in the configuration (FIG. 11) in which the aluminum pad 280 is provided in the multilayer wire layer 245 of the first semiconductor substrate 211, the shared contact may be used for electrical connections between the respective semiconductor substrates. In addition, in the

21

configuration (FIG. 12) in which the aluminum pad 280 is provided in the multilayer wire layer 255 of the second semiconductor substrate 212, the shared contact may be used for electrical connections between the respective semiconductor substrates. Further, in the configuration (FIG. 20) in which the aluminum pad 280 is provided in the insulating film layer 230, the shared contact may be used for electrical connections between the respective semiconductor substrates.

FIG. 29 is a schematic diagram of the cross-sectional view illustrating the configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

In the configuration shown in FIG. 29, in the same manner as in the case of FIG. 11, the pad hole 351 is formed in the first semiconductor substrate 211 so as to reach the aluminum pad 280a from the rear surface side (light receiving surface side) of the first semiconductor substrate 211. In addition, the aluminum pad 280 is formed in the multilayer wire layer 245 of the first semiconductor substrate 211.

In the configuration shown in FIG. 29, in the same manner as in the case of FIG. 11, the multilayer wire layer 255 of the second semiconductor substrate faces the third semiconductor substrate 213 side (the lower side of FIG. 29), and the first semiconductor substrate 211 is joined to the second semiconductor substrate 212.

In addition, in the configuration shown in FIG. 29, a contact 267 used for an electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213 is provided. The contact 267 is formed by a twin contact.

Further, in the configuration shown in FIG. 29, a metal wire 250a in the multilayer wire layer 255 of the second semiconductor substrate 212 is directly bonded to a metal wire 340a in the multilayer wire layer 345 of the third semiconductor substrate 213. Furthermore, a metal wire 250b in the multilayer wire layer 255 is directly bonded to a metal wire 340b in the multilayer wire layer 345. Thus, the second semiconductor substrate 212 is electrically connected to the third semiconductor substrate 213.

In other words, in a case of the configuration shown in FIG. 29, not a contact but direct bonding is used for electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213. Therefore, it is possible to simplify manufacturing steps and to reduce the area occupying the substrate.

In addition, the direct bonding is disclosed in detail in, for example, Japanese Unexamined Patent Application Publication No. 2013-033900 which is hereby incorporated herein by reference in its entirety for all that it teaches and for all purposes.

Next, a manufacturing process of the solid-state imaging device shown in FIG. 29 will be described.

First, as shown in FIG. 30, the first semiconductor substrate 211, the second semiconductor substrate 212, and the third semiconductor substrate 213 are prepared which are respectively provided with the multilayer wire layers. As shown in FIG. 30, the first semiconductor substrate 211 is provided with the multilayer wire layer 245, the second semiconductor substrate 212 is provided with the multilayer wire layer 255, and the third semiconductor substrate 213 is provided with the multilayer wire layer 345.

In addition, as shown in FIG. 30, the aluminum pad 280 is formed in the multilayer wire layer 245 of the first semiconductor substrate 211. Further, the metal wire 250 and the metal wire 250b are formed in the multilayer wire layer 255 of the second semiconductor substrate, and the

22

metal wire 340a and the metal wire 340b are formed in the multilayer wire layer 345 of the third semiconductor substrate.

Next, as shown in FIG. 31, the second semiconductor substrate 212 is joined to the third semiconductor substrate 213. At this time, the second semiconductor substrate 212 is joined to the third semiconductor substrate 213 such that the multilayer wire layer 255 and the multilayer wire layer 345 face each other. In addition, the metal wire 250a is directly bonded to the metal wire 340a, and the metal wire 250b is directly bonded to the metal wire 340b.

In addition, the second semiconductor substrate 212 is thinned. In FIG. 31, the width of the second semiconductor substrate 212 in the vertical direction of FIG. 31 is reduced.

Next, as shown in FIG. 32, the first semiconductor substrate 211 is joined to the second semiconductor substrate 212. At this time, the multilayer wire layer 255 of the second semiconductor substrate faces the third semiconductor substrate 213 side (the lower side of FIG. 32), and the first semiconductor substrate 211 is joined to the second semiconductor substrate 212.

In addition, the first semiconductor substrate 211 is thinned. In FIG. 32, the width of the first semiconductor substrate 211 in the vertical direction of FIG. 32 is reduced.

Next, the contact 267 is formed as shown in FIG. 33. At this time, a hole is provided which reaches the multilayer wire layer 245 from the light receiving surface of the first semiconductor substrate 211, and a hole is provided which reaches the multilayer wire layer 255 from the light receiving surface, thereby forming the contact 267.

In addition, as shown in FIG. 34, the pad hole 351 is formed so as to reach the aluminum pad 280a from the rear surface side (light receiving surface side) of the first semiconductor substrate 211.

In this way, the solid-state imaging device described with reference to FIG. 29 is manufactured. Since not a contact but direction bonding is used for electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213, it is possible to simplify manufacturing steps and to reduce the area occupying the substrate.

Although the description has been provided here of a case where the direct bonding is used as an electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213, the direct bonding may be used as an electrical connection between the first semiconductor substrate 211 and the second semiconductor substrate 212.

Similarly, in the solid-state imaging device with the configuration described with reference to FIG. 11, 12 or 20, the direct bonding may also be used as an electrical connection between the first semiconductor substrate 211 and the second semiconductor substrate 212, or as an electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213.

In other words, in the configuration (FIG. 11) in which the aluminum pad 280 is provided in the multilayer wire layer 245 of the first semiconductor substrate 211, the direct bonding may be used as an electrical connection between the respective semiconductor substrates. In addition, in the configuration (FIG. 12) in which the aluminum pad 280 is provided in the multilayer wire layer 255 of the second semiconductor substrate 212, the direct bonding may be used as an electrical connection between the respective semiconductor substrates. Further, in the configuration (FIG. 20) in which the aluminum pad 280 is provided in the

insulating film layer 230, the direct bonding may be used as an electrical connection between the respective semiconductor substrates.

FIG. 35 is a schematic diagram of the cross-sectional view illustrating the configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

In the configuration shown in FIG. 35, unlike in the case of FIG. 29, a contact 268 and a contact 316 used as an electrical connection between the first semiconductor substrate 211 and the second semiconductor substrate 212 are provided. In other words, in a case of the configuration shown in FIG. 35, the lower left end of the contact 268 in FIG. 35 is connected to the upper end of the contact 316 in FIG. 35, and thereby the first semiconductor substrate 211 is electrically connected to the second semiconductor substrate 212. In addition, the contact 268 is formed by a twin contact.

In the configuration shown in FIG. 35, it is unnecessary to provide a hole which reaches the multilayer wire layer 255 from the light receiving surface, for example, unlike in forming the contact 267 of FIG. 29. For this reason, it is possible to more simply form a contact.

A configuration of the other elements in FIG. 35 is the same as in the case of FIG. 29, and detailed description thereof will be made.

Next, a manufacturing process of the solid-state imaging device shown in FIG. 35 will be described.

First, as shown in FIG. 36, the first semiconductor substrate 211, the second semiconductor substrate 212, and the third semiconductor substrate 213 are prepared which are respectively provided with the multilayer wire layers. As shown in FIG. 36, the first semiconductor substrate 211 is provided with the multilayer wire layer 245, the second semiconductor substrate 212 is provided with the multilayer wire layer 255, and the third semiconductor substrate 213 is provided with the multilayer wire layer 345.

In addition, as shown in FIG. 36, the aluminum pad 280 is formed in the multilayer wire layer 245 of the first semiconductor substrate 211. Further, the metal wire 250a and the metal wire 250b are formed in the multilayer wire layer 255 of the second semiconductor substrate, and the metal wire 340a and the metal wire 340b are formed in the multilayer wire layer 345 of the third semiconductor substrate.

Next, as shown in FIG. 37, the second semiconductor substrate 212 is joined to the third semiconductor substrate 213. At this time, the second semiconductor substrate 212 is joined to the third semiconductor substrate 213 such that the multilayer wire layer 255 and the multilayer wire layer 345 face each other. In addition, the metal wire 250a is directly bonded to the metal wire 340a, and the metal wire 250b is directly bonded to the metal wire 340b.

In addition, the second semiconductor substrate 212 is thinned. In FIG. 37, the width of the second semiconductor substrate 212 in the vertical direction of FIG. 31 is reduced.

In addition, as shown in FIG. 38, the contact 316 is formed. At this time, a hole is provided which reaches the multilayer wire layer 255 from the upper surface of the second semiconductor substrate 212 in FIG. 38 so as to form the contact 316.

Next, as shown in FIG. 39, the first semiconductor substrate 211 is joined to the second semiconductor substrate 212. At this time, the first semiconductor substrate 211 is joined to the second semiconductor substrate 212 such that the rear surface of the first semiconductor substrate 211 becomes a light receiving surface.

In addition, the first semiconductor substrate 211 is thinned. In FIG. 39, the width of the first semiconductor substrate 211 in the vertical direction of FIG. 39 is reduced.

Further, a hole is provided which reaches the upper surface of the second semiconductor substrate in FIG. 39 from the light receiving surface of the first semiconductor substrate 211, and a hole is provided which reaches the aluminum pad 280 of the multilayer wire layer 245 from the light receiving surface, thereby forming the contact 268.

In addition, as shown in FIG. 40, the pad hole 351 is formed so as to reach the aluminum pad 280a from the light receiving surface of the first semiconductor substrate 211.

In this way, the solid-state imaging device described with reference to FIG. 35 is manufactured. In the configuration shown in FIG. 35, as described above, the contact 268 and the contact 316 are used as an electrical connection between the first semiconductor substrate 211 and the second semiconductor substrate 212. In other words, a conductor forming the contact 268 is bonded to a conductor forming the contact 316 at the bonding surface between the first semiconductor substrate 211 and the second semiconductor substrate 212. As above, in a case of the configuration shown in FIG. 35, a part of the twin contact for an electrical connection between the first semiconductor substrate 211 and the second semiconductor substrate 212 is formed in two divided steps.

In this way, it is unnecessary to provide a deep hole which reaches the multilayer wire layer 255 from the light receiving surface, for example, unlike in forming the contact 267 of FIG. 29. For this reason, it is possible to more simply form a contact.

Although the description has been provided here of a case where a part of the twin contact used for an electrical connection between the first semiconductor substrate 211 and the second semiconductor substrate 212 is formed in two divided steps, a part of a twin contact used for an electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213 may be formed in two divided steps.

In addition, similarly, in the solid-state imaging device with the configuration described with reference to FIG. 11, 12 or 20, a part of a twin contact, used for an electrical connection between the first semiconductor substrate 211 and the second semiconductor substrate 212, or electrical connection between the second semiconductor substrate 212 and the third semiconductor substrate 213, may be formed in two divided steps.

In other words, in the configuration (FIG. 11) in which the aluminum pad 280 is provided in the multilayer wire layer 245 of the first semiconductor substrate 211, a part of a twin contact used for an electrical connection between the respective semiconductor substrates may be formed in two divided steps. In addition, in the configuration (FIG. 12) in which the aluminum pad 280 is provided in the multilayer wire layer 255 of the second semiconductor substrate 212, a part of a twin contact used for an electrical connection between the respective semiconductor substrates may be formed in two divided steps. Further, in the configuration (FIG. 20) in which the aluminum pad 280 is provided in the insulating film layer 230, a part of a twin contact used for an electrical connection between the respective semiconductor substrates may be formed in two divided steps.

As described with reference to FIGS. 11 to 40, in the solid-state imaging device to which the present technology is applied, the aluminum pad 280 may be provided in the multilayer wire layer 245 of the first semiconductor substrate 211, the aluminum pad 280 may be provided in the

25

multilayer wire layer 255 of the second semiconductor substrate 212, and the aluminum pad 280 may be provided in the insulating film layer 230. In addition, as a form of electrical connection between the respective semiconductor substrates, a twin contact, a shared contact, direct bonding, and a configuration in which a part of a twin contact is formed in two divided steps may be employed.

In other words, combinations as shown in FIG. 41 may be employed as embodiments of the solid-state imaging device to which the present technology is applied.

In addition, in the above-described embodiments, the embodiments of the solid-state imaging device to which the present technology is applied have been described based on a three-layer stacked structure. However, the solid-state imaging device to which the present technology is applied may employ, for example, a four-layer structure in which a first semiconductor substrate, a second semiconductor substrate, a third semiconductor substrate, and a fourth semiconductor substrate are stacked.

FIG. 42 shows an example of a case where the solid-state imaging device to which the present technology is applied employs the four-layer structure. FIG. 42 is a schematic diagram of a cross-sectional view illustrating a configuration according to still another embodiment of the pixel portion of the solid-state imaging device to which the present technology is applied.

In the example shown in FIG. 42, a four-layer structure is employed in which a first semiconductor substrate 211, a second semiconductor substrate 212, a third semiconductor substrate 213, and a fourth semiconductor substrate 214 are stacked.

In addition, similarly, the solid-state imaging device to which the present technology is applied may employ a structure of five or more layers.

FIG. 43 is a block diagram illustrating a configuration example of a camera apparatus which is an electronic apparatus to which the present technology is applied.

A camera apparatus 600 in FIG. 43 includes an optical unit 601 including a lens group and the like, a solid-state imaging device (imaging device) 602 which employs the above-described respective configurations of the pixels 402, and a DSP circuit 603 which is a camera signal processing circuit. In addition, the camera apparatus 600 includes a frame memory 604, a display unit 605, a recording unit 606, an operation unit 607, and a power supply unit 608. The DSP circuit 603, the frame memory 604, the display unit 605, the recording unit 606, the operation unit 607, and the power supply unit 608 are connected to each other via a bus line 609.

The optical unit 601 receives incident light (image light) from a subject so as to be imaged on an imaging surface of the solid-state imaging device 602. The solid-state imaging device 602 converts a light amount of the incident light which is imaged on the imaging surface by the optical unit 601 into an electric signal in the unit of a pixel and outputs the electric signal as a pixel signal. The solid-state imaging device related to the above-described embodiments may be used as the solid-state imaging device 602.

The display unit 605 includes, for example, a panel type display device such as a liquid crystal panel or an organic electroluminescence (EL) panel, and displays moving images or still images captured by the solid-state imaging device 602. The recording unit 606 records moving images or still images captured by the solid-state imaging device 602 on a recording medium such as a video tape or a digital versatile disk (DVD).

26

The operation unit 607 issues operation commands for various functions of the camera apparatus 600 in response to an operation by a user. The power supply unit 608 appropriately supplies a variety of power which is operation power of the DSP circuit 603, the frame memory 604, the display unit 605, the recording unit 606, and the operation unit 607, to the supply targets.

In addition, the present technology is not limited to being applied to a solid-state imaging device which detects a distribution of an incident light amount of visible light so as to capture an image, and is generally applicable to a solid-state imaging device which detects a distribution of an incidence amount of infrared rays, X rays, particles, or the like so as to capture an image, or a solid-state imaging device (physical quantity distribution detection device) such as a fingerprint detection sensor which detects distributions of the other physical quantities such as a pressure or a capacitance so as to capture an image in a broad sense.

In addition, embodiments of the present technology are not limited to the above-described embodiments and may have various modifications within the scope without departing from the spirit of the present technology.

Further, the present technology may have the following configurations.

- (1) A solid-state imaging device including a first semiconductor substrate that includes a sensor circuit provided with a photoelectric conversion portion; and a second semiconductor substrate and a third semiconductor substrate that respectively include circuits different from the sensor circuit, wherein the first semiconductor substrate is located in the uppermost layer, and the first semiconductor substrate, the second semiconductor substrate, and the third semiconductor substrate are stacked in three layers, and wherein an electrode metal element forming an electrode for external connection is disposed in the first semiconductor substrate.
- (2) The solid-state imaging device according to (1), wherein the sensor circuit of the first semiconductor substrate is of a rear surface irradiation type, and wherein a hole exposing the electrode metal element is opened from a light receiving surface side of the first semiconductor substrate.
- (3) The solid-state imaging device according to (1) or (2), wherein the second semiconductor substrate or the third semiconductor substrate includes a logical circuit or a memory circuit, and wherein the logical circuit or the memory circuit is operated based on input and output of signals to and from an external device.
- (4) The solid-state imaging device according to any one of (1) to (3), wherein a light blocking mechanism which blocks light incident to the photoelectric conversion portion from an opposite side to a light receiving surface of the first semiconductor substrate is provided in at least one of the first semiconductor substrate and the second semiconductor substrate.
- (5) The solid-state imaging device according to (4), wherein the light blocking mechanism is formed by the electrode metal element.
- (6) The solid-state imaging device according to (4), wherein a wire metal element used for a wire of the second semiconductor substrate is disposed in the second semiconductor substrate, and wherein the light blocking mechanism is formed by the electrode metal element and the wire metal element.

- (7) The solid-state imaging device according to (4), wherein the light blocking mechanism is formed by a light blocking body disposed in the second semiconductor substrate.
- (8) The solid-state imaging device according to (1), wherein a wire metal element used for a wire of the first semiconductor substrate is further disposed in the first semiconductor substrate, and wherein the electrode metal element is disposed at a position closer to an adhesive surface with the second semiconductor substrate than the wire metal element.
- (9) The solid-state imaging device according to any one of (1) to (8), wherein a contact, which penetrates through the first semiconductor substrate or the second semiconductor substrate and reaches a metal wire layer of the second semiconductor substrate or the third semiconductor substrate, a part of which reaches a wire of a wire metal layer of the first semiconductor substrate or the second semiconductor substrate, is used for electrical connection between the first semiconductor substrate and the second semiconductor substrate or electrical connection between the second semiconductor substrate and the third semiconductor substrate.
- (10) The solid-state imaging device according to any one of (1) to (8), wherein a part of a contact used for electrical connection between the first semiconductor substrate and the second semiconductor substrate or electrical connection between the second semiconductor substrate and the third semiconductor substrate is formed by connecting conductors to each other at a bonding surface between the first semiconductor substrate and the second semiconductor substrate or a bonding surface between the second semiconductor substrate and the third semiconductor substrate.
- (11) The solid-state imaging device according to any one of (1) to (8), wherein wires exposed to a bonding surface between the first semiconductor substrate or the second semiconductor substrate and the second semiconductor substrate or the third semiconductor substrate are bonded to each other such that the first semiconductor substrate is electrically connected to the second semiconductor substrate.
- (12) The solid-state imaging device according to (1), wherein the first semiconductor substrate and the second semiconductor substrate are stacked such that a metal wire layer of the second semiconductor substrate comes into contact with the first semiconductor substrate, and wherein an electrode metal element forming an electrode for external connection is disposed inside a metal wire layer of the second semiconductor substrate.
- (13) The solid-state imaging device according to (1), wherein an insulating film layer is formed between the first semiconductor substrate and the second semiconductor substrate, wherein the first semiconductor substrate and the second semiconductor substrate are stacked such that a metal wire layer of the second semiconductor substrate comes into contact with the insulating film layer, and wherein an electrode metal element forming an electrode for external connection is disposed inside the insulating film layer.
- (14) An electronic apparatus including a solid-state imaging device having a first semiconductor substrate that includes a sensor circuit provided with a photoelectric conversion portion; and a second semiconductor substrate and a third semiconductor substrate that respectively include circuits different from the sensor circuit,

wherein the first semiconductor substrate is located in the uppermost layer, and the first semiconductor substrate, the second semiconductor substrate, and the third semiconductor substrate are stacked in three layers, and wherein an electrode metal element forming an electrode for external connection is disposed in the first semiconductor substrate.

<1> A semiconductor device comprising:

- a first semiconductor section including a first wiring layer at one side thereof, the first semiconductor section further including a photodiode;
- a second semiconductor section including a second wiring layer at one side thereof, the first and second semiconductor sections being secured together;
- a third semiconductor section including a third wiring layer at one side thereof, the second and the third semiconductor sections being secured together such the first semiconductor section, second semiconductor section, and the third semiconductor section are stacked together; and
- a first conductive material electrically connecting at least two of (i) the first wiring layer, (ii) the second wiring layer, and (iii) the third wiring layer such that the electrically connected wiring layers are in electrical communication.

<2> The semiconductor device of <1>, wherein the first semiconductor section, the second semiconductor section, and the third semiconductor section are stacked together in a manner such that the first wiring layer faces the second wiring layer or the second wiring layer faces the third wiring layer.

<3> The semiconductor device of <2>, further comprising: a second conductive material electrically connecting at least two of (i) the first wiring layer, (ii) the second wiring layer, and (iii) the third wiring layer such that the electrically connected wiring layers are in electrical communication.

<4> The semiconductor device of <3>, wherein at least one wiring layer electrically connected by the second conductive material is different than the wiring layers electrically connected by the first conductive material.

<5> The semiconductor device of <3> or <4>, wherein at least one of the first conductive material and the second conductive material comprises two through holes which penetrate through at least one of the first semiconductor section and the second semiconductor section in a vertical direction; and wherein a first through hole of the first conductive material electrically connects to a wiring layer that is different than a wiring layer electrically connected by a second of the two through holes.

<6> The semiconductor device of <3> or <4> wherein at least one of the first conductive material and the second conductive material comprises a single through hole which penetrates through at least one of the first semiconductor section and the second semiconductor section in a vertical direction such that the at least one of the first conductive material and the second conductive material electrically contact at least two wiring layers.

<7> The semiconductor device of any one of <1> to <6>, wherein a metal wire of at least one of (i) the first wiring layer, (ii) the second wiring layer, and (ii) the third wiring layer is directly bonded to a metal wire in another wiring layer.

<8> The semiconductor device of <7>, wherein at least one of the wiring layers having a metal wire directly

29

- bonded is different than the wiring layers electrically connected by the first conductive material.
- <9> The semiconductor device of any one of <1> to <8>, further comprising a pad electrode for external connection. 5
- <10> The semiconductor device of <9>, wherein the pad is disposed such that light is blocked from one or more transistors residing in the second semiconductor section.
- <11> The semiconductor device of any one of <1> to <10>, wherein the first semiconductor section comprises a sensor circuit, at least one of the second semiconductor section and the third semiconductor section comprises a logical circuit, and at least one of the second semiconductor section and the third semiconductor section comprises a memory circuit. 10 15
- <12> A backside illumination type solid-state imaging device comprising:
- a first semiconductor section including a first wiring layer at one side thereof, the first semiconductor section further including a circuit region and a pixel region;
 - a second semiconductor section including a second wiring layer at one side thereof, the first and second semiconductor sections being secured together; 25
 - a third semiconductor section including a third wiring layer at one side thereof, the second and the third semiconductor sections being secured together such that the first semiconductor section, second semiconductor section, and the third semiconductor section are stacked together; and 30
 - a first conductive material electrically connecting at least two of (i) the first wiring layer, (ii) the second wiring layer, and (iii) the third wiring layer such that the electrically connected wiring layers are in electrical communication. 35
- <13> The solid-state imaging device of <12>, wherein the first semiconductor section, the second semiconductor section, and the third semiconductor section are stacked together in a manner such that the first wiring layer faces the second wiring layer or the second wiring layer faces the third wiring layer. 40
- <14> The solid-state imaging device of <13>, further comprising: a second conductive material electrically connecting at least two of (i) the first wiring layer, (ii) the second wiring layer, and (iii) the third wiring layer such that the electrically connected wiring layers are in electrical communication. 45
- <15> The solid-state imaging device of <14>, wherein at least one of the first conductive material and the second conductive material comprises a single through hole which penetrates through at least one of the first semiconductor section and the second semiconductor section in a vertical direction such that the at least one of the first conductive material and the second conductive material electrically contact at least two wiring layers. 50 55
- <16> The solid-state imaging device of any one of <12> to <15>, wherein a metal wire of at least one of (i) the first wiring layer, (ii) the second wiring layer, and (ii) the third wiring layer is directly bonded to a metal wire in another wiring layer. 60
- <17> The solid-state imaging device of any one of <12> to <16>, wherein at least one of the wiring layers having a metal wire directly bonded is different than the wiring layers electrically connected by the first conductive material. 65

30

- <18> The solid-state imaging device of any one of <12> to <17>, further comprising a pad electrode for external connection, wherein the pad is disposed below the pixel region such that light is blocked from one or more transistors residing in the second semiconductor section.
- <19> The solid-state imaging device of any one of <12> to <18>, further comprising an interlayer insulating film disposed between at least two of the semiconductor sections.
- <20> The solid-state imaging device of any one of <12> to <19>, wherein the first semiconductor section comprises a sensor circuit, at least one of the second semiconductor section and the third semiconductor section comprises a logical circuit, and at least one of the second semiconductor section and the third semiconductor section comprises a memory circuit.
- <21> An electronic apparatus including:
an optical unit; and
- (a) a solid-state imaging device including: a first semiconductor section including a first wiring layer at one side thereof, the first semiconductor section further including a circuit region and a pixel region;
 - a second semiconductor section including a second wiring layer at one side thereof, the first and second semiconductor sections being secured together;
 - (b) a third semiconductor section including a third wiring layer at one side thereof, the second and the third semiconductor sections being secured together such that the first semiconductor section, second semiconductor section, and the third semiconductor section are stacked together; and
 - (c) a first conductive material electrically connecting at least two of (i) the first wiring layer, (ii) the second wiring layer, and (iii) the third wiring layer such that the electrically connected wiring layers are in electrical communication.
- <22> The electronic apparatus of <21>, wherein the first semiconductor section, the second semiconductor section, and the third semiconductor section are stacked together in a manner such that the first wiring layer faces the second wiring layer or the second wiring layer faces the third wiring layer.
- <23> The electronic apparatus of <22>, further comprising: a second conductive material electrically connecting at least two of (i) the first wiring layer, (ii) the second wiring layer, and (iii) the third wiring layer such that the electrically connected wiring layers are in electrical communication, wherein at least one wiring layer electrically connected by the second conductive material is different than the wiring layers electrically connected by the first conductive material.
- <24> The electronic apparatus of <23>, wherein at least one of the first conductive material and the second conductive material comprises two through holes which penetrate through at least one of the first semiconductor section and the second semiconductor section in a vertical direction; and wherein a first through hole of the first conductive material electrically connects to a wiring layer that is different than a wiring layer electrically connected by a second of the two through holes.
- <25> The electronic apparatus of <23>, wherein at least one of the first conductive material and the second conductive material comprises a single through hole which penetrates through at least one of the first semiconductor section and the second semiconductor

31

section in a vertical direction such that the at least one of the first conductive material and the second conductive material electrically contact at least two wiring layers.

<26> The electronic apparatus of <23>, wherein a metal wire of at least one of (i) the first wiring layer, (ii) the second wiring layer, and (ii) the third wiring layer is directly bonded to a metal wire in another wiring layer, and wherein at least one of the wiring layers having a metal wire directly bonded is different than the wiring layers electrically connected by the first conductive material.

It should be understood by those skilled in the art that various modifications, combinations, sub-combinations and alterations may occur depending on design requirements and other factors insofar as they are within the scope of the appended claims or the equivalents thereof.

REFERENCE SIGNS LIST

211 First semiconductor substrate
212 Second semiconductor substrate
213 Third semiconductor substrate
230 Insulating film layer
234 Photodiode
240 Copper wire
245 Multilayer wire layer
250 Copper wire
255 Multilayer wire layer
265 Contact
266 Contact
267 Contact
280 Aluminum pad
311 Contact
312 Contact
313 Contact
320 Aluminum pad
330 Aluminum pad
340 Copper wire
345 Multilayer wire layer
351 Pad hole
360 Light blocking body
370 Copper wire
401 Solid-state imaging device
402 Pixel
600 Camera apparatus
602 Solid-state imaging device

What is claimed is:

1. A light detecting device, comprising:
a first section including a first semiconductor layer and a first layer including a first electrode;
a second section including a second semiconductor layer, a second layer including a second electrode, and a third layer including a third electrode, wherein the second layer and the third layer are separated from another by the second semiconductor layer;
a third section including a third semiconductor layer and a fourth layer including a fourth electrode; and
a through electrode, wherein the through electrode is a unitary structure, wherein a first portion of the unitary structure of the through electrode extends into the second layer, wherein a second portion of the unitary structure of the through electrode extends through the second semiconductor layer, and wherein a third portion of the unitary structure of the through electrode extends into the third layer and terminates at an electrode disposed in the third layer,

32

wherein the first section, the second section, and the third section are stacked such that the first layer and the second layer face each other, and the third layer and the fourth layer face each other,

wherein the first electrode is directly bonded to the second electrode, and

wherein the third electrode is directly bonded to the fourth electrode.

2. The light detecting device according to claim 1, wherein the first layer, the second layer, the third layer, and the fourth layer are multilayer wire layers.

3. The light detecting device according to claim 1, wherein the first layer is joined to the second layer.

4. The light detecting device according to claim 3, wherein the third layer is joined to the fourth layer.

5. The light detecting device according to claim 1, wherein the through electrode is in contact with a wiring of the third layer.

6. The light detecting device according to claim 1, further comprising an electrode pad.

7. The light detecting device according to claim 6, wherein the electrode pad is disposed in the first section.

8. The light detecting device according to claim 7, wherein the electrode pad is disposed in the first section and is exposed by a pad hole.

9. The light detecting device according to claim 6, wherein the electrode pad is for connecting the light detecting device to an external device.

10. The light detecting device according to claim 1, wherein the first section includes a plurality of photoelectric conversion portions.

11. The light detecting device according to claim 10, wherein the second section includes a logic circuit.

12. The light detecting device according to claim 11, wherein the third section includes a memory circuit.

13. The light detecting device according to claim 1, wherein the first section is on a light incident surface side of the light detecting device.

14. An electronic apparatus, comprising:
an optical unit including a lens group; and
a light detecting device receiving incident light from the optical unit, the light detecting device including:
a first section including a first semiconductor layer and a first layer including a first electrode;
a second section including a second semiconductor layer, a second layer including a second electrode, and a third layer including a third electrode, wherein the second layer and the third layer are separated from another by the second semiconductor layer;
a third section including a third semiconductor layer and a fourth layer including a fourth electrode; and
a through electrode, wherein the through electrode is a unitary structure, wherein a first portion of the unitary structure of the through electrode extends into the second layer, wherein a second portion of the unitary structure of the through electrode extends through the second semiconductor layer, and wherein a third portion of the unitary structure of the through electrode extends into the third layer and terminates at an electrode disposed in the third layer,
wherein the first section, the second section, and the third section are stacked such that the first layer and the second layer face each other, and the third layer and the fourth layer face each other,
wherein the first electrode is directly bonded to the second electrode, and

33

wherein the third electrode is directly bonded to the fourth electrode.

15. The electronic apparatus according to claim **14**, wherein the first section is on a light incident surface side of the light detecting device.

5

* * * * *

34